Self-fulfilling Volatility, Taylor Rules, and Equilibrium Selection*

Seung Joo Lee† Marc Dordal i Carreras‡

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Abstract

We show that the canonical New Keynesian model admits multiple bounded global equilibria driven by self-fulfilling aggregate volatility: when households anticipate more volatile future consumption, they raise precautionary saving, so that output falls, and their state-contingent plans for future consumption confirm the same volatility they fear. Conventional Taylor rules restore determinacy only locally and cannot select among these global solutions, so uniqueness must instead be restored by other means. Two modifications of the baseline environment select the stabilized path as the unique bounded equilibrium: small behavioral frictions that weaken coordination on expected future outcomes, and a state-contingent fiscal intervention.

Keywords: Taylor Rules, Self-fulfilling Volatility, Precautionary Savings, Equilibrium Selection

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†Department of Finance, Saïd Business School, University of Oxford, Oxford, United Kingdom. Email: seung.lee@sbs.ox.ac.uk. Corresponding Author.

‡Department of Economics, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong SAR. Email: marcdordal@ust.hk.

Note: The ordering of the authors' names is not indicative of relative contributions; it was chosen arbitrarily (or by mutual agreement) for practical reasons.

1 Introduction

There is a large literature on equilibrium multiplicity in New Keynesian models, and Taylor rules with a sufficiently aggressive response to the output gap or inflation are often thought to restore determinacy. We show that this local uniqueness result does not survive once the models are solved globally rather than approximated around their deterministic steady state. Additional solutions then appear, driven by self-fulfilling aggregate volatility, and they satisfy the same boundedness restrictions commonly used by the literature to discard explosive paths (Cochrane, 2011).

In the canonical New Keynesian model, aggregate volatility affects consumption through the precautionary savings channel. When households anticipate higher future consumption volatility, they increase saving and reduce current spending. Because output is demand-determined under nominal rigidities, this reduction in demand lowers current output, shifting the economy to a lower baseline level of activity. State-contingent plans for future consumption then sustain that decline by validating the anticipated volatility, so the original expectation is self-fulfilling.

This intuition can be used to construct a specific equilibrium. Suppose that in Period 0 households fear higher aggregate volatility of Period 1 consumption. They increase precautionary saving and reduce current consumption, so that Period 0 output falls. In Period 1, the original fear is confirmed if each consumption realization is paired with a conditional volatility of Period 2 consumption that supports it. Higher consumption in Period 1 is supported by lower subsequent volatility and therefore less precautionary saving; lower consumption is supported by higher subsequent volatility and therefore more precautionary saving. Households' beliefs about next-period consumption volatility are then confirmed by the consumption choices specified in the contingent plan, with the fundamental shock process (in this paper, a technology shock) providing the state space on which those beliefs coordinate. That process may also generate fundamental income volatility, but this is not necessary: one of the equilibrium classes we construct below supports non-degenerate self-fulfilling volatility even as fundamental volatility vanishes, and in that limit the shock serves purely as a coordinating device. In addition, the construction does not require the incomplete information mechanisms that often generate sentiment-driven business cycles in the literature. The equilibria we present require only the nominal price rigidities commonly assumed in the New Keynesian model, and those rigidities can be arbitrarily small. We first derive analytical examples in a simplified model with perfectly rigid prices, then turn to sticky prices with a linearized Phillips curve, and finally to a numerical solution of the fully nonlinear sticky-price model.

We construct two classes of such equilibria. In the first, the output gap follows a local martingale, meaning that, on average, the economy remains at its current level from one period to the next. In this solution, the stabilized path—the flexible-price economy benchmark—acts as an attractor for almost every sample path, and both the output gap and the self-fulfilling component of volatility converge to zero almost surely. Until then, the economy remains in recession with elevated volatility. We prove in this case that almost-sure long-run stabilization is consistent with the construction because volatility can only become arbitrarily large along paths that occur with vanishing probability, which is sufficient to sustain an equilibrium that is temporarily away from the stabilized path, while satisfying the boundedness restriction.

The second class of global solutions we consider is stationary: aggregate volatility and other aggregates continue to fluctuate stochastically, but they do so according to a stable long-run distribution, and the associated long-run position is characterized by under-production (i.e., a negative output gap).

Both constructions rely on intertemporal coordination of contingent consumption plans and can be obtained when policy is described by a Taylor rule alone. Each of these features can be relaxed to recover the stabilized path as the unique bounded equilibrium. We prove that a small friction in social memory (Angeletos and Lian, 2023) or the attenuation of the general equilibrium effect under level-k reasoning (Farhi and Werning, 2019) undoes the coordination. A fiscal escape clause rules out the same alternative paths in a non-Ricardian overlapping-generations economy: the government issues a debt-financed transfer only off the equilibrium path, and the transfer is never paid along the stabilized equilibrium. Escape clauses and behavioral frictions are therefore more robust ways to restore uniqueness than Taylor rules.

Taylor rules remain useful as simple, systematic guidelines when rules are preferred to discretion (Taylor, 1993) and as a parsimonious description of the policy response that can serve as a practical benchmark (Clarida et al., 1999; Taylor, 1999; Woodford, 2001; Orphanides, 2003; Nakamura et al., 2025). That is a valuable role: a Taylor rule offers a transparent way to summarize how the central bank responds to inflation and real activity, and a useful reference point when many shocks are present. Selecting among the global equilibria we construct is a separate problem, and one for which escape clauses and behavioral frictions are better suited.

Related literature Our non-linear characterization of the model shares similarities with Caballero and Simsek (2020a,b) in terms of incorporating aggregate volatility, which affects the business cycle. However, while their framework focuses on how behavioral biases can generate intriguing crisis dynamics through the feedback loop between asset markets and business cycles,[1] our attention centers on the traditional policy rule under rational expectations and the existence of alternative equilibria arising from *endogenous* higher-order moments.

While Benhabib et al. (2002) study monetary-fiscal regimes addressing indeterminacy at the ZLB, and Obstfeld and Rogoff (2021) show how a probabilistic (and small) fiscal currency backing can rule out speculative hyperinflation, our focus is the self-fulfilling emergence of aggregate volatility outside the ZLB and the selection of the full-stabilization equilibrium by escape clauses and behavioral frictions.

There is a large macro-finance literature on the self-fulfilling nature of real and financial uncertainty: Bacchetta et al. (2017), Fajgelbaum et al. (2017),[2] Heathcote and Perri (2018), Benhabib et al. (2019, 2024), and Chan (2024). Bacchetta et al. (2017) characterize an endowment economy where current asset prices are affected by a sunspot that shifts the perceived risk of future asset prices. In Heathcote and Perri (2018), wealth-poor households who expect high unemployment cut spending through precautionary savings, leading to a self-fulfilling rise in unemployment. Benhabib et al. (2019) develop a model of "mutual learning" between financial markets and the real economy, leading to strategic complementarity and self-fulfilling uncertainty. Benhabib et al. (2024) study a model of aggregate demand externality,[3] where a positive feedback loop between aggregate output and

[1] Caballero and Simsek (2020b) present a model with optimists and pessimists who hold differing beliefs about the probability of an imminent recession or normal state. During zero lower bound (ZLB) episodes, an endogenous decline in risky asset valuation, triggered by a reduction in optimists' wealth, leads to a demand recession. We explore related ZLB issues in a separate paper, Dordal i Carreras and Lee (2025).

[2] In Fajgelbaum et al. (2017), higher uncertainty about fundamentals leads to lower investment, slowing down information flows and further discouraging investment. This results in 'uncertainty traps' characterized by self-fulfilling uncertainty and low activity.

[3] For a modern treatment of this issue, see Farhi and Werning (2016).

defaults generates a self-fulfilling default cycle.[4] We abstract from defaults and focus on the self-fulfilling appearance of aggregate volatility in a standard model with nominal rigidities and aggregate demand externalities. The literature on incomplete information and sentiments (Benhabib et al., 2015; Acharya et al., 2021; Chan, 2024) departs from the complete information benchmark in a linearized New Keynesian framework, showing that beliefs about aggregate demand can be self-fulfilling. In contrast, we assume *complete information* and solve the textbook New Keynesian model non-linearly, revealing that this approach can also generate self-fulfilling aggregate volatility.[5]

Our equilibrium multiplicity results are most closely related to Acharya and Dogra (2020) and Khorrami and Mendo (2025). While Acharya and Dogra (2020) study how determinacy conditions change when exogenous *idiosyncratic* volatilities depend on aggregate output, we construct bounded global equilibria driven by endogenous aggregate volatility in the canonical New Keynesian model. In a contemporaneous paper, Khorrami and Mendo (2025) also analyze indeterminacy linked to aggregate volatility and propose active fiscal rules as an alternative determinacy mechanism. We develop explicit martingale and stationary constructions of bounded global equilibria, characterize the intertemporal coordination that sustains them, and study both behavioral and fiscal mechanisms that select the stabilized path.

Layout Section 2 presents the model and derives the key equilibrium conditions. Section 3 constructs the bounded global equilibria that arise under a standard Taylor rule. Section 4 studies mechanisms that recover determinacy and discusses the remaining role of Taylor rules. Section 5 extends the multiplicity results to an environment with sticky prices. Section 6 concludes.

2 Standard Non-linear New Keynesian Model

This section describes the main assumptions and presents the exact non-linear optimality conditions of a standard New Keynesian model. To build analytical

[4] When aggregate output falls, it raises defaults as firms' revenues and profits decline. As defaults disrupt production, they further decrease aggregate output, ad infinitum.

[5] For other recent works on self-fulfilling fluctuations in the business cycle and financial markets, see e.g., Kaplan and Menzio (2016) and Gârleanu and Panageas (2021).

intuition for our subsequent results on equilibrium multiplicity, we first consider a simplified version of the model with perfectly rigid prices—an assumption relaxed later in Section 5. Appendix II contains detailed derivations and technical aspects underlying the results presented here.

Note that our representative agent framework assumes complete information under rational expectations, unlike previous works that introduce beauty contests and/or imperfect information, e.g., Benhabib et al. (2015), Acharya et al. (2021), Chan (2024).

2.1 Households

Consider an economy with a representative household h, whose optimization problem is given by

$$\Gamma_t^h \equiv \max_{\substack{\{C_s^h, L_s^h\}_{s \geq t} \\ \{B_s^h\}_{s>t}}} \mathbb{E}_t \int_t^\infty e^{-\rho(s-t)} \left[\log C_s^h - \frac{\left(L_s^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right] ds, \tag{1}$$

subject to the budget constraint

$$\dot{B}_t^h = i_t B_t^h - \bar{p} C_t^h + w_t L_t^h + D_t, \tag{2}$$

where C_t^h and L_t^h denote household consumption and labor supply, respectively. Here, η is the Frisch elasticity of labor supply, ρ is the time discount rate, and B_t^h represents nominal bond holdings, which are in zero net supply in equilibrium. The household receives lump-sum transfers D_t, which include firms' profits and government transfers. The equilibrium wage is w_t, and the policy rate set by the central bank is i_t. The household's value function is denoted by Γ_t^h.

We simplify the analysis by assuming a perfectly rigid price level: $p_t = \bar{p}$, $\forall t$, implying zero inflation ($\pi_t = 0$ for all t). Although not crucial, this assumption allows us to illustrate clearly the key mechanisms of interest.[6] There is no government spending, and aggregate consumption fully determines production.

[6]Section 5 relaxes the assumption of rigid prices by incorporating price stickiness à la Rotemberg (1982), showing that the multiplicity of equilibria associated with aggregate volatility persists in a model with inflation. The rigid price assumption, adopted without loss of generality, simplifies the New Keynesian Phillips curve ($\pi_t = 0$ for all t) and facilitates analytical derivations of global equilibria.

We obtain the intertemporal optimality condition of (1) as

$$-i_t dt = \mathbb{E}_t\left(\frac{d\xi_t^{N,h}}{\xi_t^{N,h}}\right), \text{ where } \xi_t^{N,h} = e^{-\rho t}\frac{1}{\bar{p}}\frac{1}{C_t^h}, \tag{3}$$

with $\frac{d\xi_t^{N,h}}{\xi_t^{N,h}}$ representing the instantaneous (nominal) stochastic discount factor, whose expected value equals the (minus) nominal risk-free rate, $-i_t dt$.[7] Due to the rigid price assumption, the real and nominal risk-free rates become equal, $r_t = i_t$, where r_t represents the real interest rate.

We can rewrite equation (3) as

$$\mathbb{E}_t\left(\frac{dC_t^h}{C_t^h}\right) = (i_t - \rho)dt + \underbrace{\mathrm{Var}_t\left(\frac{dC_t^h}{C_t^h}\right)}_{\substack{\text{Endogenous} \\ \text{precautionary savings}}}, \tag{4}$$

where the last term, $\mathrm{Var}_t(\frac{dC_t^h}{C_t^h})$, captures the *endogenous* volatility of consumption growth. Typically, this term is second-order and omitted in log-linearized models. In contrast, our non-linear characterization in (4) explicitly accounts for consumption volatility, allowing it to affect the drift. This additional term reflects the standard *precautionary savings channel*: higher business cycle volatility increases households' demand for riskless savings, lowering current consumption and raising expected consumption growth.

Finally, the household must also satisfy the intratemporal optimality condition:

$$\frac{1}{p_t C_t^h} = \frac{\left(L_t^h\right)^{\frac{1}{\eta}}}{w_t}, \tag{5}$$

[7] Appendix II provides the Hamilton-Jacobi-Bellman (HJB) equation-based derivation for equations (3) and (5), and Online Appendix B derives an analytic form of the equilibrium value function Γ_t^h of households.

and the transversality condition imposed on the value function Γ_t^h:[8]

$$\lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\rho t}\Gamma_t^h\right] = 0. \tag{7}$$

2.2 Firms

We assume the usual Dixit-Stiglitz monopolistic competition among firms, where the demand each firm $i \in [0,1]$ faces is given by

$$D_t(p_t^i, p_t) = \left(\frac{p_t^i}{p_t}\right)^{-\varepsilon} Y_t, \;\; \text{with} \;\; p_t = \left(\int_0^1 \left(p_t^i\right)^{1-\varepsilon} di\right)^{\frac{1}{1-\varepsilon}},$$

where p_t^i is an individual firm i's price, p_t is the price aggregator, and Y_t is the aggregate output. In the assumed rigid price equilibrium, firms never change their prices so $p_t^i = p_t = \bar{p}$ and $D_t(p_t^i, p_t) = D_t(\bar{p}, \bar{p}) = Y_t$ for all $i \in [0,1]$ and $\forall t$, i.e., each firm i produces to meet the aggregate demand Y_t.

An individual firm i produces with the linear production function: $Y_t^i = A_t L_t^i$, taking the aggregate price $p_t = \bar{p}$, wage w_t, and the aggregate output Y_t as given, where L_t^i is firm i's labor hiring, and A_t is the economy's total factor productivity assumed to be exogenous and to follow a geometric Brownian motion with drift:

$$\frac{dA_t}{A_t} = g\,dt + \sigma dZ_t, \tag{8}$$

where g is its expected growth rate and σ is what we call *fundamental* volatility, assumed to be constant. Firms' profits to be rebated are given by $D_t = \bar{p}Y_t - w_t L_t$, with $L_t = \int L_t^i di$.

[8] Note that without the transversality condition (7), the value function Γ_t^h in equation (1) is not uniquely pinned down: Wiszniewska-Matyszkiel (2011) suggests (7) as a sufficient condition for optimality. In Online Appendix B, we show that under our constructed equilibria, (7) is equivalent to

$$\lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\rho t} u'(C_t^h)\frac{B_t^h}{p_t}\right] = 0, \tag{6}$$

which is a conventional form of the transversality condition (Acemoglu, 2009). With bonds in zero net supply, condition (6) holds trivially. Online Appendix P assumes that the government issues a positive net supply of bonds and studies the household transversality condition together with a rate-compounded No-Ponzi-Game restriction under Ricardian fiscal policy.

We assume that all the aggregate variables are adapted to the filtration $(\mathcal{F}_t)_{t\in\mathbb{R}}$ generated by the process in (8) in a given *filtered* probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\mathbb{R}}, \mathbb{P})$. In other words, for any aggregate variable X_t, it follows a stochastic process of the form of $dX_t = \mu_t^X dt + \sigma_t^X dZ_t$ in equilibrium, where μ_t^X and σ_t^X are endogenous drift and volatility of X_t process.

Definition 1 (Equilibrium) *An equilibrium is defined by the following sets of variables: household-specific quantities $\{B_t^h, C_t^h, L_t^h\}_h$, firm-specific quantities $\{Y_t^i, L_t^i\}_i$, aggregate quantities $\{B_t, C_t, Y_t, L_t\}$, and prices $\{w_t, \bar{p}\}$. These variables satisfy the household optimality conditions (i.e., equations (3) and (5) together with the transversality condition (7)), the aggregation conditions $B_t^h = B_t = 0$ for all h, $C_t^h = C_t = Y_t$ for all h, $L_t^h = L_t$ for all h, $Y_t^i = Y_t$ and $L_t^i = L_t$ for all i,[9] and a monetary policy rule introduced in Section 2.3. All aggregate quantity and price variables are adapted to a given filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\mathbb{R}}, \mathbb{P})$ generated by (8).*

Henceforth, we adopt aggregate notation by omitting the superscripts h and i, whenever the distinction between household- or firm-specific and aggregate variables is not essential.

Flexible Price Equilibrium We characterize the counterfactual flexible price equilibrium as the equilibrium in which firms freely set prices. The outcomes under this equilibrium are termed 'natural' because central banks, facing price rigidity, target these outcomes using monetary policy instruments. As proven in Appendix II.2.2, the natural output Y_t^n follows:

$$\frac{dY_t^n}{Y_t^n} = \Big(\underbrace{r^n}_{\text{Natural rate}} - \rho + \sigma^2 \Big) dt + \underbrace{\sigma}_{\text{Natural volatility}} dZ_t. \tag{9}$$

where $r^n = \rho + g - \sigma^2$ denotes the natural rate of interest. From the central bank's perspective, the process (9) is exogenous and thus beyond the influence of monetary policy. Notice that Y_t^n follows a geometric Brownian motion with volatility σ, matching the volatility of the A_t process in (8).

[9] Due to rigid prices, firm optimization problems can be abstracted away.

Rigid Price Equilibrium and the 'Gap' Economy Returning to the rigid-price economy, we introduce σ_t^s as the *excess* volatility of the growth rate of output $\{Y_t\}$ relative to the benchmark flexible-price output in (9). By definition, we have:

$$\mathrm{Var}_t\left(\frac{dY_t}{Y_t}\right) = (\sigma + \sigma_t^s)^2 dt, \tag{10}$$

where σ_t^s is an *endogenous* volatility term determined in equilibrium. Substituting equation (10) into the nonlinear Euler equation (4), and using $C_t^h = C_t = Y_t$ for all h, yields:

$$\frac{dY_t}{Y_t} = \left(i_t - \rho + (\sigma + \sigma_t^s)^2\right) dt + (\sigma + \sigma_t^s) dZ_t. \tag{11}$$

Using the standard definition of the output gap $\hat{Y}_t = \ln\left(\frac{Y_t}{Y_t^n}\right)$, we obtain:[10]

$$\begin{aligned} d\hat{Y}_t = &\left(i_t - \left(r^n \overbrace{-\frac{1}{2}(\sigma + \sigma_t^s)^2 + \frac{1}{2}\sigma^2}^{\text{New terms}}\right)\right)dt \\ &+ \sigma_t^s dZ_t. \end{aligned} \tag{12}$$

Note that equation (12) includes a feedback effect that is absent in log-linearized models.[11] Given the policy rate i_t, an increase in σ_t^s raises the drift term and reduces $\hat{Y}_t$. The intuition follows directly from the precautionary savings mechanism in (4), as higher volatility induces households to save more and consume less, triggering a demand recession. This volatility feedback is essential in generating the multiplicity of equilibria to be discussed in Section 3.

Finally, define the *risk-adjusted* natural rate as:

$$r_t^T = r^n - \frac{1}{2}(\sigma + \sigma_t^s)^2 + \frac{1}{2}\sigma^2. \tag{13}$$

Note that the rate r_t^T is endogenous and negatively related to the aggregate excess volatility σ_t^s. This risk-adjusted natural rate r_t^T represents the economy's effective *reference* risk-free rate, at which setting $i_t = r_t^T$ completely eliminates the drift of the

[10] In equation (11), we assume that the output Y_t is adapted to the filtration $(\mathcal{F}_t)_{t\in\mathbb{R}}$ generated by the technology process in (8). Thus, σ_t^s can be interpreted as *fundamental* excess volatility.

[11] For comparison, see the log-linearized IS equation (17), where endogenous excess volatility σ_t^s does not affect the drift.

output gap.

2.3 Taylor Rule

In closing the model, we assume the central bank sets the risk-free interest rate i_t according to the following Taylor rule:

$$i_t = r^n + \phi_y \hat{Y}_t, \quad \text{with} \quad \phi_y > 0. \tag{14}$$

Condition $\phi_y > 0$, known as the *Taylor principle*, ensures a unique bounded equilibrium in the log-linearized version of the model. Substituting (14) into (12), we derive the following dynamics for $\hat{Y}_t$:

$$d\hat{Y}_t = \left(\phi_y \hat{Y}_t - \frac{\sigma^2}{2} + \frac{(\sigma + \sigma_t^s)^2}{2}\right) dt + \sigma_t^s dZ_t. \tag{15}$$

Definition 2 (Bounded Equilibrium) *Within the set of equilibria defined in Definition 1, we restrict our attention to a particular class of bounded equilibria that satisfies:*

$$\lim_{t \to \infty} \mathbb{E}_0 \left|\hat{Y}_t\right| < \infty. \tag{16}$$

In particular, a bounded equilibrium is a class of equilibria satisfying (15) and (16).[12]

Henceforth, the term *equilibrium* will specifically refer to *bounded equilibrium*.

2.4 Log-Linear Approximation

We first analyze the log-linearized version of the model as a benchmark case. Omitting the volatility terms from the drift of equation (12), we obtain a linearized IS equation:

$$d\hat{Y}_t = (i_t - r^n)\, dt + \sigma_t^s dZ_t. \tag{17}$$

[12]As Cochrane (2011) points out, the standard New Keynesian model often determines equilibrium via explosive dynamics: unless the economy converges to a particular point in each period, monetary policy—such as Taylor rules—drives the level of the economy (e.g., output gap) to escalate over time. When these explosive paths are taken into account, the model already admits multiple equilibria, most of which are unbounded. In line with the literature, we impose the boundedness condition (16) to rule out those uninteresting unbounded equilibria. We then demonstrate that additional *bounded* equilibria arise in a non-linear framework.

Proposition 1 (Benchmark Equilibrium) *The log-linearized model, defined by the Taylor rule (14), the linearized IS equation (17), and the transversality condition (7), admits a unique rational expectations equilibrium characterized by perfect stabilization of the output gap $\hat{Y}_t = 0$ and zero excess volatility $\sigma_t^s = 0$ for all t.*

Proof. Substituting (14) into (17), we obtain the standard log-linear approximation of the output gap dynamics:

$$d\hat{Y}_t = \left(\phi_y \hat{Y}_t\right) dt + \sigma_t^s dZ_t. \tag{18}$$

With the local dynamics around the flexible-price equilibrium given by (18), Blanchard and Kahn (1980) establish the existence of a *unique* linear rational expectations equilibrium under the Taylor principle $\phi_y > 0$ in (14) and the boundedness condition (16). The resulting equilibrium with $\hat{Y}_t = \sigma_t^s = 0$ for all t corresponds to a perfectly stabilized economy. ■

The benchmark equilibrium described in Proposition 1, characterized by perfect stabilization (i.e., $\sigma_t^s = \hat{Y}_t = 0$ for all t), remains an equilibrium of the exact non-linear model defined by the IS equation (12) and the Taylor rule (14). However, equilibrium uniqueness is no longer guaranteed in a non-linear framework. The next section analyzes how multiple equilibria emerge when moving beyond the log-linear approximation.

3 Multiple Equilibria

This section illustrates alternative global solutions of the model driven by aggregate volatility and studies the properties of the resulting business cycle dynamics. We proceed in two steps. First, in Section 3.1, we construct a non-stationary equilibrium that allows aggregate volatility to temporarily deviate from the perfectly stabilized path (i.e., benchmark equilibrium) and graphically explore the underlying economic mechanisms. Next, in Section 3.2, we introduce a broader class of stationary equilibria with permanent deviations from the benchmark equilibrium. Appendix I and Appendix II provide a detailed characterization and formal proofs of the results presented here.

3.1 Martingale equilibrium

We begin by presenting a rational expectations equilibrium that supports the emergence of an initial excess volatility $\sigma_0^s > 0$ by explicitly constructing an equilibrium path in which $\hat{Y}_t$ follows a local martingale. Our martingale equilibrium construction (i) supports an initial jump in excess volatility, $\sigma_0^s > 0$, which arises in a self-fulfilling manner;[13] (ii) satisfies the process defined by the dynamic IS equation (15); and (iii) does not diverge in expectations in the long run, consistent with the boundedness condition given in Definition 2. We will show that this particular equilibrium is non-stationary by design. It is explicitly constructed through the following steps, with derivation details provided in Appendix I.

Step 1 Assume that $\hat{Y}_t$ is a local martingale consistent with the dynamics in (15). Therefore, the drift of the $\{\hat{Y}_t\}$ process in (15) must be zero, resulting in:

$$\hat{Y}_t = -\frac{\left(\sigma + \sigma_t^s\right)^2}{2\phi_y} + \frac{\sigma^2}{2\phi_y}. \tag{19}$$

Step 2 Second, we show the existence of a stochastic process for $\{\sigma_t^s\}$ starting from σ_0^s that supports the equilibrium expression for the output gap in (19). Using (15) and (19), we obtain an expression for that process as follows:

$$\begin{aligned} d\sigma_t^s = -(\phi_y)^2 \frac{(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3} dt \\ - \phi_y \frac{\sigma_t^s}{\sigma + \sigma_t^s} dZ_t. \end{aligned} \tag{20}$$

Equations (19) and (20) describe the dynamics of our constructed rational expectations equilibrium that supports initial excess volatility $\sigma_0^s > 0$. Online Appendix D shows that the stochastic differential equation (20) admits a strong solution, as required in the filtered probability space assumed in the paper.

Proposition 2 (**Martingale Equilibrium**) *The model admits a rational expectations equilibrium which supports initial excess volatility $\sigma_0^s > 0$ and is represented by the $\hat{Y}_t$*

[13] The emergence of the initial excess volatility σ_0^s is not part of the economy's filtration $(\mathcal{F}_t)_{t\in\mathbb{R}}$. This can be regarded as a *sunspot* shock to the aggregate excess volatility σ_t^s, with aggregate variables responding to its appearance.

dynamics in equation (19)*, and the* σ_t^s *process in equation* (20)*. We refer to this class of equilibria as the "Martingale" equilibrium, which has the following properties:*

Property 1 *Excess volatility* σ_t^s *converges to zero almost surely, i.e.,* $\sigma_t^s \xrightarrow{a.s} \sigma_\infty^s = 0$.

Property 2 *Output gap* $\hat{Y}_t$ *converges to zero almost surely, i.e.,* $\hat{Y}_t \xrightarrow{a.s} \hat{Y}_\infty = 0$.

Property 3 *Non-uniform integrability: aggregate variance* $\left(\sigma + \sigma_t^s\right)^2$ *satisfies*

$$\mathbb{E}_0\left(\sup_{t\geq 0}\left(\sigma + \sigma_t^s\right)^2\right) = \infty,$$

$$\lim_{K\to\infty}\sup_{t\geq 0}\left(\mathbb{E}_0\left(\sigma + \sigma_t^s\right)^2 \mathbb{1}_{\left\{\left(\sigma+\sigma_t^s\right)^2 \geq K\right\}}\right) > 0.$$

Property 4 *Boundedness:* $\lim_{t\to\infty}\mathbb{E}_0\left|\hat{Y}_t\right| < \infty$.

Proof. See Appendix I.2. ∎

The results that $\sigma_t^s \xrightarrow{a.s} \sigma_\infty^s = 0$ and $\hat{Y}_t \xrightarrow{a.s} \hat{Y}_\infty = 0$ imply that equilibrium paths originating from an initial excess volatility $\sigma_0^s > 0$ are almost surely stabilized in the long run. Nevertheless, this almost-sure stabilization remains compatible with the self-fulfilling emergence of $\sigma_0^s > 0$. Specifically, by Property 3, we have $\mathbb{E}_0\left(\sup_{t\geq 0}\left(\sigma + \sigma_t^s\right)^2\right) = \infty$, indicating that an initial volatility shock σ_0^s is sustained by a vanishing probability of infinitely large equilibrium aggregate variance occurring along some future paths.[14] Finally, on equilibrium paths with $\sigma_0^s > 0$, $\sigma_t^s > 0$ and $\hat{Y}_t < 0$ almost surely.

As Online Appendix F shows, the martingale equilibrium with an initial negative excess volatility shock, $\sigma_0^s < 0$ is not feasible, implying the recessionary nature of the martingale equilibrium.

Corollary 1 (Vanishing Fundamental Volatility) *As* $\sigma \to 0$*, the martingale equilibrium of Proposition* 2 *is described by* $\hat{Y}_t = -\frac{\left(\sigma_t^s\right)^2}{2\phi_y}$ *and by the Bessel process*

$$d\sigma_t^s = -\frac{\left(\phi_y\right)^2}{2\sigma_t^s}dt - \phi_y dZ_t, \tag{21}$$

[14]While this property is a mathematical consequence of this particular equilibrium construction, we interpret it as an *endogenously generated* rare disaster event arising in a self-fulfilling manner. For the litearature on rare-disaster risk, see e.g., Barro (2006); Gabaix (2012); Gourio (2012); Martin (2012).

which is absorbed at zero.

Proof. See Appendix I.2. ■

When fundamental volatility vanishes, dZ_t no longer moves technology and therefore generates no income risk. It continues to serve as a coordinating device for the contingent consumption plans that sustain self-fulfilling excess volatility. The construction therefore does not rely on fundamental shocks.

Intuition We explain the economic intuition for (i) how an initial excess volatility σ_0^s can appear, and (ii) the four properties in Proposition 2. To that end, we simplify the setting and make the following assumptions:

A.1 A shock dZ_t in each period takes one of two possible values: $\{+1, -1\}$, with equal probability.

A.2 Martingale equilibrium: output gap $\hat{Y}_t$ equals the conditional expected value of the next-period output gap, $\hat{Y}_{t+1}$. Thus, if $\hat{Y}_{t+1}$ takes either $\hat{Y}_{t+1}^{(1)}$ or $\hat{Y}_{t+1}^{(2)}$ when $dZ_{t+1} = 1$ or -1, respectively, then

$$\hat{Y}_t = \frac{1}{2}\left(\hat{Y}_{t+1}^{(1)} + \hat{Y}_{t+1}^{(2)}\right).$$

A.3 Aggregate demand (i.e., output gap) $\hat{Y}_t$ falls as the conditional variance of the next period's $\hat{Y}_{t+1}$ rises due to precautionary savings. Both $\hat{Y}_t$ and σ_t^s become zero on the stabilized path (i.e., flexible-price economy).

Since there are two possible realizations of the shock dZ_t in each period, we can represent this with a tree diagram, as depicted in Figure 1. The thick vertical line represents the stabilized path, with areas to its left and right representing recessions and booms, respectively. To build a rational expectations equilibrium that supports a self-fulfilling jump in excess volatility $\sigma_0^s > 0$, we construct a path-dependent consumption strategy for agents with time-varying conditional volatilities.[15]

First, let us imagine that the the current period agents (Agents$_0$) suddenly believe that the future themselves will choose a path-dependent consumption demand so

[15] Note that agents' demand determines output in this environment with rigid prices. Note also that since households in each period are the same (i.e., representative agent), this stochastic consumption path becomes a common knowledge among the agents.

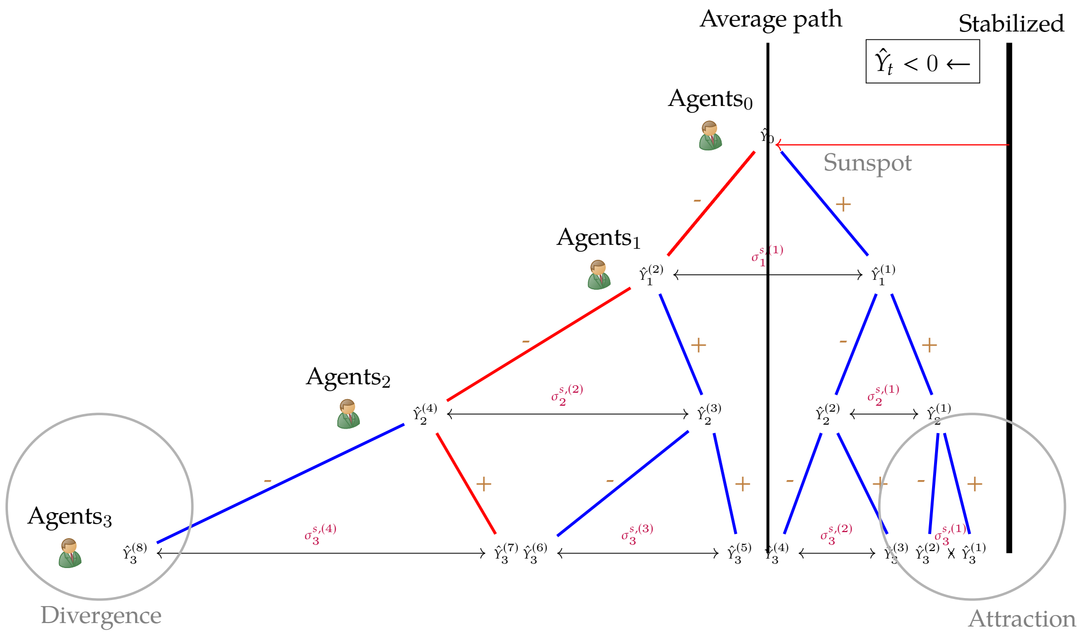


Figure 1: A rise in σ_0^s as a rational expectations equilibrium.

that the next-period's $\hat{Y}_1$ becomes $\hat{Y}_1^{(1)}$ after $dZ_1 = +1$ is realized and $\hat{Y}_1^{(2)}$ if $dZ_1 = -1$ is realized, with $\hat{Y}_1^{(1)} > \hat{Y}_1^{(2)}$. Then the current output $\hat{Y}_0$ becomes $\hat{Y}_0 = \frac{1}{2}\left(\hat{Y}_1^{(1)} + \hat{Y}_1^{(2)}\right)$ with $\hat{Y}_0$ below the stabilized path, as Agents$_0$ believe that there exists dispersion in next-period outputs, which is given by $\sigma_1^{s,(1)} = \frac{\hat{Y}_1^{(1)} - \hat{Y}_1^{(2)}}{2}$, and which leads to lower consumption through precautionary savings at $t = 0$. Now imagine $dZ_1 = -1$ is realized in period 1. For Agents$_0$'s belief in $\hat{Y}_1 = \hat{Y}_1^{(2)}$ to be consistent, the same agents in period 1 (Agents$_1$) must believe that future themselves (Agents$_2$) will choose their consumption in a way that, at time 2, $\hat{Y}_2$ becomes $\hat{Y}_2^{(3)}$ with $dZ_2 = +1$ and $\hat{Y}_2^{(4)}$ with $dZ_2 = -1$, with conditional volatility $\sigma_2^{s,(2)} = \frac{\hat{Y}_2^{(3)} - \hat{Y}_2^{(4)}}{2}$ higher than $\sigma_1^{s,(1)}$, since $\hat{Y}_1^{(2)}$ is lower than the initial output, $\hat{Y}_0$.

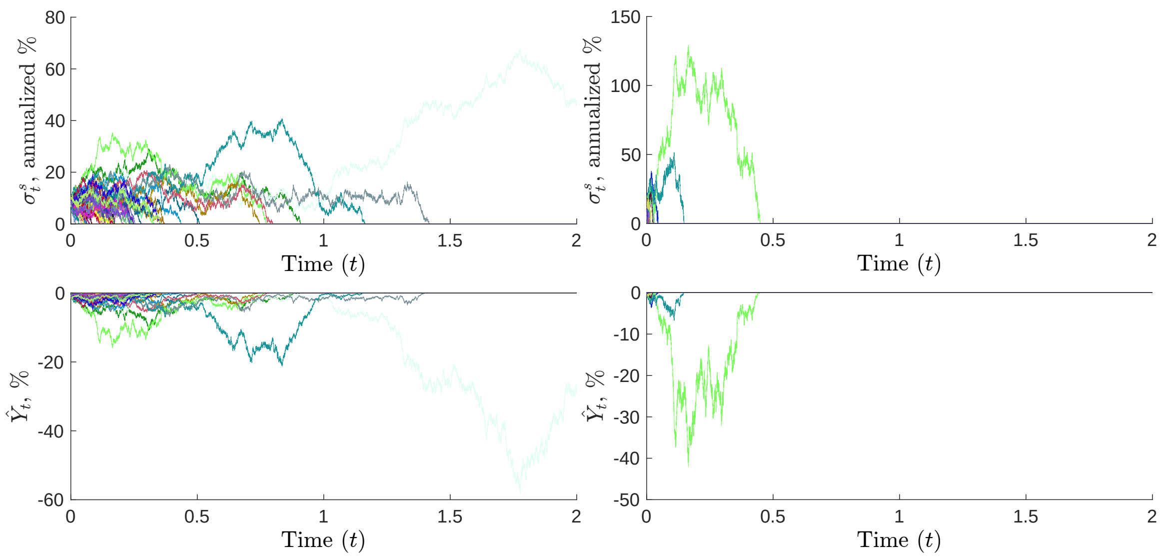
After dZ_2 is realized, Agents$_1$'s belief about $\hat{Y}_2$ can be made consistent through future agents $\{\text{Agents}_{n\geq 2}\}$'s coordination in a forward looking fashion. Observe that all the nodes in Figure 1 satisfy assumptions A.2 and A.3, with distance between adjacent nodes getting progressively narrower (wider) as output gap gets closer (farther) to the stabilization. This results in divergent and attraction paths offsetting each other, making the output gap $\{\hat{Y}_t\}$ follow a local martingale process in expectation. In summary, Agents$_0$'s initial doubt about volatility in the next-period outcome is validated through coordination among themselves (i.e., the representative household) at each node. This construction of equilibrium is feasible because agents in different periods have common knowledge of their stochastic consumption strategies (i.e., policy functions) and there is frictionless communication across intertemporal periods (i.e., perfect recall). Also, this self-fulfilling volatility is attached to the fundamental shock, dZ_t, which can be distinguished from the previous literature on sentiment-driven fluctuations under incomplete information (Acharya et al., 2021).

Note that (i) we obtain an equilibrium with a *stochastic* aggregate volatility: i.e., σ_t^s is dependent on the path of shocks, as output gap $\{\hat{Y}_t\}$ is stochastic and negatively depends on the conditional volatility of its next-period level. Equation (20) specifies the exact stochastic process of $\{\sigma_t^s\}$ starting from $\sigma_0^s > 0$; (ii) Since excess volatility σ_t^s decreases as output gap $\hat{Y}_t$ approaches the stabilized path, this path becomes an attraction point for the set of alternative paths in its neighborhood, justifying the result of Proposition 2 that σ_t^s almost surely converges to zero over time. Nonetheless, as excess volatility σ_t^s rises whenever output $\hat{Y}_t$ deviates farther from the stabilized level, this also aligns with the result of Proposition 2 that maximal σ_t^s diverges, $\mathbb{E}_0(\sup(\sigma + \sigma_t^s)^2) = \infty$. However, this divergent behavior only happens with vanishingly small probability as $\sigma_t^s \xrightarrow{a.s} 0$.

In summary, conventional Taylor rules almost surely stabilize the disruption caused by an initial excess volatility shock $\sigma_0^s > 0$ in the long-run, while not preventing the economy from entering a crisis phase with a positive $\{\sigma_t^s\}$ path starting from $\sigma_0^s > 0$.

Simulation Figure 2 illustrates the dynamic paths of $\{\sigma_t^s\}$ under the martingale equilibrium, starting from $\sigma_0^s = 0.0436$, and shows how they change with the policy response to the output gap, ϕ_y. Panel 2a uses a calibration with a weak output-gap response, $\phi_y = 0.1$, while Panel 2b reports the baseline calibration with a stronger response, $\phi_y = 0.5$. Given $\sigma_0^s > 0$, a higher ϕ_y speeds up convergence to perfect stabilization, but increases the likelihood of a more severe crisis path over a finite horizon. Put differently, for a fixed initial excess volatility to persist under a more responsive monetary policy, future endogenous volatility must be higher (that is, σ_t^s must rise) and recessions must be deeper (that is, $\hat{Y}_t$ must fall), although such paths occur with vanishing probability.

In summary, our model's non-linear characterization yields two key predictions:



(a) $\phi_y = 0.1$ (b) $\phi_y = 0.5$

Figure 2: Simulated paths for the excess volatility $\{\sigma_t^s\}$ and output gap $\{\hat{Y}_t\}$ processes under the martingale equilibrium, starting from $\sigma_0^s = 0.0436$, with 50 sample paths in each panel. The calibration for the remaining parameters used to construct Figure 2 is reported in Online Appendix L.

(i) the emergence of crisis phases driven by self-fulfilling volatility shocks; and (ii) the joint evolution of the first (output level) and second (conditional volatility) moments of the model during crises.

3.2 Ornstein-Uhlenbeck equilibria

This section introduces a broader class of equilibria with several noteworthy properties: (i) initial *excess* volatility σ_0^s adapted to the economy's filtration (i.e., equilibria that do not require initial sunspot volatility shocks in σ_0^s), (ii) non-degenerate and stationary stochastic processes for equilibrium variables in the long run,[16] and (iii) the potential for alternative deterministic steady states characterized by underproduction.

For that purpose, we conjecture an alternative class of equilibria where the output

[16]In our martingale equilibrium, once σ_t^s reaches zero, it is absorbed at zero, resulting in a non-stationary solution to the model.

gap $\{\hat{Y}_t\}$ dynamics follow a process of the form

$$d\hat{Y}_t = -\theta \cdot \left[\hat{Y}_t - \mu\right] \mathrm{d}t + \sigma_t^s \mathrm{d}Z_t, \tag{22}$$

where θ and μ are constant parameters.[17] When $\theta = 0$, the process (22) becomes the martingale equilibrium in Section 3.1. Note that with $\theta > 0$, the $\hat{Y}_t$ process features mean reversion to μ. To close the model, we equate the drift terms in equation (15) and equation (22) and obtain[18]

$$\hat{Y}_t = \frac{\theta\mu}{\theta + \phi_y} - \frac{1}{2(\theta + \phi_y)} \left[(\sigma + \sigma_t^s)^2 - \sigma^2\right], \tag{23}$$

with

$$\begin{aligned} \mathrm{d}(\sigma + \sigma_t^s)^2 = &-\theta \left[2\mu\phi_y + (\sigma + \sigma_t^s)^2 - \sigma^2\right] \mathrm{d}t \\ &- 2(\theta + \phi_y)\sigma_t^s \mathrm{d}Z_t. \end{aligned} \tag{24}$$

Proposition 3 (Ornstein-Uhlenbeck Equilibrium) *The model admits a rational expectations equilibrium characterized by the $\hat{Y}_t$ dynamics in equation (23) and the σ_t^s process in equation (24). We refer to this class of equilibria as the "Ornstein-Uhlenbeck" equilibrium, with the following properties:*

Property 1 *For $\theta > 0$, $\mu < 0$, the process (24) of σ_t^s is stable and admits a unique stationary distribution. In the limit $\sigma \longrightarrow 0$, if $(\theta + \phi_y) \neq 0$ and $\mu\phi_y < 0$, this stationary distribution coincides with the generalized gamma distribution $GGD(a, d, p)$, given by*[19]

$$a = \sqrt{\frac{2(\theta + \phi_y)^2}{\theta}}, \quad d = -\frac{2\theta\mu\phi_y}{(\theta + \phi_y)^2}, \quad \textit{and} \quad p = 2, \tag{25}$$

where a is the scale parameter, d is the power-law shape parameter, and p is the exponential shape parameter.

[17] Note that (22) resembles an Ornstein-Uhlenbeck process, with one major difference: the process features an endogenous volatility σ_t^s, which is determined in equilibrium, whereas typical Ornstein-Uhlenbeck processes have constant volatility associated with the diffusion component.

[18] Online Appendix B verifies the transversality condition—equation (7)—under the Ornstein-Uhlenbeck equilibria represented by (23) and (24). Online Appendix D verifies that the stochastic differential equation (24) admits a strong solution, as required in a given *filtered* probability space.

[19] See, e.g., Boukai (2022) for an application of the generalized gamma distribution as a benchmark risk-neutral distribution in stochastic volatility models.

Property 2 *For $\theta > 0$ and $\mu = 0$, the process of σ_t^s defined in (24) is non-stationary and its distribution degenerates to the perfectly stabilized equilibrium with $\sigma_t^s = 0$ as $t \to \infty$.*

Property 3 *The long-run expectations of the output gap $\hat{Y}_t$ and excess variance $(\sigma+\sigma_t^s)^2-\sigma^2$ are given by*

$$\lim_{t\to\infty} \mathbb{E}_0 \left[\hat{Y}_t\right] = \mu,$$
$$\lim_{t\to\infty} \mathbb{E}_0 \left[(\sigma + \sigma_t^s)^2 - \sigma^2\right] = -2\mu\phi_y.$$

Property 4 *Boundedness:* $\lim_{t\to\infty} \mathbb{E}_0 \left|\hat{Y}_t\right| < \infty$.

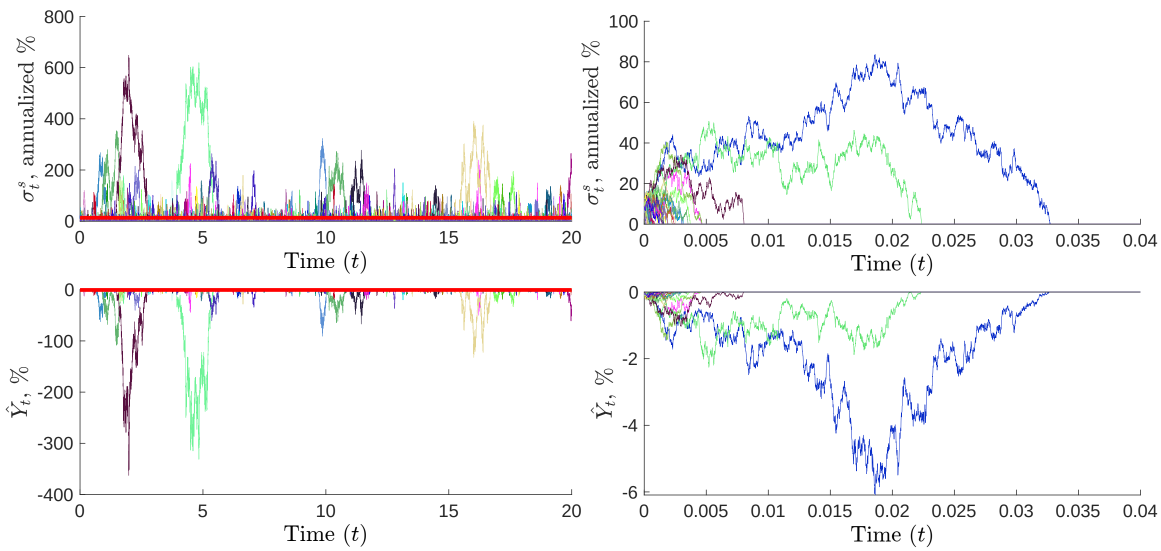
Proof. See Online Appendices B, C, and F. ■

Property 3 of Proposition 3 implies that solutions with $\mu < 0$ exhibit a long-run expected excess variance given by $-2\mu\phi_y > 0$. Together with Property 1, this result indicates that the volatility process $\{\sigma_t^s\}$ is never permanently fixed at any particular value, including zero. For example, even if the economy initially has $\sigma_0^s = 0$, the aggregate variance $(\sigma + \sigma_t^s)^2$ immediately starts drifting toward $\sigma^2 - 2\mu\phi_y$, since the drift term in (24) is positive when excess volatility is zero. Consequently, the model admits alternative equilibria characterized by an endogenous, stationary stochastic process $\{\sigma_t^s\}$ arising from arbitrary initial conditions. Note from Property 3 that the $\hat{Y}_t$ process does not diverge in expectations in the long run. Property 4 records the boundedness condition in Definition 2 is satisfied.

In contrast, the equilibrium under $\mu = 0$ does not support a stationary stochastic process in the long-run but still allows for temporary deviations from the perfectly stabilized equilibrium, analogous to the martingale equilibria discussed in Section 3.1. Figure 3 illustrates representative sample paths of the volatility process $\{\sigma_t^s\}$ under both calibrations.[20]

Property 3 also implies that the Ornstein-Uhlenbeck equilibria feature a deterministic steady state for the output gap $\hat{Y}_t$, equal to the parameter μ. Therefore,

[20]In Online Appendix G, we demonstrate that our result of multiple bounded global equilibria is generally not unique to the continuous-time version of the model, even though the discrete-time formulation can sometimes be challenging. We then outline a way to link the two specifications: as the time interval *dt* approaches zero, the discrete-time solution converges to its continuous-time counterpart.



(a) $\mu < 0$ (b) $\mu = 0$

Figure 3: Simulated paths for the excess volatility $\{\sigma_t^s\}$ and output gap $\{\hat{Y}_t\}$ processes under the Ornstein-Uhlenbeck solution with $\theta = 0.95$ and $\phi_y = 0.5$. Panel 3a starts from $\sigma_0^s = 0$; the red line denotes the long-run expected excess volatility, $\sigma^{s,*}$. Panel 3b starts from $\sigma_0^s = 0.0436$. Each panel shows 50 sample paths. The calibration for the remaining parameters used to construct Figure 3 is reported in Online Appendix L.

equilibria with $\theta > 0$ and $\mu < 0$ exhibit steady-state under-production, where output is $|100 \times \mu|$ percent below its natural level. The intuition for this result is that a process for $\{\sigma_t^s\}$ with long-run expected excess variance $-2\mu\phi_y > 0$ induces a stronger precautionary response in household consumption to aggregate volatility relative to the flexible-price benchmark.

Implications As shown in Online Appendix F, these constructions are admissible only when excess variance remains non-negative. That restriction rules out an initial negative excess-volatility shock in the martingale equilibrium and rules out $\mu > 0$ in the Ornstein-Uhlenbeck equilibrium represented by (23) and (24). Combined with equations (19) and (23), this implies $\hat{Y}_t \leq 0$ along both classes of equilibria, which is consistent with Khorrami and Mendo (2025) who show that, every non-explosive non-fundamental equilibrium is necessarily recessionary under active Taylor rules. The martingale and Ornstein-Uhlenbeck solutions fall in that class. A related conclusion appears when multiplicity is generated by precautionary saving against countercyclical idiosyncratic risk: the additional steady state is recessionary (Acharya and Dogra, 2020; Ravn and Sterk, 2021; Bilbiie, 2025; Acharya and Benhabib, 2024).

4 Equilibrium Selection

The alternative global equilibria of Section 3 appear when policy is described by a Taylor rule alone and households can coordinate contingent consumption plans on the history of realized shocks. This section studies how the full-stabilization path can be recovered as the unique bounded equilibrium. Behavioral frictions, including a small friction in social memory and Level-k reasoning, restrict that coordination. A fiscal escape clause, specified as a state-contingent off-equilibrium threat, selects the same path without being triggered along it.

4.1 Behavioral Frictions

Our equilibrium construction featuring time-varying endogenous aggregate excess volatility σ_t^s is feasible because households are able to coordinate their intertemporal consumption choices—potentially over an infinite horizon—based on the history of realized shocks. Accordingly, even a small degree of social memory loss can render such consumption coordination infeasible, whether the implied output gap paths are non-stationary—the martingale equilibrium—or stationary in the long run, as in the case of Ornstein–Uhlenbeck equilibria.

Online Appendix I formally shows that introducing a small friction in social memory removes the alternative equilibria and uniquely selects the full-stabilization equilibrium—i.e., $\hat{Y}_t = \sigma_t^s = 0$, for all t—as the only bounded equilibrium. This result is achieved by extending Angeletos and Lian (2023) to a potentially non-stationary framework that incorporates stochastic conditional volatility.[21]

A social memory loss is not the only mechanism capable of ruling out alternative equilibria driven by self-fulfilling volatility. Level-k reasoning (Farhi and Werning, 2019) may also eliminate such equilibria, because it attenuates households' consideration of the full general equilibrium effects. While we present a formal mathematical

[21] The full-stabilization equilibrium corresponds to the minimum-state variable (MSV) equilibrium, which survives as the unique equilibrium in Angeletos and Lian (2023).

representation in Online Appendix I, this can be easily seen as follows:

Level-1 reasoning assumes that households expect the path for future output and volatility to remain as in the original perfect stabilization equilibrium, where they choose optimal consumption and savings based on these expectations.[22] Their optimal consumption (output) path up to level-1 then corresponds to the perfect stabilization equilibrium.[23] In level-k iteration, households assume that the future paths of output and volatility coincide with the equilibrium paths determined in the previous round. Because the level-1 path remains identical to the full stabilization path, the level-k path, for any finite k, will continue to coincide with the same path, thereby ruling out our alternative equilibria.

We conjecture that other behavioral frictions—such as lack of common knowledge or sparse thinking (Gabaix, 2014, 2020)— will similarly reduce the dependence of current outcomes on far-future contingencies, and therefore undermine the equilibrium construction underlying the volatility-driven solutions.

4.2 Fiscal Intervention

Due to *Ricardian equivalence*, a debt-financed transfer cannot select among equilibria in our baseline economy: households fully capitalize the future taxes that service any current deficit, so aggregate demand is unchanged.

Online Appendix H studies the escape clause in a non-Ricardian overlapping-generations extension of the model. The demand block uses perpetual-youth households as in Blanchard (1985); Yaari (1965); Angeletos et al. (2024), and the rest of the environment of Section 2 is unchanged. A living household dies with Poisson intensity $\lambda \geq 0$ and is replaced by a newborn who does not inherit the incumbents' bonds. Let d_t be public debt in the hands of households, in units of current output, and write excess consumption variance as $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2$. Because current cohorts do not live forever, they do not fully internalize taxes levied on future households. A deficit today is then spent in part by those who receive it, and demand rises even though goods-market clearing remains $C_t = Y_t$. Under the Taylor rule

[22] Our constructed equilibria arise in a self-fulfilling way, not requiring any additional exogenous disturbance such as a monetary policy shock to bring the economy into them. Therefore, we need not analyze how households would optimally respond to such a shock in a partial-equilibrium setting (Farhi and Werning, 2019).

[23] It is because the flexible price equilibrium is an equilibrium. See Definition 1.

(14), Online Appendix H derives the output-gap IS equation:

$$d\hat{Y}_t = \left(\phi_y \hat{Y}_t + \frac{1}{2}\mathcal{E}_t - \psi d_t\right)dt + \sigma_t^s \, dZ_t, \tag{26}$$

where $\psi \equiv \lambda(\rho + \lambda)$. When $\lambda = 0$ or $d_t = 0$, equation (26) reduces to the baseline IS equation (15). The new term $-\psi d_t$ corresponds to the generational-turnover effect: a higher d_t raises current consumption out of wealth and therefore lowers expected output-gap growth.

The government activates the fiscal transfer only off the equilibrium path. If a recessionary volatility path starts with $\hat{Y}_t < 0$, the government jumps d_t so that

$$\psi d_t = \lambda_v \mathcal{E}_t + \varepsilon, \qquad \lambda_v \geq \frac{1}{2}, \quad \varepsilon > 0. \tag{27}$$

The jump in d_t is a debt-financed transfer to living households, who spend part of it. Newborns enter with no bonds, and future cohorts service the debt. The resulting generational-turnover term $-\psi d_t$ more than offsets the precautionary drift $\frac{1}{2}\mathcal{E}_t$ in (26), so expected output-gap growth becomes strictly negative. Remaining on that path would send $\hat{Y}_t$ to $-\infty$, which violates boundedness in Definition 2. Because the threat is believed, the path is not started. Along the unique bounded equilibrium, $\hat{Y}_t = \mathcal{E}_t = d_t = 0$.

4.3 Why Taylor Rules?

Since Taylor rules inherently give rise to multiple equilibria related to self-fulfilling volatility, we conclude that their primary role in the real world is not to eliminate equilibrium multiplicity, but rather to offer simple, systematic guidelines when rules are favored over discretion (Taylor, 1993) and a parsimonious description of the policy response that serves as a benchmark in practice (Clarida et al., 1999; Taylor, 1999; Woodford, 2001; Orphanides, 2003). A Taylor rule remains a transparent way to summarize how the central bank responds to inflation and real activity, and a useful reference point when many shocks are present. Selecting among the equilibria we construct is a separate problem.

This reading is consistent with Nakamura et al. (2025), who argue that determinacy depends more on firmly anchored long-run inflation expectations than on adherence to the Taylor principle alone, underpinned by a credible commitment to respond when inflation is driven by non-fundamental shocks or self-reinforcing spirals. A small behavioral friction that prevents a full coordination of consumption plans and a fiscal escape clause are more robust ways to restore determinacy than Taylor rules themselves in our setting.

Thus far, we have abstracted from inflation by assuming perfectly rigid prices, which isolates the source of global multiplicity in household intertemporal demand and the central bank's policy rule. The next section shows that this result extends to standard sticky-price models.

5 A Model with Inflation

We briefly summarize the key assumptions underlying the derivation of the New Keynesian Phillips curve with sticky prices following Rotemberg (1982), and present the main results regarding equilibrium multiplicity in this framework. Detailed derivations are provided in Online Appendix K.

Rotemberg (1982) assumes a continuum of identical firms indexed by the interval $[0, 1]$, operating under monopolistic competition. The price process for firm i is given by

$$dp_t^i = \pi_t^i p_t^i \, dt, \tag{28}$$

where each firm can adjust its price p_t^i by choosing an inflation rate π_t^i. These adjustments incur convex adjustment costs $\Theta(\pi_t^i)$, specified by

$$\Theta(\pi_t^i) = \frac{\tau}{2}(\pi_t^i)^2 p_t Y_t,$$

with $\tau \geq 0$ determining the penalty on the speed of adjustment. In equilibrium, the model yields symmetric inflation rates across firms, and the demand block remains the same.

For analytical tractability, we first consider the case in which inflation dynamics can be approximated by their log-linearized form.[24] Under the log-linearized New Keynesian Phillips curve, the model continues to admit multiple bounded equilibria.

[24]We nevertheless keep the demand block (the IS equation) fully non-linear.

In particular, the martingale equilibrium and the Ornstein–Uhlenbeck equilibrium can arise as bounded equilibria. We adopt the following Taylor rule:[25]

$$i_t = r^n + \phi_y \hat{Y}_t + \phi_\pi \pi_t. \tag{29}$$

As shown in Online Appendix K, the dynamic IS equation and log-linearized New Keynesian Phillips curve under sticky prices à la Rotemberg (1982) are given by

$$d\hat{Y}_t = \left[i_t - \pi_t - r_t^T\right] dt + \sigma_t^s \, dZ_t, \tag{30}$$

and

$$\mathbb{E}_t(d\pi_t) = \left(\kappa^\pi \pi_t - \kappa^y \hat{Y}_t\right) dt, \tag{31}$$

where κ^π and κ^y are positive constants.[26]

Equations (29), (30), and (31) comprise our non-linear system. Plugging equation (29) into (30) leads to

$$d\hat{Y}_t = \left(\phi_y \hat{Y}_t + \left(\phi_\pi - 1\right) \pi_t + \frac{1}{2}\left[\left(\sigma + \sigma_t^s\right)^2 - \sigma^2\right]\right) dt + \sigma_t^s dZ_t. \tag{32}$$

5.1 Multiple Equilibria with the Log-Linearized Phillips Curve

This section shows that the multiple equilibria derived under rigid prices continue to emerge when inflation dynamics are approximated by the log-linearized Phillips curve (31). The key reason is that these results—our equilibrium constructions—are driven by the model's demand block, in particular by the intertemporal coordination

[25] The Online Appendix derives the constructions of this section under a more general interest-rate rule that also targets excess variance,

$$i_t = r^n + \phi_y \hat{Y}_t + \phi_\pi \pi_t - \phi_{vol}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right],$$

where ϕ_{vol} is the response to excess variance. The Taylor rule used here is the special case $\phi_{vol} = 0$. For $\phi_{vol} < \frac{1}{2}$, the qualitative results remain the same. At $\phi_{vol} = \frac{1}{2}$, the extra volatility-targeting term offsets the precautionary-saving feedback and can select the full-stabilization path as the unique bounded equilibrium, but that case is knife-edge. Online Appendix J shows that any small departure from an exact one-half response restores multiplicity.

[26] For derivation of (31), see equation (K.29) in Online Appendix K in the case of sticky prices à la Rotemberg (1982). For robustness, Online Appendix M replicates an identical log-linearized New Keynesian Phillips curve (31) under the alternative assumptions of sticky price adjustments à la Calvo (1983), à la Taylor (1980), and pricing based on menu costs, e.g., Gertler and Leahy (2008).

in household consumption.

5.1.1 Benchmark: Log-Linear Approximation of the IS equation

As in the baseline model with rigid prices, we first analyze the case where the IS equation can be approximated by its log-linearized form. Omitting the volatility terms from the drift of equation (30), we obtain the following linear IS equation:

$$d\hat{Y}_t = (i_t - \pi_t - r^n)\, dt + \sigma_t^s dZ_t. \tag{33}$$

Henceforth, we assume $\phi_y > 0$ and

$$\phi \equiv \phi_y + \frac{\kappa^y}{\kappa^\pi}\left(\phi_\pi - 1\right) > 0, \tag{34}$$

which corresponds to the Taylor principle in the presence of inflation, as seen in the following proposition.

Proposition 4 (Benchmark Equilibrium) *The log-linearized model, defined by the Taylor rule (29) satisfying the Taylor principle (34), the linear New Keynesian Phillips curve (31), and the linear IS equation (33), admits unique rational expectations equilibrium characterized by the perfect stabilization of the output gap and inflation, i.e., $\hat{Y}_t = \pi_t = 0$, with zero excess volatility $\sigma_t^s = 0$ for all t.*

Proof. Substituting (29) into (33), we obtain the standard log-linear approximation of the output gap dynamics:

$$d\hat{Y}_t = \left(\phi_y \hat{Y}_t + \left(\phi_\pi - 1\right) \pi_t\right) dt + \sigma_t^s dZ_t. \tag{35}$$

With local dynamics around the flexible-price equilibrium given by (31) and (35), Blanchard and Kahn (1980) and Buiter (1984) establish the existence of a *unique* linear, bounded, rational expectations equilibrium under the Taylor principle (34). The resulting equilibrium with $\hat{Y}_t = \pi_t = \sigma_t^s = 0$ for all t corresponds to a perfectly stabilized economy. ■

Note that the benchmark equilibrium described in Proposition 4, i.e., $\pi_t = \hat{Y}_t = \sigma_t^s = 0$ for all t, remains an equilibrium of the exact non-linear model defined by the IS equation (30), the linear Phillips curve (31) and the Taylor rule (29). However, equilibrium uniqueness is no longer guaranteed there.

5.1.2 Martingale Equilibrium

This section shows that the dynamic system summarized by the non-linear IS equation (30), the linearized Phillips curve (31), and the Taylor rule (29) features the martingale equilibrium as a viable rational expectations equilibrium.

To show this, we guess and verify that inflation π_t is given by:

$$\pi_t = \frac{\kappa^y}{\kappa^\pi}\hat{Y}_t, \tag{36}$$

which with (31) implies $\mathbb{E}_t\left(d\pi_t\right) = 0$, i.e., π_t becomes a local martingale. Plugging (36) into the IS equation—equation (32)—and imposing that $\hat{Y}_t$ is a local martingale,[27] i.e., $\mathbb{E}_t\left(d\hat{Y}_t\right) = 0$, we obtain

$$\hat{Y}_t = -\frac{1}{2\phi}\left[\left(\sigma + \sigma_t^s\right)^2 - \sigma^2\right], \tag{37}$$

where ϕ is defined in (34). For (36) and (37) to be equilibrium conditions, excess volatility σ_t^s must follow:

$$d\sigma_t^s = -(\phi)^2\frac{(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3}dt - \phi\frac{\sigma_t^s}{\sigma + \sigma_t^s}dZ_t, \tag{38}$$

where ϕ_y in equation (20) is replaced by ϕ in equation (38). Due to the isomorphic mathematical structure between (20) and (38), Proposition 2 holds in this case as well.[28]

5.1.3 Ornstein-Uhlenbeck Equilibria

We now show that the same dynamic system summarized by equations (30), (31), and (29) admits the Ornstein-Uhlenbeck equilibrium where $\hat{Y}_t$ follows

$$d\hat{Y}_t = -\theta\left(\hat{Y}_t - \mu\right)dt + \sigma_t^s dZ_t. \tag{39}$$

[27] Since the conjectured π_t is a local martingale proportional to $\hat{Y}_t$, $\hat{Y}_t$ must also be a local martingale for consistency.

[28] In this case, we have $\pi_t \to 0$ almost surely.

To show this, we guess and verify that inflation π_t is given by:

$$\pi_t = \frac{\kappa^y}{\kappa^\pi}\mu + \left(\frac{\kappa^y}{\kappa^\pi + \theta}\right)\left(\hat{Y}_t - \mu\right). \tag{40}$$

Note that our conjectured solution (40) satisfies the linear New Keynesian Phillips curve (31).[29] For our conjectured solution to be a viable equilibrium, equations (32) and (39) must hold jointly under (40), yielding

$$\hat{Y}_t = \mu - \frac{1}{2\phi_2}\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi\right]\right]. \tag{41}$$

and

$$\pi_t = \frac{\kappa^y}{\kappa^\pi}\mu - \frac{1}{2\phi_2}\left(\frac{\kappa^y}{\kappa^\pi + \theta}\right)\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi\right]\right], \tag{42}$$

where ϕ_2 is defined as

$$\phi_2 \equiv \phi_y + \theta + (\phi_\pi - 1)\frac{\kappa^y}{\kappa^\pi + \theta}, \tag{43}$$

satisfying $\phi_2 > 0$ if $\phi_\pi > 1$, and ϕ is defined as in (34).

Plugging equation (41) into (39), we obtain

$$\mathrm{d}(\sigma + \sigma_t^s)^2 = -\theta\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi\right]\right]\mathrm{d}t - 2\phi_2\sigma_t^s\mathrm{d}Z_t, \tag{44}$$

which leads to the following stochastic differential equation for $\{\sigma_t^s\}$:

$$\begin{aligned}\mathrm{d}\sigma_t^s = &-\frac{1}{2}\left(\frac{1}{\sigma + \sigma_t^s}\right)\left[\theta\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi\right]\right] + \left(\phi_2\right)^2\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)^2\right]\mathrm{d}t \\ &- \phi_2\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)\mathrm{d}Z_t.\end{aligned} \tag{45}$$

[29] It can be easily seen as equation (40) implies

$$\mathbb{E}_t\left(d\pi_t\right) = \left(\frac{\kappa^y}{\kappa^\pi + \theta}\right)\mathbb{E}_t\left(d\hat{Y}_t\right) = \left(\frac{\kappa^y}{\kappa^\pi + \theta}\right)\theta\left(\mu - \hat{Y}_t\right)dt = \left(\kappa^\pi\pi_t - \kappa^y\hat{Y}_t\right)dt.$$

Because equations (44) and (24) have similar structures, we can establish the following proposition characterizing the Ornstein-Uhlenbeck equilibrium with inflation using the same arguments as in Proposition 3.[30]

Proposition 5 (Ornstein-Uhlenbeck Equilibrium) *The model with a linear New Keynesian Phillips curve (31) admits a rational expectations equilibrium characterized by $\hat{Y}_t$ in (39), π_t given in (40), and the σ_t^s process in (44). This "Ornstein-Uhlenbeck" equilibrium satisfies the following properties:*

Property 1 *For $\theta > 0$, $\mu < 0$,[31] the process of σ_t^s in (44) is stable and admits a unique stationary distribution. If $\phi_2 \neq 0$ and $\phi > 0$, in the limit $\sigma \to 0$, this stationary distribution coincides with the generalized gamma distribution $GGD(a, d, p)$, given by*[32]

$$a = \sqrt{\frac{2\left(\phi_2\right)^2}{\theta}}, \quad d = -\frac{2\theta\mu\phi}{\left(\phi_2\right)^2}, \quad \textit{and} \quad p = 2, \tag{46}$$

where ϕ_2 is defined in (43). a is the scale parameter, d is the power-law shape parameter, and p is the exponential shape parameter.

Property 2 *For $\theta > 0$ and $\mu = 0$, the process of σ_t^s defined in (44) is non-stationary and its distribution degenerates to the perfectly stabilized equilibrium with $\sigma_t^s = 0$ as $t \to \infty$.*

Property 3 *The long-run expectations of the output gap $\hat{Y}_t$, excess variance $(\sigma + \sigma_t^s)^2 - \sigma^2$, and inflation π_t are given by*

$$\begin{aligned} \lim_{t\to\infty} \mathbb{E}_0\left[\hat{Y}_t\right] &= \mu, \\ \lim_{t\to\infty} \mathbb{E}_0\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right] &= -2\mu\phi, \\ \lim_{t\to\infty} \mathbb{E}_0\left[\pi_t\right] &= \frac{\kappa^y}{\kappa^\pi}\mu. \end{aligned}$$

Property 4 *Boundedness:* $\lim_{t\to\infty} \mathbb{E}_0\left|\hat{Y}_t\right| < \infty$.

[30] The difference is that the drift and diffusion terms in (45) now depend on two policy parameters, ϕ and ϕ_2, which enter the statement of Proposition 5.

[31] Due to the same reason as in Proposition 3, $\mu > 0$ is not allowed. See Online Appendix F.

[32] See, e.g., Boukai (2022) for an application of the generalized gamma distribution as a benchmark risk-neutral distribution in stochastic volatility models.

Flexible price approximation The flexible-price limit corresponds to letting $\kappa^y \to \infty$ in the linearized Phillips curve (31). As $\kappa^y \to \infty$, both ϕ in (34) and ϕ_2 in (43) diverge. In this limit, both classes of equilibria become trivial: $\hat{Y}_t$ in (37) (martingale) and in (41) (Ornstein-Uhlenbeck) converges to zero,[33] and therefore $\sigma_t^s = 0$ and $\pi_t = 0$. Thus, the excess-volatility equilibria we construct vanish under the flexible-price approximation.

Value function and transversality condition Online Appendix N characterizes the household's equilibrium value function and verifies the transversality condition (7) when inflation dynamics is governed by (31). Since the value function takes the same functional form as (B.7), the transversality condition follows by the same argument as in the rigid price model. Online Appendix P then considers an economy in which the government supplies a positive net quantity of bonds. It derives conditions on bond issuance under Ricardian fiscal policy that ensure the household transversality condition and a textbook No-Ponzi-Game condition that discounts remaining debt by the compounded short rate hold in our alternative equilibria. The two conditions need not coincide, as they discount debt with different factors (Bohn, 1995). The household condition is the relevant no-bubble restriction (Santos and Woodford, 1997).

Long-run targets of monetary policy In the Ornstein-Uhlenbeck equilibria, the Taylor rule (29) is written as if the central bank targets a long-run zero output gap and zero inflation. Consistent with this interpretation, the linearized New Keynesian Phillips curve (31) is derived around a zero-inflation steady state.

Yet, Proposition 5 implies that in the Ornstein-Uhlenbeck equilibria, the economy converges to negative long-run output gaps and inflation when $\mu < 0$. Moreover, the long-run mean of total variance $(\sigma + \sigma_t^s)^2$ differs from σ^2 and depends on μ and σ. As $\mu < 0$ moves further away from zero, the log-linearized Phillips curve (31) becomes a poorer approximation to the underlying non-linear Phillips curve.

Online Appendix O relaxes these implicit zero targets and also allows the monetary authority to choose a long-run target for aggregate output variance $x_t \equiv (\sigma + \sigma_t^s)^2$.

[33] For the Ornstein-Uhlenbeck equilibria, (41) implies $\hat{Y}_t \to -(\theta/\kappa^\pi)\mu$. Combined with (39), this requires $\theta\mu = 0$, implying that the admissible set of (θ, μ) for the Ornstein-Uhlenbeck equilibria is given by $\theta\mu = 0$ given $\kappa^y \to \infty$. It implies $\hat{Y}_t \to 0$. Moreover, since $\theta\mu = 0$, we have $\theta\mu\kappa^y = 0$ even as $\kappa^y \to \infty$, which implies $\pi_t \to 0$ via (42).

The three long-run targets for the output gap $\hat{Y}_t$, inflation π_t, and x_t are denoted by μ, $\bar{\pi}$, and $\bar{x}$, respectively.[34] We consider the Taylor rule

$$i_t = r^T + \bar{\pi} + \phi_\pi(\pi_t - \bar{\pi}) + \phi_y(\hat{Y}_t - \mu) - \phi_{vol}(x_t - \bar{x}), \tag{47}$$

where $r^T \equiv r^n - \frac{1}{2}(\bar{x} - \sigma^2)$ is the risk-adjusted steady-state real rate. Under this specification, the equilibrium construction is essentially unchanged relative to Section 5. Moreover, centering the policy rate at $r^T + \bar{\pi}$ ensures that the steady-state values of the output gap, aggregate output variance, and inflation coincide with their respective long-run targets.

5.2 Multiple Equilibria with the Non-linear Phillips Curve

This section shows that multiple bounded equilibria arise in the exact non-linear model with Rotemberg price adjustment costs, without relying on either the rigid-price simplification in Section 3 or the log-linear approximation of the New Keynesian Phillips curve in Section 5.1.

Online Appendix K derives the non-linear Phillips curve under Rotemberg price adjustment costs:

$$\mathbb{E}_t d\pi_t = \left[\left[2(\rho + \pi_t) - i_t - x_t\right](\pi_t - \bar{\pi}) - \sigma_t^\pi \sqrt{x_t} + \left(\frac{\epsilon - 1}{\tau}\right)\left[1 - e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}\right]\right] dt, \tag{48}$$

where σ_t^π is inflation volatility, $\bar{\pi}$ is the long-run (steady-state) inflation target, and $x_t \equiv (\sigma + \sigma_t^s)^2$ is aggregate output variance.

Since equation (48) is highly nonlinear, the system formed by (48) and the IS equation (30) does not admit a closed-form analytical solution. Thus, we solve the system numerically. Online Appendix K.6 provides the full description of the numerical procedure.

The method proceeds by conjecturing an analytical structure for the solution and then checking numerically that the conjecture is internally consistent within the system. For the output gap, we use the Ornstein–Uhlenbeck specification from

[34] In this case, the New Keynesian Phillips curve is log-linearized around the long-run targets for the output gap (μ) and inflation ($\bar{\pi}$), with aggregate output variance evaluated at $\bar{x}$. See equation (K.29) in Online Appendix K.4.

previous sections:

$$d\hat{Y}_t = -\theta\left[\hat{Y}_t - \mu\right]dt + \sigma_t^s dZ_t, \quad \text{where} \quad \theta \geq 0, \ \text{ and } \ \mu \leq 0.$$

Since inflation is now part of the system, we impose the additional restriction that inflation is a smooth function of aggregate output variance, $\pi_t \equiv \pi(x_t)$. Under these conjectures, the system reduces to a second-order ordinary differential equation (ODE) for $\pi(\cdot)$, which we solve numerically. The equilibrium output gap can then also be expressed as a function of x_t, that is, $\hat{Y}_t \equiv \hat{Y}(x_t)$.

Figure 4 plots the numerical solution, mapping from aggregate variance x_t to inflation π_t and the output gap $\hat{Y}_t$. Figures 4–6 set $\mu = -0.0047$ to the OLS point estimate of the long-run mean of the output gap constructed from CBO potential output, and set $\theta = 0.95$ as a highly autoregressive value for the numerical illustrations. The policy-rule coefficients are $\phi_y = 0.5$ and $\phi_\pi = 1.5$ (Taylor, 1993). Online Appendix L reports the remaining parameters and the construction of these values. In line with the model's precautionary savings channel, a higher x_t (greater uncertainty about future output and consumption) is associated with lower π_t and a lower $\hat{Y}_t$. This outcome reflects depressed demand driven by a stronger precautionary saving motive of households.

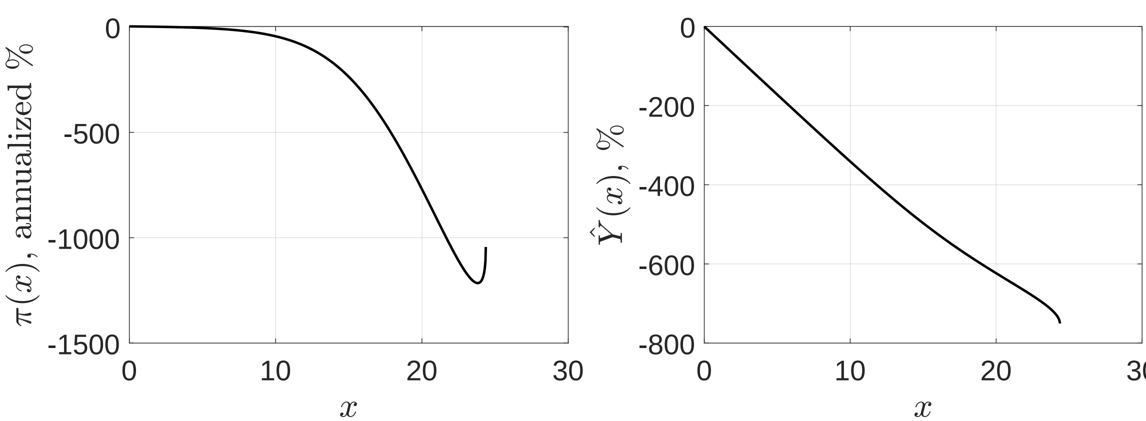


Figure 4: Mapping from aggregate volatility to inflation and output gap. The figure calibrates the persistence and mean of the output-gap process to $\theta = 0.95$ and $\mu = -0.0047$, with Taylor-rule coefficients $\phi_y = 0.5$ and $\phi_\pi = 1.5$. Remaining parameters are reported in Online Appendix L.

Stationarity The diffusion process for x_t is given by

$$dx_t = \mu_t^x \, dt + \sigma_t^x \, dZ_t,$$

where both drift and diffusion depend on x_t, i.e., $\mu_t^x \equiv \mu^x(x_t)$ and $\sigma_t^x \equiv \sigma^x(x_t)$. Figure 5 plots these functions over the numerically relevant range of x_t. The diffusion term becomes zero at two points: $x_t = \sigma^2$ (zero excess volatility, $\sigma_t^s = 0$) and at a finite upper level of x_t. The drift is positive at the lower point and negative at the upper point, so it points inward at both ends of the domain. Together, inward drift and vanishing diffusion at the endpoints imply a bounded state space with mean-reverting dynamics for x_t. Starting from an interior point, the process remains within this range and is unable to leave the boundaries. Since the remaining equilibrium variables are functions of x_t, they inherit bounded, mean-reverting dynamics and therefore admit stationary distributions.

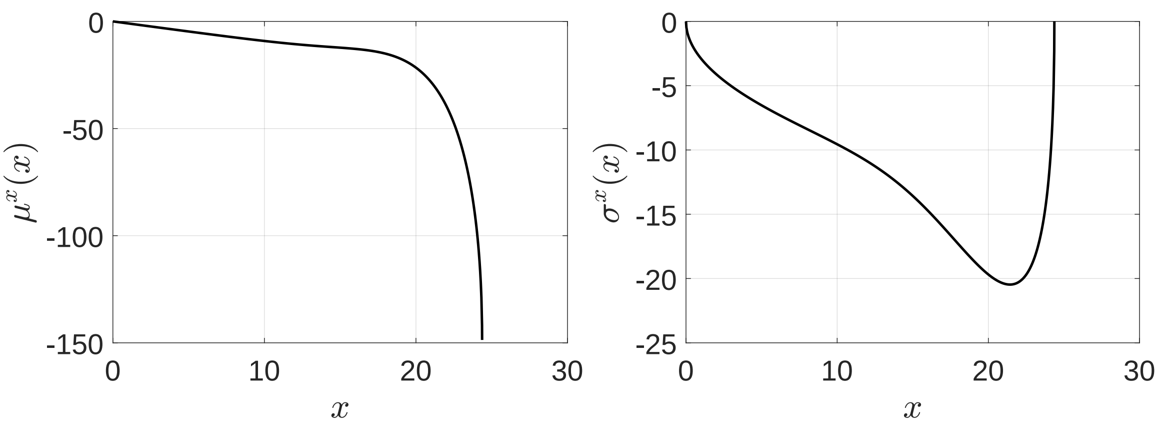


Figure 5: Drift and diffusion of total variance in the non-linear Rotemberg model. The figure calibrates the persistence and mean of the output-gap process to $\theta = 0.95$ and $\mu = -0.0047$, with Taylor-rule coefficients $\phi_y = 0.5$ and $\phi_\pi = 1.5$. Remaining parameters are reported in Online Appendix L.

Figure 6 reports a representative simulated path under that same calibration. The dynamics are close to the rigid-price baseline in Figure 3a: the system remains near the steady state most of the time, with occasional episodes of high excess volatility, strongly negative output gaps, and deflation, consistent with the mapping in Figure 4. One time unit corresponds to a quarter. Inflation and excess volatility are plotted in annualized units, and the path is a continuous-time discretization with small time steps, so instantaneous movements can be large even when the implied quarterly or annual changes are more moderate for most of the sample. The large spikes in the figures are difficult to eliminate even under a different calibration of the output-gap process, because the model is a simple New Keynesian specification without the additional adjustment frictions of a medium-scale DSGE model that could dampen their magnitude. Those spikes are infrequent, however, and Figure K.2 in Online Appendix K trims the tail observations and shows that typical realizations remain more moderate.

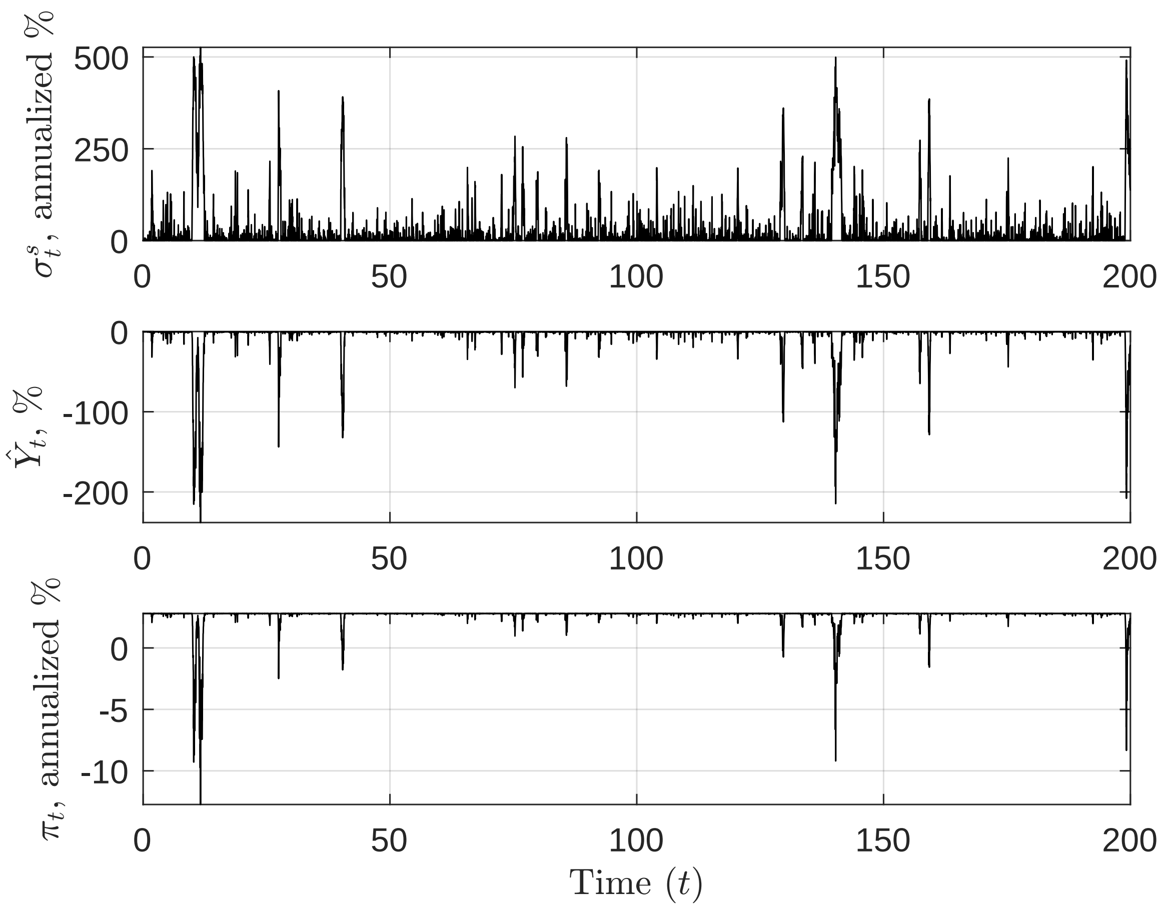


Figure 6: Simulated path of excess volatility $\{\sigma_t^s\}$, output gap $\{\hat{Y}_t\}$, and inflation $\{\pi_t\}$. The figure calibrates the persistence and mean of the output-gap process to $\theta = 0.95$ and $\mu = -0.0047$, with Taylor-rule coefficients $\phi_y = 0.5$ and $\phi_\pi = 1.5$. Remaining parameters are reported in Online Appendix L.

Finally, the solutions presented here are valid for a wide range of (μ, θ) combinations in the output-gap process satisfying $\mu \leq 0$ and $\theta \geq 0$. In practice, this means that these equilibria are not only distinct from the traditional perfect-stabilization solution in Proposition 1, but also flexible enough to accommodate many alternative specifications of output-gap dynamics and/or other targeted moments. We leave for future work the analysis of possible equilibria outside the Ornstein–Uhlenbeck class considered here.

6 Conclusion

The canonical New Keynesian model features additional classes of bounded global solutions in which aggregate volatility is sustained by households' contingent consumption plans. Some converge almost surely to the stabilized equilibrium; others remain stochastic in the long run. Both classes differ from the unique local solution found around the deterministic steady state when the Taylor principle is satisfied. The Taylor principle therefore fails as a global equilibrium selector, and uniqueness requires additional assumptions beyond a specific interest-rate response. The Taylor rule remains useful, however, as a reduced-form summary of central bank policy.

Model determinacy and stabilization can be restored by restricting the coordination that sustains these alternative paths. A small friction in social memory prevents households from coordinating contingent consumption plans on the history of realized shocks, and selects the stabilized equilibrium as the unique bounded solution. Level-k reasoning has a similar effect because it attenuates households' consideration of the full general equilibrium consequences of those plans. Finally, a fiscal escape clause rules out the same alternative paths in a non-Ricardian overlapping-generations economy: the government issues a debt-financed transfer only off the equilibrium path, and the transfer is never paid along the stabilized equilibrium.

Which of these mechanisms is more relevant in practice remains an open question.

Declaration of Interest

The authors have no conflicts of interest to disclose.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to assist with coding the simulation exercises and to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of this publication.

I Proofs and Derivations

I.1 Derivations in Section 2

Derivation of Equation (4) From the definition of (nominal) state-price density $\xi_t^N = e^{-\rho t}\frac{1}{C_t}\frac{1}{p_t}$, we obtain

$$\frac{d\xi_t^N}{\xi_t^N} = -\rho dt - \frac{dC_t}{C_t} - \frac{dp_t}{p_t} + \frac{1}{2}\left(\frac{dC_t}{C_t}\right)^2 + \frac{1}{2}\left(\frac{dp_t}{p_t}\right)^2 + \frac{dC_t}{C_t}\frac{dp_t}{p_t}. \tag{I.1}$$

Since we have a perfectly rigid price (i.e., $p_t = \bar{p}$ for all t), the above (I.1) becomes

$$\begin{aligned}\frac{d\xi_t^N}{\xi_t^N} &= -\rho dt - \frac{dC_t}{C_t} + \left(\frac{dC_t}{C_t}\right)^2 \\ &= -\rho dt - \frac{dC_t}{C_t} + \mathrm{Var}_t\left(\frac{dC_t}{C_t}\right).\end{aligned} \tag{I.2}$$

Plugging equation (I.2) into equation (3), we obtain

$$\mathbb{E}_t\left(\frac{dC_t}{C_t}\right) = \left(i_t - \rho\right)dt + \mathrm{Var}_t\left(\frac{dC_t}{C_t}\right).$$

Derivation of Equation (12) From equation (11), we obtain

$$d\ln Y_t = \left(i_t - \rho + \frac{1}{2}\left(\sigma + \sigma_t^s\right)^2\right)dt + (\sigma + \sigma_t^s)dZ_t. \tag{I.3}$$

From (9), we obtain

$$d\ln Y_t^n = \left(r^n - \rho + \frac{1}{2}\sigma^2\right)dt + \sigma dZ_t. \tag{I.4}$$

Therefore, by subtracting (I.4) from (I.3), we obtain (12):

$$d\hat{Y}_t = \left(i_t - \left(r^n - \frac{1}{2}\left(\sigma + \sigma_t^s\right)^2 + \frac{1}{2}\sigma^2\right)\right)dt + \sigma_t^s dZ_t.$$

I.2 Proofs of Section 3.1

The Construction of the Martingale Equilibrium in Section 3.1 and the Proof of Proposition 2 and Corollary 1. Setting the drift of the $\hat{Y}_t$ process in (15) to zero, i.e.,

$$d\hat{Y}_t = \left(\underbrace{\phi_y \hat{Y}_t - \frac{\sigma^2}{2} + \frac{(\sigma + \sigma_t^s)^2}{2}}_{=0} \right) dt + \sigma_t^s dZ_t = \sigma_t^s dZ_t, \tag{I.5}$$

leads to

$$\hat{Y}_t = -\frac{\left(\sigma + \sigma_t^s\right)^2}{2\phi_y} + \frac{\sigma^2}{2\phi_y}, \tag{I.6}$$

proving equation (19). From equations (I.5) and (I.6), we obtain

$$d\hat{Y}_t = -\frac{1}{\phi_y}(\sigma + \sigma_t^s) d\sigma_t^s - \frac{1}{2\phi_y}\left(d\sigma_t^s\right)^2 = \sigma_t^s dZ_t,$$

which leads to

$$d\sigma_t^s = -(\phi_y)^2 \frac{(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3} dt - \phi_y \frac{\sigma_t^s}{\sigma + \sigma_t^s} dZ_t,$$

proving equation (20).

From equations (I.5) and (I.6), it is evident that $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2$ has no drift, thereby qualifying as a local martingale. This can also be demonstrated as follows:

$$\begin{aligned} d\mathcal{E}_t &= 2\left(\sigma + \sigma_t^s\right) d\sigma_t^s + \left(d\sigma_t^s\right)^2 \\ &= 2(\sigma + \sigma_t^s)\left(\cancel{-\frac{(\phi_y)^2 (\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3} dt} - \phi_y \frac{\sigma_t^s}{\sigma + \sigma_t^s} dZ_t \right) + \cancel{(\phi_y)^2 \frac{(\sigma_t^s)^2}{(\sigma + \sigma_t^s)^2} dt} \\ &= -2\phi_y(\sigma_t^s) dZ_t = 2\phi_y \left(\sigma - \sqrt{\sigma^2 + \mathcal{E}_t} \right) dZ_t. \end{aligned} \tag{I.7}$$

Our logic proceeds as follows:

Step 1 Starting from $\mathcal{E}_0 > 0$, $\mathcal{E}_t > 0$ for all t almost surely, i.e., 0 is a natural boundary of the $\mathcal{E}_t$ process.

Proof. We use the methodology of Linetsky (2007): by defining a function

$m(\mathcal{E})$ that measures the speed of convergence in the process (I.7) as follows:

$$m(\mathcal{E}) \equiv \frac{1}{\left(\sqrt{\sigma^2 + \mathcal{E}} - \sigma\right)^2},$$

which determines the behavior of $\mathcal{E}_t$ process.

For small $\Delta > 0$, we calculate the following two integrals:

$$\begin{aligned} I_0 &\equiv \int_0^{\Delta} \mathcal{E} \cdot m(\mathcal{E})\, d\mathcal{E} = \int_0^{\Delta} \mathcal{E} \cdot \frac{1}{\left(\sqrt{\sigma^2 + \mathcal{E}} - \sigma\right)^2} d\mathcal{E} \\ &= \int_0^{\sqrt{\Delta+\sigma^2}-\sigma} \left(\frac{t + 2\sigma}{t}\right) \cdot 2(t+\sigma) dt \longrightarrow \infty, \end{aligned} \tag{I.8}$$

with $t \equiv \sqrt{\mathcal{E} + \sigma^2} - \sigma$ as a change of variable. Similarly,

$$\begin{aligned} J_0 &\equiv \int_0^{\Delta} (\Delta - \mathcal{E}) \cdot m(\mathcal{E})\, d\mathcal{E} = \int_0^{\Delta} (\Delta - \mathcal{E}) \cdot \frac{1}{\left(\sqrt{\sigma^2 + \mathcal{E}} - \sigma\right)^2} d\mathcal{E} \\ &= \int_0^{\sqrt{\Delta+\sigma^2}-\sigma} \left[\frac{(\Delta + \sigma^2) - (t+\sigma)^2}{t^2}\right] \cdot 2(t+\sigma) dt \longrightarrow \infty. \end{aligned} \tag{I.9}$$

With $I_0 = \infty$ and $J_0 = \infty$, process $\mathcal{E}_t$ has zero as a natural boundary, i.e., $\mathcal{E}_t$ never reaches the boundary at zero if it starts in the interior of the state space. In other words, if $\mathcal{E}_0 > 0$, then $\mathcal{E}_t \geq 0$ almost surely. ■

Step 2 As $\mathcal{E}_t$ is a local martingale that is non-negative due to Step 1, it becomes a supermartingale: see e.g., Le Gall (2016) about how to use Fatou's lemma in proving this statement.[1] The supermartingale property gives $\mathbb{E}_0[\mathcal{E}_t] \leq \mathcal{E}_0$ for all t, and therefore $0 \leq \lim_{t\to\infty} \mathbb{E}_0[\mathcal{E}_t] \leq \mathcal{E}_0 < \infty$. From (I.6), $\hat{Y}_t = -\mathcal{E}_t/(2\phi_y)$. Combined with $\mathcal{E}_t \geq 0$, this yields $|\hat{Y}_t| = \mathcal{E}_t/(2\phi_y)$, and

$$\lim_{t\to\infty} \mathbb{E}_0 \left|\hat{Y}_t\right| \leq |\hat{Y}_0| < \infty,$$

which is the boundedness condition (16).

Step 3 Since $\mathcal{E}_t$ is a supermartingale that is non-negative (or more generally, bounded

[1] $\mathcal{E}_t$ might not be a true martingale as $\sigma - \sqrt{\sigma^2 + \mathcal{E}_t}$ in equation (I.7) is not bounded.

from below), we can apply the famous martingale convergence theorem (see e.g., Williams (1991) and Le Gall (2016)), that implies:

$$\mathcal{E}_t \xrightarrow{a.s} \mathcal{E}_\infty,$$

point-wise, where $\mathcal{E}_\infty$ exists and is finite almost surely.

Step 4 Now, we define a function that is globally concave:[2]

$$\Phi(x) \equiv 4\left[\sigma \log\left(\sqrt{\sigma^2 + x} - \sigma\right) + \sqrt{\sigma^2 + x}\right],$$

which yields

$$\Phi'(x) = \frac{1}{\sqrt{\sigma^2 + x} - \sigma},$$

and $\Phi(x) \longrightarrow -\infty$ as $x \longrightarrow 0$. Additionally, we can obtain

$$\begin{aligned} d\left(\Phi\left(\mathcal{E}_t\right)\right) &= \Phi'(\mathcal{E}_t)d\mathcal{E}_t + \frac{1}{2}\Phi''(\mathcal{E}_t)(d\mathcal{E}_t)^2 \\ &= -\phi_y^2 \frac{1}{\sqrt{\sigma^2 + \mathcal{E}_t}} dt - 2\phi_y dZ_t. \end{aligned} \tag{I.10}$$

From Step 3, we know that $\mathcal{E}_\infty$ is finite with probability one, which implies that the drift $\frac{1}{\sqrt{\sigma^2+\mathcal{E}_\infty}}$ of (I.10) is finite and positive as well. Then, the only way to satisfy (I.10) in the long run is for $\Phi(\mathcal{E}_t) \longrightarrow -\infty$, implying $\mathcal{E}_\infty = 0$.

Step 5 Finally, $\mathcal{E}_t \longrightarrow 0$ implies that $\sigma_t^s \longrightarrow 0$ almost surely. It can be easily shown that this satisfies our stochastic process (20) as follows:

$$\underbrace{d\sigma_t^s}_{\xrightarrow{a.s} 0} = -\underbrace{\frac{(\phi_y)^2(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3}}_{\xrightarrow{a.s} 0} dt - \phi_y \underbrace{\frac{\sigma_t^s}{\sigma + \sigma_t^s}}_{\xrightarrow{a.s} 0} dZ_t. \tag{I.11}$$

[2]We appreciate Victor Kleptsyn at CNRS à Institut de Recherche Mathématique de Rennes for suggesting function $\Phi(x)$.

Step 6 Finally from Le Gall (2016), we know that if

$$\mathbb{E}_0\left(\sup_{t\geq 0}|\mathcal{E}_t|\right) < \infty,$$
$$\text{or} \quad \lim_{K\to\infty}\sup_{t\geq 0}\left(\mathbb{E}_0\left(|\mathcal{E}_t|\,\mathbb{1}_{\{|\mathcal{E}_t|\geq K\}}\right)\right) > 0,$$

then $\mathcal{E}_t$, which is a local martingale, becomes an uniformly integrable martingale. But then, if $\mathcal{E}_t$ is an uniformly integrable martingale,

$$0 < \mathcal{E}_0 = \lim_{t\to\infty}\mathbb{E}_0\mathcal{E}_t = \mathbb{E}_0\underbrace{\mathcal{E}_\infty}_{=0} = 0,$$

which is a contradiction. Therefore, Property 3 of Proposition 2 is proved.

Special case Corollary 1 follows by setting $\sigma = 0$ in equation (19) and equation (20), which yields the Bessel process equation (21). The process stops when σ_t^s reaches zero. In this case, we can observe that equation (I.8) becomes[3]

$$I_0 \equiv \int_0^{\Delta}\mathcal{E}\cdot\frac{1}{\left(\sqrt{0^2+\mathcal{E}}-0\right)^2}d\mathcal{E} = \Delta < \infty,$$

with $J_0 = \infty$. This implies that the $\mathcal{E}_t$ process has zero as an exit boundary, meaning the $\mathcal{E}_t$ process is instantaneously terminated the first time this boundary is reached. The behavior of the Bessel process (21) hitting time (i.e., the first time it reaches zero) is well known. For example, its hitting time τ has a well-defined distribution, as derived in Lawler (2019).

■

Boundedness Under the martingale equilibrium, $\mathcal{E}_t$ is a non-negative local martingale, which, by Step 2, is also a supermartingale. Therefore, by definition,

$$\mathbb{E}_0[\mathcal{E}_t] \leq \mathcal{E}_0.$$

[3]We appreciate an anonymous referee for pointing out this discontinuity.

Taking the limit as $t \to \infty$, we obtain

$$0 \leq \lim_{t\to\infty} \mathbb{E}_0[\mathcal{E}_t] \leq \mathcal{E}_0. \tag{I.12}$$

Using the relation $\hat{Y}_t = -\frac{1}{2\phi_y}\mathcal{E}_t$ from (I.6), equation (I.12) implies that

$$\lim_{t\to\infty} \mathbb{E}_0 \left|\hat{Y}_t\right| \leq \left|\hat{Y}_0\right| < \infty,$$

which establishes the boundedness condition in (16).

I.3 Derivations of Section 3.2

The equilibrium output gap $\hat{Y}_t$ follows:

$$\mathrm{d}\hat{Y}_t = \left[\phi_y \hat{Y}_t - \frac{\sigma^2}{2} + \frac{\left(\sigma + \sigma_t^s\right)^2}{2}\right] \mathrm{d}t + \sigma_t^s \mathrm{d}Z_t. \tag{I.13}$$

Now we guess that the solution of the model represented by equation (I.13) has the following form:

$$\mathrm{d}\hat{Y}_t = \theta \cdot \left[\mu - \hat{Y}_t\right] \mathrm{d}t + \sigma_t^s \mathrm{d}Z_t, \tag{I.14}$$

where θ and μ are constant parameters. The process I.14 is similar to the Ornstein-Uhlenbeck process, except for the fact that it has an endogenous volatility σ_t^s which is to be determined in equilibrium.

Note that when $\theta = 0$, the process (I.14) becomes the martingale conjectured in Section 3.1. For this new conjectured solution to be valid, the drift term in (I.13) and (I.14) should be equal, implying that the output gap under this conjecture is:

$$\hat{Y}_t = \frac{\theta\mu}{\theta + \phi_y} - \frac{1}{2(\theta + \phi_y)} \left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]. \tag{I.15}$$

We know that σ_t^s follows a process of the form:[4]

$$\mathrm{d}\sigma_t^s = \mu_t^\sigma \mathrm{d}t + \tilde{\sigma}_t \mathrm{d}Z_t,$$

where μ_t^σ and $\tilde{\sigma}_t$ are unknown variables. Applying Ito's Lemma to I.15 we obtain:

$$\mathrm{d}\hat{Y}_t = -\left(\frac{1}{\theta + \phi_y}\right)\left[(\sigma + \sigma_t^s)\cdot \mu^\sigma + \frac{\tilde{\sigma}_t^2}{2}\right]\mathrm{d}t - \tilde{\sigma}_t \cdot \left(\frac{\sigma + \sigma_t^s}{\theta + \phi_y}\right)\mathrm{d}Z_t. \tag{I.16}$$

By equating the drift and volatility terms of equations I.16 and I.15, and based on equation (23), we obtain the unknown variables μ_t^σ and $\tilde{\sigma}_t$ consistent with our guessed solution as follows:

$$\begin{aligned}
\tilde{\sigma}_t &= -(\theta + \phi_y)\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right), \\
\mu_t^\sigma &= -\left(\frac{\theta}{\sigma + \sigma_t^s}\right)\left[\mu\phi_y + \frac{1}{2}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]\right] - (\theta + \phi_y)^2 \frac{(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3}.
\end{aligned}$$

Therefore, the σ_t^s process consistent with our guessed solution in (22) can be written as:

$$\begin{aligned}
\mathrm{d}\sigma_t^s = &-\left[\left(\frac{\theta}{\sigma + \sigma_t^s}\right)\left[\mu\phi_y + \frac{1}{2}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]\right] + (\theta + \phi_y)^2 \frac{(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3}\right]\mathrm{d}t \\
&- (\theta + \phi_y)\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)\mathrm{d}Z_t,
\end{aligned} \tag{I.17}$$

leading to

$$\mathrm{d}(\sigma + \sigma_t^s)^2 = -\theta\left[2\mu\phi_y + (\sigma + \sigma_t^s)^2 - \sigma^2\right]\mathrm{d}t - 2(\theta + \phi_y)\sigma_t^s \mathrm{d}Z_t.$$

Notice the two following important observations:

Observation 1 When $\underline{\theta = 0}$, the solution (I.17) becomes the martingale equilibrium solution of Section 3.1.

Observation 2 We obtain solutions with different values of θ and μ parameters, so there is a chance that different parametrization of this pair of parameters can be

[4] As in Section 2, we assume that all the aggregate variables are adapted to the given filtration $\{\mathcal{F}_t\}_{t\geq 0}$ generated by the Brownian motion dZ_t.

consistent with a rational expectations equilibrium solution considered in Section 3.2.

II Detailed Derivations in Section 2

II.1 Model Setup

A representative household h solves the following intertemporal consumption-savings optimization problem:

$$\max_{\substack{\{C_s^h, L_s^h\}_{s\geq t} \\ \{B_s^h\}_{s>t}}} \mathbb{E}_t \int_t^{\infty} e^{-\rho(s-t)} \left[\log C_s^h - \frac{\left(L_s^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right] \mathrm{d}s,$$
$$\text{s.t. } dB_t^h = \left[i_t B_t^h - p_t C_t^h + w_t L_t^h + D_t \right] dt,$$

where C_t^h is consumption, L_t^h is aggregate labor, w_t is the equilibrium wage level, B_t^h are risk-free bonds held by the household h at the beginning of t (hence, B_t^h at t is taken as given for each household), i_t is the nominal interest rate, D_t is a lump-sum transfer of any firm profits/losses towards the household, p_t the nominal price of consumption goods and ρ is the subjective discount rate of the household.[5]

An individual firm i produces in this economy with the following production function:

$$Y_t^i = A_t L_t^i, \text{ with } \frac{\mathrm{d}A_t}{A_t} = g\mathrm{d}t + \underbrace{\sigma}_{\text{Fundamental risk}} \mathrm{d}Z_t,$$

where A_t is the economy's total factor productivity, assumed to be exogenous and to follow a geometric Brownian motion with drift g and volatility σ, which we assume to be constant over time and define as the *fundamental* volatility, and Z_t is a standard Brownian motion process. It follows that firms' profits are defined as:

$$D_t = p_t Y_t - w_t L_t.$$

Finally, we assume bonds are in zero net supply in equilibrium (i.e., $B_t = 0, \ \forall t$), and

[5] Later, we will impose the equilibrium condition: $C_t^h = C_t, \forall h$, $L_t^h = L_t, \forall h$, $B_t^h = B_t = 0, \forall h$, where C_t, L_t, and B_t are aggregate variables.

that there is no government spending, so market clearing in this economy results in $C_t = Y_t$.

II.2 Flexible Price Economy

We first solve the flexible price economy as our benchmark economy. For that purpose, we assume the usual Dixit-Stiglitz monopolistic competition among firms, where the demand each firm i faces is given by

$$D(p_t^i, p_t) = \left(\frac{p_t^i}{p_t}\right)^{-\varepsilon} Y_t,$$

where p_t^i is an individual firm i's price, p_t is the price aggregator, and Y_t is the aggregate output. Each firm i takes the aggregate price p_t, wage w_t, and the aggregate output Y_t as given.

II.2.1 Household problem

In the flexible price economy, each household takes the $\{A_t, p_t, i_t\}$ processes as given:

$$\frac{dp_t}{p_t} = \pi_t dt + \sigma_t^p dZ_t,$$

and

$$di_t = \mu_t^i dt + \sigma_t^i dZ_t,$$

where π_t, σ_t^p, μ_t^i, and σ_t^i are all endogenous, so the state variables for each household would become $\{B_t^h, A_t, p_t, i_t\}$.[6]

Hamilton-Jacobi-Bellman (HJB) formulation of the households' problem We define the value function of household h as:

[6]This is conjectural but correct due to the classical dichotomy between real and nominal sectors: output, consumption, and labor in equilibrium depend on A_t only, and p_t and i_t do not affect the real economy and welfare of the households.

$$\Gamma^h \equiv \Gamma^h\left(B_t^h, A_t, p_t, i_t, t\right) = \max_{\substack{\{C_s^h, L_s^h\}_{s\geq t}\\ \{B_s^h\}_{s>t}}} \mathbb{E}_t \int_t^{\infty} e^{-\rho(s-t)} \left[\log C_s^h - \frac{\left(L_s^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}}\right] \mathrm{d}s.$$

subject to $dB_t^h = \left[i_t B_t^h - p_t C_t^h + w_t L_t^h + D_t\right] dt$. The HJB equation is given by

$$\rho \cdot \Gamma^h = \max_{C_t^h, L_t^h} \left\{ \log C_t^h - \frac{\left(L_t^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \frac{\mathbb{E}_t\left[\mathrm{d}\Gamma^h\right]}{\mathrm{d}t} \right\}. \tag{II.1}$$

Using Ito's Lemma, we compute:

$$d\Gamma^h = \mu_t^{\Gamma,h} dt + \sigma_t^{\Gamma,h} dZ_t, \tag{II.2}$$

where

$$\begin{aligned}
\mu_t^{\Gamma,h} =& \Gamma_t^h + \Gamma_B^h \cdot \left(i_t B_t^h - p_t C_t^h + w_t L_t^h + D_t\right) + \Gamma_A^h \cdot A_t g + \Gamma_p^h \cdot p_t \pi_t + \Gamma_i^h \cdot \mu_t^i \\
&+ \frac{1}{2}\Gamma_{AA}^h \cdot (A_t\sigma)^2 + \frac{1}{2}\Gamma_{pp}^h \cdot (p_t\sigma_t^p)^2 + \frac{1}{2}\Gamma_{ii}^h \cdot (\sigma_t^i)^2 \\
&+ \Gamma_{Ap}^h \cdot (\sigma A_t)(p_t\sigma_t^p) + \Gamma_{Ai}^h \cdot (\sigma A_t)\sigma_t^i + \Gamma_{pi}^h \cdot (p_t\sigma_t^p)\sigma_t^i,
\end{aligned}$$

and $\sigma_t^{\Gamma,h} = \Gamma_A^h(\sigma A_t) + \Gamma_p^h(p_t\sigma_t^p) + \Gamma_i^h(\sigma_t^i)$. In the same way, we compute $d\Gamma_B^h = \mu_t^{\Gamma_B,h} dt + \sigma_t^{\Gamma_B,h} dZ_t$, where

$$\begin{aligned}
\mu_t^{\Gamma_B,h} =& \Gamma_{Bt}^h + \Gamma_{BB}^h \cdot \left(i_t B_t^h - p_t C_t^h + w_t L_t^h + D_t\right) + \Gamma_{BA}^h \cdot A_t g + \Gamma_{Bp}^h \cdot p_t \pi_t + \Gamma_{Bi}^h \cdot \mu_t^i \\
&+ \frac{1}{2}\Gamma_{BAA}^h \cdot (A_t\sigma)^2 + \frac{1}{2}\Gamma_{Bpp}^h \cdot (p_t\sigma_t^p)^2 + \frac{1}{2}\Gamma_{Bii}^h \cdot (\sigma_t^i)^2 \\
&+ \Gamma_{BAp}^h \cdot (\sigma A_t)(p_t\sigma_t^p) + \Gamma_{BAi}^h \cdot (\sigma A_t)\sigma_t^i + \Gamma_{Bpi}^h \cdot (p_t\sigma_t^p)\sigma_t^i,
\end{aligned} \tag{II.3}$$

and $\sigma_t^{\Gamma_B,h} = \Gamma_{BA}^h(\sigma A_t) + \Gamma_{Bp}^h(p_t\sigma_t^p) + \Gamma_{Bi}^h(\sigma_t^i)$. Note that $\Gamma_\Delta^h = \frac{\partial \Gamma^h}{\partial \Delta}$ is defined as the derivative with respect to any subindex variable $\Delta = \{t, B^h, A, p, i\}$. Now plug equation (II.2) into equation (II.1) to obtain:

$$\rho \cdot \Gamma^h = \max_{C_t^h, L_t^h} \left\{ \log C_t^h - \frac{\left(L_t^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h} \right\}. \tag{II.4}$$

Households' first-order conditions (FOC) Computing the first-order conditions with respect to C_t^h and L_t^h from equation (II.4), we obtain:

$$\Gamma_B^h = \frac{1}{p_t C_t^h}, \tag{II.5}$$

$$\Gamma_B^h = \frac{\left(L_t^h\right)^{\frac{1}{\eta}}}{w_t}. \tag{II.6}$$

Finally, merging (II.5) with (II.6) gives us the intratemporal optimality condition.

State price density and pricing kernel We know the state price density and the stochastic discount factor between two adjacent periods are given by $\zeta_t^{N,h} = e^{-\rho t}\frac{1}{p_t C_t^h}$, and $dQ_t^h = \frac{\mathrm{d}\zeta_t^{N,h}}{\zeta_t^{N,h}}$, respectively. Let us use a star superscript to denote the choice variables evaluated at the optimum, that is $C_t^{h,*}$ and $L_t^{h,*}$. Then, we can express equation (II.4) as:

$$\rho \cdot \Gamma^h = \log C_t^{h,*} - \frac{\left(L_t^{h,*}\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h,*}. \tag{II.7}$$

Taking the derivative of both sides of equation (II.7) with respect to B_t, using the envelope theorem and rearranging, we obtain:

$$(\rho - i_t) \cdot \Gamma_B^h = \mu_t^{\Gamma_B,h,*}, \tag{II.8}$$

where $\mu_t^{\Gamma_B,h,*}$ is from equation (II.3) and it is evaluated at the optimum. Plugging (II.8) into the process for Γ_B^h, we obtain a simplified expression:

$$\mathrm{d}\Gamma_B^h = (\rho - i_t) \cdot \Gamma_B^h \mathrm{d}t + \underbrace{\left(\Gamma_{BA}^h (A_t \sigma) + \Gamma_{Bp}^h (p_t \sigma_t^p) + \Gamma_{Bi}^h \left(\sigma_t^i\right)\right)}_{\equiv \sigma_t^{\Gamma_B,h}} \mathrm{d}Z_t. \tag{II.9}$$

Note that $\zeta_t^{N,h} = e^{-\rho t}\Gamma_B^h$, then, using equation (II.9) and applying Ito's Lemma, we obtain:

$$\mathrm{d}\zeta_t^{N,h} = -\,\zeta_t^N \cdot i_t \mathrm{d}t + \zeta_t^N \cdot \left[\frac{\sigma_t^{\Gamma_B,h}}{\Gamma_B^h}\right] \mathrm{d}Z_t.$$

From the definition of $\mathrm{d}Q_t$, we obtain:

$$dQ_t^h \equiv \frac{\mathrm{d}\zeta_t^{N,h}}{\zeta_t^{N,h}} = -i_t \mathrm{d}t + \left[\frac{\sigma_t^{\Gamma_B,h}}{\Gamma_B^h}\right] \mathrm{d}Z_t, \tag{II.10}$$

and $\mathbb{E}_t\left[\mathrm{d}Q_t^h\right] = -i_t \mathrm{d}t$ follows by taking expectations, which proves (3) in the flexible price equilibrium.

Equilibrium is defined as in Definition 1 and Definition 2. From now, we interchangeably use variables with and without h superscript.

Nominal and real interest rates Prices and consumption should be adapted to the filtration generated by the Brownian motion Z_t process. Let us express the processes for consumption and price as:

$$\begin{aligned} \mathrm{d}p_t &= \pi_t p_t \mathrm{d}t + \sigma_t^p p_t \mathrm{d}Z_t, \\ \mathrm{d}C_t &= g_t^C C_t \mathrm{d}t + \sigma_t^C C_t \mathrm{d}Z_t, \end{aligned} \tag{II.11}$$

where π_t, g_t^C, σ_t^p and σ_t^C are inflation, expected consumption growth, and the volatility of the price and consumption processes, respectively.[7] As the real state density is defined as $\zeta_t^r = e^{-\rho t}\frac{1}{C_t}$, the real interest rate r_t is defined by the relation $\mathbb{E}_t\left[\frac{d\zeta_t^r}{\zeta_t^r}\right] = -r_t dt$, similarly to (3).

With (II.11), applying Ito's Lemma to the real state density $\zeta_t^r = e^{-\rho t}\frac{1}{C_t}$ results in

$$\frac{d\zeta_t^r}{\zeta_t^r} = -\underbrace{\left[\rho + g_t^C - \left(\sigma_t^C\right)^2\right]}_{\equiv r_t} \mathrm{d}t - \sigma_t^C \mathrm{d}Z_t, \tag{II.12}$$

which determines the real interest rate $r_t = \rho + g_t^C - (\sigma_t^C)^2$. We also apply Ito's Lemma to $\zeta_t^N = e^{-\rho t}\frac{1}{p_t C_t}$ and use the above processes for p_t and C_t to obtain:

[7] Note that these variables will be determined in equilibrium.

$$dQ_t \equiv \frac{\mathrm{d}\zeta_t^N}{\zeta_t^N} = -\left[\rho + g_t^C + \pi_t - (\sigma_t^p)^2 - \left(\sigma_t^C\right)^2 - \sigma_t^p \sigma_t^C\right] \mathrm{d}t - \left[\sigma_t^p + \sigma_t^C\right] \mathrm{d}Z_t,$$

which can be rearranged as:

$$dQ_t \equiv \frac{\mathrm{d}\zeta_t^N}{\zeta_t^N} = -\underbrace{\left[r_t + \pi_t - \sigma_t^p \left(\sigma_t^C + \sigma_t^p\right)\right]}_{=i_t} \mathrm{d}t - \left[\sigma_t^p + \sigma_t^C\right] \mathrm{d}Z_t. \tag{II.13}$$

Comparing equation (II.10) and equation (II.13), we obtain

$$i_t = r_t + \pi_t - \sigma_t^p \left(\sigma_t^C + \sigma_t^p\right), \text{ where: } r_t = \rho + g_t^C - \left(\sigma_t^C\right)^2.$$

II.2.2 Firm problem and equilibrium

Firm optimization The demand faced by each firm i is given by

$$D(p_t^i, p_t) = \left(\frac{p_t^i}{p_t}\right)^{-\varepsilon} Y_t,$$

where p_t^i is an individual firm's price, p_t is the price aggregator, and Y_t is the aggregate output. Each firm i solves the following problem:

$$\max_{p_t^i} \quad p_t^i \left(\frac{p_t^i}{p_t}\right)^{-\varepsilon} Y_t - \frac{w_t}{A_t}\left(\frac{p_t^i}{p_t}\right)^{-\varepsilon} Y_t,$$

which results in the following first-order condition for the firm:[8]

$$p_t = \left(\frac{\varepsilon}{\varepsilon - 1}\right) \frac{w_t}{A_t}, \tag{II.14}$$

which is intuitive since it tells us that in equilibrium, price is equal to the marginal cost of production multiplied by the constant mark-up, due to the constant elasticity of demand $\varepsilon > 1$. Using (II.14) and the equilibrium condition $C_t = Y_t = A_t L_t$ in the first-order condition of the household in (II.5) and (II.6), we obtain $L_t^n = \left(\frac{\varepsilon-1}{\varepsilon}\right)^{\frac{\eta}{\eta+1}}$,[9]

[8] In equilibrium, $p_t^i = p_t$ as every firm chooses the same price level.

[9] We impose the superscript n (i.e., natural) in variables to denote that those are the equilibrium values in the flexible price economy.

which is a constant. This implies that in the flexible price equilibrium, we have $C_t^n = Y_t^n = A_t \left(\frac{\varepsilon-1}{\varepsilon}\right)^{\frac{\eta}{\eta+1}}$. It follows that the stochastic process for Y_t^n is the same as that for A_t, as follows:

$$\frac{\mathrm{d}Y_t^n}{Y_t^n} = \frac{\mathrm{d}C_t^n}{C_t^n} = g\mathrm{d}t + \sigma \mathrm{d}Z_t. \tag{II.15}$$

Equation (II.15) implies that the growth rate of consumption and its volatility are $g_t^C = g$ and $\sigma_t^C = \sigma$, so the real interest rate in the flexible price economy, i.e., the natural rate of interest, can be expressed as $r_t^n \equiv r^n = \rho + g - \sigma^2$ from (II.12), which finally gives

$$\frac{\mathrm{d}Y_t^n}{Y_t^n} = \left(\underbrace{r^n}_{\text{Natural rate}} - \rho + \sigma^2 \right) \mathrm{d}t + \sigma \mathrm{d}Z_t,$$

which proves equation (9).

II.3 Rigid Price Economy

We now solve the equilibrium of the rigid price economy with $p_t = \bar{p}$ for all t. The rigid price economy's consumption volatility, which we define as σ_t^C, is given by $\sigma_t^C = \sigma + \sigma_t^s$ (i.e. volatility of the flexible price equilibrium in (II.15), plus excess volatility of rigid price equilibrium). Therefore, the consumption process can be written as:

$$\mathrm{d}C_t = g_t^C C_t \mathrm{d}t + (\sigma + \sigma_t^s) C_t \mathrm{d}Z_t. \tag{II.16}$$

We conjecture that this endogenous 'excess' volatility σ_t^s, which is one of the state variables in the rigid price economy, follows the process $d\sigma_t^s = \mu_t^\sigma dt + \sigma_t^\sigma dZ_t$. Under price rigidity (i.e., $p_t = \bar{p}$ for all t), the agent takes the $\{A_t, \sigma_t^s\}$ processes as given, so the state variables for each household become $\{B_t^h, A_t, \sigma_t^s\}$.[10]

Hamilton-Jacobi-Bellman (HJB) formulation of the households' problem We define the value function of household h as:

[10] This is conjectural (but correct) as the actual output (thereby, consumption and other variables including inflation, nominal interest rate (that follows the Taylor rule), etc) would turn out to only depend on A_t and σ_t^s under our equilibrium construction.

$$\Gamma^h \equiv \Gamma^h\left(B_t^h, A_t, \sigma_t^s, t\right) = \max_{\substack{\{C_s^h, L_s^h\}_{s\geq t}\\ \{B_s^h\}_{s>t}}} \mathbb{E}_t \int_s^\infty e^{-\rho(s-t)}\left[\log C_s^h - \frac{\left(L_s^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}}\right] \mathrm{d}s.$$

subject to $dB_t^h = \left[i_t B_t^h - p_t C_t^h + w_t L_t^h + D_t\right] dt$. The HJB equation can be written as:

$$\rho \cdot \Gamma^h = \max_{C_t^h, L_t^h}\left\{\log C_t^h - \frac{\left(L_t^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \frac{\mathbb{E}_t\left[\mathrm{d}\Gamma^h\right]}{\mathrm{d}t}\right\}, \tag{II.17}$$

Using Ito's Lemma, we compute:

$$d\Gamma^h = \mu_t^{\Gamma,h} dt + \sigma_t^{\Gamma,h} dZ_t, \tag{II.18}$$

where

$$\begin{aligned}\mu_t^{\Gamma,h} =& \Gamma_t^h + \Gamma_B^h \cdot \left(i_t B_t^h - \bar{p} \cdot C_t^h + w_t L_t^h + D_t\right) + \Gamma_A^h \cdot A_t g + \Gamma_\sigma^h \cdot \mu_t^\sigma \\ &+ \frac{1}{2}\Gamma_{AA}^h \cdot (A_t\sigma)^2 + \frac{1}{2}\Gamma_{\sigma\sigma}^h \cdot (\sigma_t^\sigma)^2 + \Gamma_{A\sigma}^h \cdot (A_t\sigma)(\sigma_t^\sigma),\end{aligned}$$

and $\sigma_t^{\Gamma,h} = \Gamma_A^h(\sigma A_t) + \Gamma_\sigma^h(\sigma_t^\sigma)$. Applying Ito's Lemma to Γ_B^h, we compute $d\Gamma_B^h = \mu_t^{\Gamma_B,h} dt + \sigma_t^{\Gamma_B,h} dZ_t$, where

$$\begin{aligned}\mu_t^{\Gamma_B,h} =& \Gamma_{Bt}^h + \Gamma_{BB}^h \cdot \left(i_t B_t^h - \bar{p} \cdot C_t^h + w_t L_t^h + D_t\right) + \Gamma_{BA}^h \cdot A_t g + \Gamma_{B\sigma}^h \cdot \mu_t^\sigma \\ &+ \frac{1}{2}\Gamma_{BAA}^h \cdot (A_t\sigma)^2 + \frac{1}{2}\Gamma_{B\sigma\sigma}^h \cdot (\sigma_t^\sigma)^2 + \Gamma_{BA\sigma}^h \cdot (A_t\sigma)(\sigma_t^\sigma),\end{aligned} \tag{II.19}$$

and $\sigma_t^{\Gamma_B,h} = \Gamma_{BA}^h \cdot (\sigma A_t) + \Gamma_{B\sigma}^h \cdot \sigma_t^\sigma$. Note $\Gamma_\Delta^h = \frac{\partial \Gamma^h}{\partial \Delta}$ is the derivative with respect to any subindex variable $\Delta = \{t, B^h, A, \sigma^s\}$. Now plug equation (II.18) into equation (II.17) to obtain:

$$\rho \cdot \Gamma^h = \max_{C_t^h, L_t^h}\left\{\log C_t^h - \frac{\left(L_t^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h}\right\}. \tag{II.20}$$

Households' first-order conditions (FOC) Computing the first-order conditions with respect to C_t^h and L_t^h from equation (II.20), we obtain:

$$\Gamma_B^h = \frac{1}{\bar{p}C_t^h}, \tag{II.21}$$

$$\Gamma_B^h = \frac{\left(L_t^h\right)^{\frac{1}{\eta}}}{w_t}. \tag{II.22}$$

Finally, merging (II.21) with (II.22) gives us the intratemporal condition of the problem.

State price density and pricing kernel We know that the state price density and stochastic discount factor between two adjacent periods are given by $\zeta_t^{N,h} = e^{-\rho t}\frac{1}{\bar{p}C_t^h}$, and $dQ_t^h = \frac{\mathrm{d}\zeta_t^{N,h}}{\zeta_t^{N,h}}$, respectively. We use a star superscript to denote the choice variables evaluated at the optimum, that is $C_t^{h,*}$ and $L_t^{h,*}$. Then, we can express equation (II.20) as:

$$\rho \cdot \Gamma^h = \log C_t^{h,*} - \frac{\left(L_t^{h,*}\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h,*}. \tag{II.23}$$

Taking the derivative of both sides of equation (II.23) with respect to B_t, using the envelop theorem and rearranging, we obtain:

$$(\rho - i_t) \cdot \Gamma_B^h = \mu_t^{\Gamma_B,h,*}, \tag{II.24}$$

where $\mu_t^{\Gamma_B,h,*}$ follows from (II.19) evaluated at the optimum. Plugging (II.24) into the process for Γ_B^h, we obtain a simplified expression at the optimum:

$$\mathrm{d}\Gamma_B^h = (\rho - i_t) \cdot \Gamma_B^h \mathrm{d}t + \underbrace{\left(\Gamma_{BA}^h \cdot (A_t\sigma) + \Gamma_{B\sigma}^h \cdot (\sigma_t^\sigma)\right)}_{\equiv \sigma_t^{\Gamma_B,h}} \mathrm{d}Z_t. \tag{II.25}$$

Notice that $\zeta_t^{N,h} = e^{-\rho t}\Gamma_B^h$, then using equation (II.25) and applying Ito's Lemma, we obtain:

$$\mathrm{d}\zeta_t^{N,h} = -\,\zeta_t^N \cdot i_t \mathrm{d}t + \zeta_t^{N,h} \cdot \left[\frac{\sigma_t^{\Gamma_B,h}}{\Gamma_B^h}\right] \mathrm{d}Z_t.$$

From the previous equation, we obtain:

$$dQ_t^h \equiv \frac{\mathrm{d}\zeta_t^{N,h}}{\zeta_t^{N,h}} = -i_t \mathrm{d}t + \left[\frac{\sigma_t^{\Gamma_B,h}}{\Gamma_B^h}\right] \mathrm{d}Z_t, \tag{II.26}$$

and $\mathbb{E}_t\left[\mathrm{d}Q_t^h\right] = -i_t \mathrm{d}t$ follows in this rigid price economy by taking conditional expectations.

Again, the equilibrium is defined in Definition 1. Henceforth, we interchangeably use variables with and without h superscript.

Verification of Constructed Equilibria We now can verify that our Ornstein-Uhlenbeck equilibrium,[11] characterized by (I.15) and (I.17), satisfies the equilibrium conditions derived above. From (I.15) and (I.17),

$$\hat{Y}_t = \frac{\theta\mu}{\theta+\phi_y} - \frac{1}{2(\theta+\phi_y)}\left[(\sigma+\sigma_t^s)^2 - \sigma^2\right], \tag{II.27}$$

$$\begin{aligned} \mathrm{d}\sigma_t^s = &-\left[\left(\frac{\theta}{\sigma+\sigma_t^s}\right)\left[\mu\phi_y + \frac{1}{2}\left[(\sigma+\sigma_t^s)^2 - \sigma^2\right]\right] + (\theta+\phi_y)^2 \frac{(\sigma_t^s)^2}{2(\sigma+\sigma_t^s)^3}\right]\mathrm{d}t \\ &-(\theta+\phi_y)\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)\mathrm{d}Z_t, \end{aligned} \tag{II.28}$$

(II.27) and (II.28) will form a solution to the model, as long as there is no contradiction with the equilibrium conditions. In order to check if (II.27) and (II.28) satisfy the equilibrium conditions, first, the output gap is defined as:

$$\hat{Y}_t = \log\left(\frac{Y_t}{Y_t^n}\right) = \log\left(\frac{C_t}{C_t^n}\right) = \log\left(\frac{C_t}{A_t}\right) - \frac{\eta}{\eta+1}\log\left(\frac{\varepsilon-1}{\varepsilon}\right), \tag{II.29}$$

where the last equality follows from $C_t^n = A_t\left(\frac{\varepsilon-1}{\varepsilon}\right)^{\frac{\eta}{\eta+1}}$ for the flexible price equilibrium. Combining (II.27) and (II.29), we obtain:

$$C_t = A_t\left(\frac{\varepsilon-1}{\varepsilon}\right)^{\frac{\eta}{\eta+1}} \cdot \exp\left\{\frac{\theta\mu}{\theta+\phi_y} - \frac{1}{2(\theta+\phi_y)}\left[(\sigma+\sigma_t^s)^2 - \sigma^2\right]\right\},$$

[11] With $\theta = 0$, it becomes the martingale equilibrium of Section 3.1.

which is a function of A_t and σ_t^s. We can compute the derivative of (II.21) with respect to A_t and σ_t^s as:

$$\Gamma_{BA} = -\frac{\Gamma_B}{C_t} \cdot \frac{\partial C_t}{\partial A_t}, \tag{II.30}$$

$$\Gamma_{B\sigma} = -\frac{\Gamma_B}{C_t} \cdot \frac{\partial C_t}{\partial \sigma_t^s}. \tag{II.31}$$

Plugging equations (II.30) and (II.31) into equation (II.25), we obtain:

$$\mathrm{d}\Gamma_B = (\rho - i_t) \cdot \Gamma_B \mathrm{d}t - \Gamma_B \left[\frac{A_t}{C_t} \cdot \frac{\partial C_t}{\partial A_t} \cdot \sigma + \frac{1}{C_t} \cdot \frac{\partial C_t}{\partial \sigma_t^s} \cdot \sigma_t^\sigma \right] \mathrm{d}Z_t. \tag{II.32}$$

Using Ito's Lemma in equation (II.21) together with equation (II.16), we obtain

$$\mathrm{d}\Gamma_B = -\Gamma_B \left(g_t^C - (\sigma_t^C)^2 \right) \mathrm{d}t - \Gamma_B (\sigma + \sigma_t^s) \mathrm{d}Z_t. \tag{II.33}$$

Comparing the diffusion terms in (II.32) and (II.33) (i.e., the terms with $\mathrm{d}Z_t$), it must follow that:

$$\sigma + \sigma_t^s = \frac{A_t}{C_t} \cdot \frac{\partial C_t}{\partial A_t} \cdot \sigma + \frac{1}{C_t} \cdot \frac{\partial C_t}{\partial \sigma_t^s} \cdot \sigma_t^\sigma. \tag{II.34}$$

We can now compute the derivative of C_t with respect to A_t and σ_t^s as:

$$\frac{\partial C_t}{\partial A_t} = \frac{C_t}{A_t},$$

and

$$\frac{\partial C_t}{\partial \sigma_t^s} = C_t \cdot \left(\frac{-(\sigma + \sigma_t^s)}{\theta + \phi_y} \right) = C_t \cdot (\sigma_t^\sigma)^{-1} \cdot \sigma_t^s,$$

which satisfies (II.34). Therefore, the conjectured martingale solution is verified as a valid equilibrium of the model.

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A Equilibrium Dynamics: Summary

As in the main text, we use the superscript h to denote household-specific variables (e.g., C_t^h); variables without the superscript refer to aggregates. We assume a representative-agent economy with a unit measure of households, where each household solves an identical optimization problem and selects the same consumption, labor, and bond investment in equilibrium. This results in an equilibrium given by: $C_t^h = C_t = Y_t$ for all h, $L_t^h = L_t$ for all h, $D_t^h = D_t$ for all h, $Y_t = A_t L_t$ and $B_t = 0$.

Throughout the Online Appendix, we consider a more general version of the model, with an extended Taylor rule that also targets excess variance with responsiveness ϕ_{vol},[1]

$$i_t = r^n + \phi_y \hat{Y}_t - \phi_{vol}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]. \tag{A.1}$$

With this rule, individual optimality conditions, after imposing equilibrium conditions $C_t^h = C_t = Y_t$ for all h, $L_t^h = L_t$ for all h, $D_t^h = D_t$ for all h, $Y_t = A_t L_t$ and $B_t = 0$, are characterized by

$$\begin{aligned}
\dot{B}_t &= i_t B_t - \bar{p} C_t + w_t L_t + D_t, \\
B_t &= \dot{B}_t = 0, \\
D_t &= \bar{p} Y_t - w_t L_t, \\
\frac{dC_t}{C_t} &= \left(i_t - \rho + \left(\sigma + \sigma_t^s\right)^2\right) dt + \left(\sigma + \sigma_t^s\right) dZ_t, \\
L_t &= \left(\frac{w_t}{\bar{p}} \frac{1}{C_t^h}\right)^\eta.
\end{aligned}$$

Ornstein-Uhlehbeck equilibrium Under the Ornstein-Uhlenbeck equilibrium described in Section 3.2, output gap $\hat{Y}_t$ follows $d\hat{Y}_t = \theta\left(\mu - \hat{Y}_t\right) dt + \sigma_t^s dZ_t$. Under the extended Taylor rule (A.1), the equilibrium is characterized by the following

[1] In particular, we assume $\phi_{vol} < \frac{1}{2}$. The textbook Taylor rule corresponds to the case of $\phi_{vol} = 0$.

conditions:

$$
\begin{aligned}
Y_t &= A_t L_t, \\
A_t &= \underbrace{A_0}_{=1} e^{\left(g-\frac{1}{2}\sigma^2\right)\cdot t+\sigma Z_t}, \\
Y_t^n &= A_t \left(\frac{\varepsilon-1}{\varepsilon}\right)^{\frac{\eta}{\eta+1}} = \left(\frac{\varepsilon-1}{\varepsilon}\right)^{\frac{\eta}{\eta+1}} e^{\left(g-\frac{1}{2}\sigma^2\right)\cdot t+\sigma Z_t}.
\end{aligned}
\tag{A.2}
$$

with

$$
r^n = \rho + g - \sigma^2,
$$

$$
\begin{aligned}
\hat{Y}_t &= \mu - \left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\left[\left(\sigma+\sigma_t^s\right)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right] \\
&= \frac{\mu\theta}{\theta+\phi_y} - \left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\left[\left(\sigma+\sigma_t^s\right)^2 - \sigma^2\right],
\end{aligned}
\tag{A.3}
$$

where the endogenous excess volatility σ_t^s follows a stochastic differential equation given by

$$
\begin{aligned}
\mathrm{d}\sigma_t^s &= \mu_t^\sigma \mathrm{d}t + \sigma_t^\sigma \mathrm{d}Z_t, \\
\mu_t^\sigma &\equiv \tilde{\mu}\left(\sigma_t^s\right) \\
&= -\frac{1}{2}\left(\frac{1}{\sigma+\sigma_t^s}\right)\Bigg[\theta\left[(\sigma+\sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right] \\
&\qquad + \left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\Bigg], \\
\sigma_t^\sigma &\equiv \tilde{\sigma}\left(\sigma_t^s\right) \\
&= -\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right).
\end{aligned}
\tag{A.4}
$$

which leads to

$$
\begin{aligned}
&\mathrm{d}\left[(\sigma+\sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right] \\
&= -\theta\left[(\sigma+\sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\mathrm{d}t \\
&\quad - 2\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)\sigma_t^s \mathrm{d}Z_t.
\end{aligned}
\tag{A.5}
$$

Note that with $\phi_{vol} = 0$, equation (A.4) becomes (24) of Section 3.2. With $\theta = 0$, we return to the martingale equilibrium of Section 3.1 and equation (A.4) becomes equation (20).

B Equilibrium Value Function for the Representative Household and the Transversality Condition

In this section, we characterize the equilibrium value function of the household under the general equilibrium described in Online Appendix A, and prove the transversality condition for individual households.

The Hamilton-Jacobi-Bellman (HJB) equation of household h is given by:

$$\rho \cdot \Gamma^h = \max_{C_t^h, L_t^h} \left\{ \log C_t^h - \frac{\left(L_t^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h} \right\}. \tag{B.1}$$

where we assume B_t^h as the only endogenous state variable,[2] and $\left\{A_t, \sigma_t^s\right\}$ as exogenous state variables from individual household's perspective, as σ_t^s is an aggregate variable that affects aggregate price and quantity variables. We calculate $\mu_t^{\Gamma,h}$ as follows:

$$\begin{aligned} \mu_t^{\Gamma,h} =& \Gamma_t^h + \Gamma_B^h \cdot \left(i_t B_t^h - \bar{p} \cdot C_t^h + w_t L_t^h + D_t \right) + \Gamma_A^h \cdot A_t g + \Gamma_\sigma^h \cdot \mu_t^\sigma \\ &+ \frac{1}{2}\Gamma_{AA}^h \cdot (A_t \sigma)^2 + \frac{1}{2}\Gamma_{\sigma\sigma}^h \cdot (\sigma_t^\sigma)^2 + \Gamma_{A\sigma}^h \cdot (A_t \sigma)(\sigma_t^\sigma), \end{aligned}$$

where subscripts denote differentiation with respect to the corresponding variable, yielding the following first-order conditions:

$$\Gamma_B^h = \frac{1}{\bar{p} C_t} = \frac{L_t^{\frac{1}{\eta}}}{w_t},$$

where we impose equilibrium conditions $C_t^h = C_t$ and $L_t^h = L_t$.

Under the general Ornstein-Uhlenbeck equilibrium described in Online Appendix A, we can characterize the exact functional form of $\Gamma^h(B_t^h, A_t, \sigma_t^s, t)$. From

[2] Eventually, we will impose the bond market equilibrium condition, i.e., $B_t = 0$.

(A.4), together with[3]

$$i_t = \rho + g - \sigma^2 + \frac{\theta\phi_y\mu}{\theta+\phi_y} - \frac{\phi_y + 2\theta\phi_{vol}}{2(\theta+\phi_y)} \cdot \left[\left(\sigma+\sigma_t^s\right)^2 - \sigma^2\right], \tag{B.2}$$

and using the fact that $w_t L_t - \bar{p}\cdot C_t + D_t = 0$, together with $\log(C_t) = \left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \log(A_t) + \hat{Y}_t$ and equation (A.2) to substitute for labor L_t, we can express the value function Γ^h of the household evaluated at the optimum as:[4]

$$\begin{aligned}
\Gamma^h = & \frac{1}{\rho}\cdot\left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{\rho}\cdot\Gamma_B^h \cdot i_t B_t^h \\
& + \frac{1}{\rho}\cdot\log A_t + \frac{1}{\rho}\cdot\Gamma_A^h\cdot A_t g + \frac{1}{2\rho}\Gamma_{AA}^h\cdot(A_t\sigma)^2 \\
& - \frac{1}{\rho}\Gamma_{A\sigma}^h\cdot(A_t\sigma)\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right) + \frac{1}{\rho}\cdot\Gamma_t^h \\
& + \frac{1}{\rho}\cdot\left[\mu - \left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\left[\left(\sigma+\sigma_t^s\right)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right] \\
& - \frac{1}{\rho}\cdot\left(\frac{\varepsilon-1}{\varepsilon}\right)\frac{1}{1+\frac{1}{\eta}} \\
& \qquad \times\exp\left\{\left(\frac{\eta+1}{\eta}\right)\left[\mu - \left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\left[\left(\sigma+\sigma_t^s\right)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right]\right\} \\
& - \frac{1}{2\rho}\Gamma_\sigma^h\cdot\left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right. \\
& \qquad \left. + \left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
& + \frac{1}{2\rho}\Gamma_{\sigma\sigma}^h\cdot\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2.
\end{aligned} \tag{B.3}$$

To further simplify the problem, we adopt a guess-and-verify approach by assuming

[3] In equilibrium, $B_t^h = 0$. We characterize the individual value Γ^h as a function of B_t^h, which is specific to an individual household, and (A_t, σ_t^s), which are exogenous to an individual household, under the equilibrium dynamics of Online Appendix A.

[4] Note that i_t following the extended Taylor rule will be a sole function of σ_t^s in equilibrium, as shown in (B.2).

that the value function Γ^h takes the following form:

$$\Gamma^{guess} = \frac{1}{\rho} \cdot \log(A_t) + h\left(\sigma_t^s\right) \cdot \frac{B_t^h}{A_t} + \tilde{\Gamma}\left(\sigma_t^s\right).$$

Therefore, we assume that the value function depends on $\log(A_t)$; on a term that is a multiple of a function of σ_t^s and the ratio $\frac{B_t^h}{A_t}$; and on a function solely of σ_t^s (plus constants). Differentiating yields:

$$\begin{aligned}
\Gamma_B^{guess} &= \frac{h\left(\sigma_t^s\right)}{A_t}, \quad \Gamma_A^{guess} = \frac{1}{\rho A_t} - h\left(\sigma_t^s\right) \cdot \frac{B_t^h}{A_t^2}, \\
\Gamma_{AA}^{guess} &= -\frac{1}{\rho\left(A_t\right)^2} + 2h\left(\sigma_t^s\right) \cdot \frac{B_t^h}{A_t^3}, \quad \Gamma_{A\sigma}^{guess} = -h'\left(\sigma_t^s\right) \cdot \frac{B_t^h}{A_t^2}, \\
\Gamma_\sigma^{guess} &= h'\left(\sigma_t^s\right) \cdot \frac{B_t^h}{A_t} + \tilde{\Gamma}'\left(\sigma_t^s\right), \quad \Gamma_{\sigma\sigma}^{guess} = h''\left(\sigma_t^s\right) \cdot \frac{B_t^h}{A_t} + \tilde{\Gamma}''\left(\sigma_t^s\right).
\end{aligned}$$

Finding $h(\sigma_t^s)$ First, collecting all the terms that have $\frac{B_t^h}{A_t}$, we see that $h(\sigma_t^s)$ satisfies

$$\begin{aligned}
h(\sigma_t^s) =& \frac{1}{\rho} i_t h(\sigma_t^s) - \frac{1}{\rho} g h(\sigma_t^s) + \frac{1}{\cancel{2}\rho}\sigma^2 \cdot \cancel{2} h(\sigma_t^s) \\
&- \frac{1}{\rho} h'(\sigma_t^s)\sigma\left[-\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)\right] \\
&- \frac{1}{2\rho} h'(\sigma_t^s)\left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right. \\
&\qquad \left. + \left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
&+ \frac{1}{2\rho} h''(\sigma_t^s)\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2.
\end{aligned} \tag{B.4}$$

We conjecture

$$h(\sigma_t^s) \propto \exp\left[\left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\left[\left(\sigma+\sigma_t^s\right)^2\right]\right]. \tag{B.5}$$

If the conjecture (B.5) is true, we obtain

$$
\begin{aligned}
h'(\sigma_t^s) &= h(\sigma_t^s)\cdot\frac{1-2\phi_{vol}}{(\theta+\phi_y)}(\sigma+\sigma_t^s),\\
h''(\sigma_t^s) &= h(\sigma_t^s)\cdot\left[\frac{1-2\phi_{vol}}{(\theta+\phi_y)}+\left(\frac{1-2\phi_{vol}}{(\theta+\phi_y)}\right)^2(\sigma+\sigma_t^s)^2\right],
\end{aligned}
$$

which can be plugged into (B.4) and lead to

$$
\begin{aligned}
\rho =&(i_t-g+\sigma^2)+\sigma\sigma_t^s\\
&-\frac{1}{2}\frac{1-2\phi_{vol}}{(\theta+\phi_y)}\cancel{(\sigma+\sigma_t^s)}\left(\frac{1}{\cancel{\sigma+\sigma_t^s}}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2-\sigma^2+\frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right.\\
&\quad\left.+\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right]\\
&+\frac{1}{2}\left[\frac{1-2\phi_{vol}}{(\theta+\phi_y)}+\left(\frac{1-2\phi_{vol}}{(\theta+\phi_y)}\right)^2(\sigma+\sigma_t^s)^2\right]\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2,
\end{aligned}
$$

leading to the equilibrium interest rate (B.2), thereby confirming our conjectural functional form (B.5). Finally, from the optimality condition

$$
\Gamma_B^h=\Gamma_B=h(\sigma_t^s)\frac{1}{A_t}=\frac{1}{\bar{p}C_t}=\frac{1}{\bar{p}Y_t},
$$

and with the help of (A.3), we obtain

$$
h\left(\sigma_t^s\right)=\frac{1}{\bar{p}\left(\frac{\varepsilon-1}{\varepsilon}\right)^{\left(\frac{\eta}{\eta+1}\right)}}\cdot e^{-\left[\mu-\left(\frac{1-2\phi_{vol}}{2(\theta+\phi_y)}\right)\left[\left(\sigma+\sigma_t^s\right)^2-\sigma^2+\frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right]}.
$$

Finding $\tilde{\Gamma}(\sigma_t^s)$ The final term $\tilde{\Gamma}(\sigma_t^s)$ should satisfy the following ordinary differential equation (ODE):

$$
\begin{aligned}
\tilde{\Gamma}\left(\sigma_t^s\right)=&\frac{1}{\rho}\cdot\left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right)+\frac{1}{\rho^2}\left(g-\frac{1}{2}\sigma^2\right)\\
&+\frac{1}{\rho}\cdot\left[\mu-\left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\left[\left(\sigma+\sigma_t^s\right)^2-\sigma^2+\frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right]
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{1}{\rho}\left(\frac{\eta}{\eta+1}\right)\left(\frac{\varepsilon-1}{\varepsilon}\right) \\
&\quad \cdot \exp\left\{\left(\frac{\eta+1}{\eta}\right)\left[\mu - \left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\right.\right. \\
&\qquad\qquad \left.\left.\times\left[\left(\sigma+\sigma_t^s\right)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right]\right\} \\
&- \frac{1}{2\rho}\tilde{\Gamma}'\left(\sigma_t^s\right)\cdot\left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right. \\
&\quad \left. + \left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
&+ \frac{1}{2\rho}\tilde{\Gamma}''\left(\sigma_t^s\right)\cdot\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2.
\end{aligned}
$$

We can rearrange the previous equation as

$$
\tilde{\Gamma}\left(\sigma_t^s\right) + \mathcal{A}\left(\sigma_t^s\right)\tilde{\Gamma}'\left(\sigma_t^s\right) + \mathcal{B}\left(\sigma_t^s\right)\tilde{\Gamma}''\left(\sigma_t^s\right) = \mathcal{R}\left(\sigma_t^s\right) \tag{B.6}
$$

where:

$$
\begin{aligned}
\mathcal{A}\left(\sigma_t^s\right) =& \frac{1}{2\rho}\left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right. \\
& \left. + \left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
\mathcal{B}\left(\sigma_t^s\right) =& -\frac{1}{2\rho}\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2 \\
\mathcal{R}\left(\sigma_t^s\right) =& \frac{1}{\rho}\cdot\left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{\rho^2}\left(g - \frac{1}{2}\sigma^2\right) \\
& + \frac{1}{\rho}\cdot\left[\mu - \left(\frac{1-2\phi_{vol}}{2\left(\theta+\phi_y\right)}\right)\left[\left(\sigma+\sigma_t^s\right)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right] \\
& - \frac{1}{\rho}\left(\frac{\eta}{\eta+1}\right)\left(\frac{\varepsilon-1}{\varepsilon}\right)\cdot e^{\left(\frac{\eta+1}{\eta}\right)\left[\mu-\left(\frac{1-2\phi_{vol}}{2(\theta+\phi_y)}\right)\left[\left(\sigma+\sigma_t^s\right)^2-\sigma^2+\frac{2\mu\phi_y}{1-2\phi_{vol}}\right]\right]}.
\end{aligned}
$$

Then, by Peano's Theorem (Walter, 1973), we know that a function $\tilde{\Gamma}\left(\sigma_t^s\right)$ solving the ODE in equation (B.6) locally exists. It becomes global as all three functions

$\mathcal{A}\left(\sigma_t^s\right)$, $\mathcal{B}\left(\sigma_t^s\right)$, and $\mathcal{R}\left(\sigma_t^s\right)$ are globally Lipschitz in $\mathcal{E}_t \equiv (\sigma+\sigma_t^s)^2-\sigma^2$,[5] and σ_t^s converges to a stationary distribution from Proposition 3, covering the full support up to ∞ on the equilibrium paths (Murray and Miller, 2007).

Several points are worth noting. First, the initial assumption that Γ depends on $\{B_t^h, A_t, \sigma_t^s, t\}$, i.e. $\Gamma = \Gamma(B_t^h, A_t, \sigma_t^s, t)$, can be simplified to $\Gamma(B_t, A_t, \sigma_t^s)$ by dropping the explicit time dependence (and hence eliminating Γ_t from the derivations). Second, with aggregate bonds in zero net supply, $B_t = 0$ for all t, the representative household's value function simplifies to:

$$\Gamma\left(A_t, \sigma_t^s\right) = \frac{1}{\rho}\cdot\log(A_t) + \tilde{\Gamma}\left(\sigma_t^s\right). \tag{B.7}$$

B.1 Transversality condition

Using equation (B.7), the transversality condition of the representative household is given by:

$$\lim_{t\to\infty}\mathbb{E}_0\left[e^{-\rho t}\cdot\Gamma\left(A_t,\sigma_t^s\right)\right] = \lim_{t\to\infty}\mathbb{E}_0\left[e^{-\rho t}\frac{1}{\rho}\log(A_t)\right] + \lim_{t\to\infty}\mathbb{E}_0\left[e^{-\rho t}\tilde{\Gamma}\left(\sigma_t^s\right)\right] = 0.$$

The first limit tends to zero, as shown by

$$\begin{aligned}\lim_{t\to\infty}\mathbb{E}_0\left[e^{-\rho t}\frac{1}{\rho}\log(A_t)\right] &= \lim_{t\to\infty}\mathbb{E}_0\left[e^{-\rho t}\frac{1}{\rho}\left[\left(g-\frac{1}{2}\sigma^2\right)t+\sigma Z_t\right]\right] \\ &= \frac{1}{\rho}\lim_{t\to\infty}e^{-\rho t}\left(g-\frac{1}{2}\sigma^2\right)t = 0.\end{aligned}$$

The second limit requires

$$\lim_{t\to\infty}\mathbb{E}_0\left[e^{-\rho t}\tilde{\Gamma}\left(\sigma_t^s\right)\right] = 0,$$

which is trivially satisfied by Oksendal (1995) as (i) $\tilde{\Gamma}(\sigma_t^s)$ is well-defined by the ODE in (B.6), (ii) the distribution of σ_t^s is stationary in the long run, as proven in Online Appendix C.3; (iii) As shown in Online Appendix C.2, the process (A.4) is irreducible and stable, thus converges to the stationary distribution.[6]

[5] Note that $\mathcal{E}_t \geq 0$ almost surely for $\theta \geq 0$ and $\mu \leq 0$.

[6] Solutions converging to degenerate distributions with zero excess volatility (i.e., $\sigma_t^s = 0$ for all t)—e.g., the perfect stabilization, martingale equilibrium, or Ornstein-Uhlenbeck process with

Sufficiency Bertsekas (2005) (Proposition 3.2.1), Wiszniewska-Matyszkiel (2011), and Liberzon (2012) (Section 5.1.4) find that the solution to the Hamilton-Jacobi-Bellman equation (B.1) is both necessary and sufficient, as it satisfies the transversality condition and the the utility function is concave in consumption and labor, with the budget constraint linear in bond holdings.[7] Therefore, our conjectured class of solutions, which follows an Ornstein-Uhlenbeck process, satisfies the optimality conditions for the individual household.

$\mu = 0$—trivially satisfy these conditions as well.

[7] It leads to a weakly concave stochastic Hamiltonian function. See Liberzon (2012) for details.

C Characterizing Stability and Stationary Properties of the Distribution: the Ornstein-Uhlenbeck Process

As explained in Online Appendix A, we examine the most general case of the model under rigid prices, which incorporates the policy rule in (A.1) and determines the output gap as in (A.3).

C.1 Limit Behavior

Define

$$u_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1 - 2\phi_{vol}},$$

and $m_t \equiv \mathbb{E}_0\left(u_t\right)$, then due to (A.5), u_t follows

$$du_t = -\theta u_t dt - 2\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right)\sigma_t^s \mathrm{d}Z_t,$$

leading to

$$u_t = -\theta \int_0^t u_h dh - 2\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right)\int_0^t \sigma_h^s dZ_h + p_0.$$

Imposing the expectations operator $\mathbb{E}_0$, we obtain

$$\frac{dm_t}{dt} + \theta m_t = 0.$$

Solving the equation above:

$$m_T \equiv \mathbb{E}_0\left[\left(\sigma + \sigma_T^s\right)^2 - \sigma^2 + \frac{2\mu\phi_y}{1 - 2\phi_{vol}}\right] = e^{-\theta T} m_0, \tag{C.1}$$

which can be rewritten as

$$\mathbb{E}_0\left[(\sigma + \sigma_T^s)^2\right] = \sigma^2 - \frac{2\mu\phi_y}{1 - 2\phi_{vol}} + e^{-\theta T}\mathbb{E}_0\left[(\sigma + \sigma_0^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1 - 2\phi_{vol}}\right].$$

Taking the limit as $T \to \infty$, assuming that $\theta > 0$, and using equation (C.1), we obtain:

$$\lim_{T\to\infty} \mathbb{E}_0 \left[(\sigma + \sigma_T^s)^2\right] = \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}} + \lim_{T\to\infty} e^{-\theta T}\mathbb{E}_0 \left[(\sigma + \sigma_0^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]$$
$$= \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}.$$

Similarly, taking the limit of equation (A.3), assuming that $\theta > 0$, and using equation (C.1), we obtain:

$$\lim_{T\to\infty} \mathbb{E}_0 \left[\hat{Y}_T\right] = \mu - \frac{1-2\phi_{vol}}{2(\theta+\phi_y)} \lim_{T\to\infty} e^{-\theta T}\mathbb{E}_0 \left[(\sigma + \sigma_0^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}\right]$$
$$= \mu.$$

Therefore, parameter μ in our Ornstein-Uhlenbeck process can be regarded as the long-run expectation of $\hat{Y}_t$.

Online Appendix F shows that admissible solutions require $\mu \leq 0$ and $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2 \geq 0$ almost surely. Combined with (A.3), this implies $\hat{Y}_t \leq 0$, so $\mathbb{E}_0 \left|\hat{Y}_t\right| = -\mathbb{E}_0[\hat{Y}_t]$. The limit above then yields

$$\lim_{T\to\infty} \mathbb{E}_0 \left|\hat{Y}_T\right| = -\mu < \infty,$$

which is the boundedness condition (16).

C.2 Process Stability: Foster-Lyapunov Drift Condition

We now demonstrate that the process $\{\sigma_t^s\}$ defined in (A.4) is stable and converges to the stationary distribution described in Online Appendix C. Following the techniques of Meyn and Tweedie (1993, 1998, 2012), we select the Lyapunov function $\Phi(\sigma_t^s) = (\sigma_t^s)^2$, which is positive definite and radially unbounded. The Lyapunov generator is given by:

$$\mathcal{L}\Phi(\sigma_t^s) = \Phi'(\sigma_t^s)\tilde{\mu}(\sigma_t^s) + \frac{1}{2}(\tilde{\sigma}(\sigma_t^s))^2\Phi''(\sigma_t^s).$$

The Foster-Lyapunov drift condition requires for there to exist some constants $c > 0$ and $d \leq 0$ such that

$$\mathcal{L}\Phi(\sigma_t^s) \leq -c\Phi(\sigma_t^s) + d, \text{ for all } |\sigma_t^s| \geq R, \tag{C.2}$$

for some $R > 0$. This condition implies that outside a compact set $|\sigma_t^s| \leq R$, the process exhibits a negative drift relative to Φ. This negative drift is critical for establishing both the tightness and stability of the process, ensuring convergence to its invariant measure (i.e., the stationary distribution).

Substituting the drift and diffusion terms into the Lyapunov generator yields:

$$\mathcal{L}\Phi(\sigma_t^s) = -\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)\theta\left[(\sigma + \sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1 - 2\phi_{vol}}\right] + \sigma\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right)^2 \frac{(\sigma_t^s)^2}{(\sigma + \sigma_t^s)^3}. \tag{C.3}$$

It is easy to notice that as $\sigma_t^s \to \infty$, we obtain:[8]

$$\lim_{\sigma_t^s \to \infty} \mathcal{L}\Phi(\sigma_t^s) \simeq -\theta(\sigma_t^s)^2.$$

Thus, for some constants $0 < c \leq \theta$ ($\theta > 0$) and $d = 0$, there exists an R sufficiently large to satisfy the Foster-Lyapunov condition in equation (C.2). Since the process is smooth and irreducible, the results in Meyn and Tweedie (1993, 1998, 2012) ensure its long-run stability.

C.3 Stationary Distribution

We define $n(\sigma_t^s, t)$ as the probability density function of σ_t^s, which satisfies the following Kolmogorov Forward Equation:

$$\frac{\partial n\left(\sigma^s, t\right)}{\partial t} = -\frac{d}{d\sigma^s}\left[\tilde{\mu}\left(\sigma^s\right) n\left(\sigma^s, t\right)\right] + \frac{d^2}{d(\sigma^s)^2}\left[\frac{1}{2}\tilde{\sigma}(\sigma^s)^2 n\left(\sigma^s, t\right)\right], \tag{C.4}$$

where $\tilde{\mu}(\sigma^s)$ and $\tilde{\sigma}(\sigma^s)$ are defined in process (A.4).

Setting $\frac{\partial n(\sigma^s,t)}{\partial t} = 0$ to obtain the stationary distribution, denoted $n(\sigma^s)$, and

[8] More specifically, we obtain from (C.3) that $\lim_{\sigma_t^s \to \infty} \frac{\mathcal{L}\Phi(\sigma_t^s)}{(\sigma_t^s)^2} = -\theta$.

integrating (C.4) once:

$$\tilde{\mu}\left(\sigma^{s}\right) n\left(\sigma^{s}\right)=\frac{d}{d \sigma^{s}}\left[\frac{1}{2} \tilde{\sigma}(\sigma^{s})^{2} n\left(\sigma^{s}\right)\right]+C_{1},$$

where C_1 is the integration constant. Imposing the no-flux condition, $C_1 = 0$, and defining $D(\sigma^s) \equiv \tilde{\sigma}(\sigma^s)^2 n(\sigma^s)$, we have

$$\frac{dD(\sigma^s)}{d\sigma^s} = \frac{2\tilde{\mu}(\sigma^s)}{\tilde{\sigma}(\sigma^s)^2} D(\sigma^s),$$

which leads to

$$D(\sigma^s) \propto \exp\left(\int \frac{2\tilde{\mu}(\sigma^s)}{\tilde{\sigma}(\sigma^s)^2} d\sigma^s\right),$$

and results in a probability density function for the stationary distribution proportional to

$$n(\sigma^s) \propto \frac{1}{\tilde{\sigma}(\sigma^s)^2} \exp\left(\int \frac{2\tilde{\mu}(\sigma^s)}{\tilde{\sigma}(\sigma^s)^2} d\sigma^s\right). \tag{C.5}$$

Substituting the expressions for $\tilde{\mu}(\sigma^s)$ and $\tilde{\sigma}(\sigma^s)$ defined in (A.4) into equation (C.5) and evaluating the integral yields:

$$\begin{aligned} n\left(\sigma^{s}\right)= & \frac{C_{2}}{\left(\frac{\theta+\phi_{y}}{1-2 \phi_{vol}}\right)^{2}\left(\frac{\sigma_{t}^{s}}{\sigma+\sigma_{t}^{s}}\right)^{2}} \\ & \cdot \exp \left\{-\left[\begin{array}{l} \theta\left(\frac{1-2 \phi_{vol}}{\theta+\phi_{y}}\right)^{2}\left[\frac{(\sigma^{s})^{2}}{2}+3 \sigma^{2} \log \left|\sigma^{s}\right|+3 \sigma \sigma^{s}-\frac{\sigma^{3}}{\sigma^{s}}\right] \\ +\theta\left(\frac{1-2 \phi_{vol}}{\theta+\phi_{y}}\right)^{2}\left(\frac{2 \mu \phi_{y}}{1-2 \phi_{vol}}-\sigma\right)\left[\log \left|\sigma^{s}\right|-\frac{\sigma}{\sigma^{s}}\right] \\ +\log \left|\sigma+\sigma^{s}\right| \end{array}\right]\right\}, \end{aligned} \tag{C.6}$$

where C_2 is the integration constant for the density to integrate to one over the domain of σ^s. Equation (C.6) defines the stationary distribution of σ^s_t, demonstrating that the Ornstein–Uhlenbeck solution attains a stationary stochastic equilibrium. Note that this distribution does not belong to any standard family.

Note that for $\mu = 0$ and $\theta > 0$ (see Property 2 of Proposition 3), we obtain $C_2 \to 0$ to ensure that $\eta(\sigma^s_t)$ is a proper probability density function. This is consistent with the result that the distribution degenerates at $\sigma^s_\infty = 0$.

C.3.1 Limiting Case: Zero Fundamental Volatility

As fundamental volatility tends to zero ($\sigma \longrightarrow 0$), the long-run stationary distribution of $\{\sigma^s_t\}$ becomes

$$n(\sigma^s) = \tilde{C}_2 \cdot (\sigma^s)^{-\left(1+\frac{2\theta\mu\phi_y}{(\theta+\phi_y)^2}(1-2\phi_{vol})\right)} e^{-\left(\frac{\theta(1-2\phi_{vol})^2(\sigma^s)^2}{2(\theta+\phi_y)^2}\right)}. \tag{C.7}$$

Moreover, if (i) the Ornstein-Uhlenbeck parameters satisfy $\theta > 0$ and $\mu < 0$, and (ii) the Taylor rule coefficients satisfy $\phi_y > 0$ and $\phi_{vol} < \frac{1}{2}$, this distribution corresponds to the generalized gamma distribution.

Generalized Gamma Distribution (GGD) The generalized gamma distribution, $GGD(a, d, p)$, is defined by the probability density function

$$\tilde{n}(x) = \frac{p/a^d}{\Gamma(d/p)}\, x^{d-1} \exp\left[-\left(\frac{x}{a}\right)^p\right].$$

where $a > 0$ is the scale parameter, $d > 0$ is the power-law shape parameter, $p > 0$ is the exponential shape parameter, and $\Gamma(\cdot)$ denotes the gamma function. It is straightforward to verify that equation (C.7) conforms to this definition with the parametrization

$$\begin{aligned} a &= \sqrt{\frac{2(\theta+\phi_y)^2}{\theta\left(1-2\phi_{vol}\right)^2}}, \\ d &= -\frac{2\theta\mu\phi_y}{(\theta+\phi_y)^2}\left(1-2\phi_{vol}\right), \\ p &= 2. \end{aligned}$$

Thus, in the long run, σ^s_t follows a generalized gamma distribution when $\sigma \longrightarrow 0$ and the additional parameter restrictions hold.

Furthermore, by the properties of the generalized gamma distribution, the stationary distribution of $(\sigma^s_\infty)^2$ is given by $GGD(a^2,\ d/2,\ p/2)$. Since $p/2 = 1$ in this case, the distribution reduces to a standard Gamma distribution, $(\sigma^s_\infty)^2 \sim Gamma\,(a', d')$, where $a' = a^2$ is the scale parameter and $d' = d/2$ is the shape parameter.

D Strong Solution

Proving that our solution is strong requires verifying that the proposed solution for σ_t^s (or a transformation of σ_t^s such as $(\sigma + \sigma_t^s)^2$) satisfies both the *Lipschitz continuity* and the *linear growth conditions*.[9] Here, we focus on the case where $\mu < 0$ and $\theta > 0$;[10] proofs for other feasible cases are analogous.

To verify these conditions, define $y_t = (\sigma + \sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}$ and rearrange equation (A.5) as follows:

$$dy_t = \underbrace{-\theta y_t}_{\equiv v(y_t)} dt + \underbrace{\left[2\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)\sigma - 2\left(\frac{\theta+\phi_y}{1-2\phi_{vol}}\right)\sqrt{y_t + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}}\right]}_{\equiv g(y_t)} dZ_t,$$

where we defined $v(y_t)$ as the drift, and $g(y_t)$ as the volatility of the $\{y_t\}$ process.

D.1 Lipschitz Continuity

Lipschitz Continuity requires that for any y_1, y_2:

$$|v(y_1) - v(y_2)| + |g(y_1) - g(y_2)| \leq \mathcal{L} \cdot |y_1 - y_2|,$$

which we prove by showing that the following conditions separately hold:

$$|v(y_1) - v(y_2)| \leq \mathcal{L}_v \cdot |y_1 - y_2|,$$
$$|g(y_1) - g(y_2)| \leq \mathcal{L}_g \cdot |y_1 - y_2|,$$

which then allows us to trivially prove the original Lipschitz Continuity with $\mathcal{L} = \mathcal{L}_v + \mathcal{L}_g$.

Drift term Since

$$|v(y_1) - v(y_2)| = |\theta| \cdot |y_1 - y_2|,$$

[9] See Chapter 5.3, Weak and Strong Solutions of Øksendal (2003).
[10] In this case, $\sigma_t^s \geq 0$ almost surely for all t, assuming $\sigma_0^s = 0$.

which leads to

$$|v(y_1) - v(y_2)| \leq \mathcal{L}_v \cdot |y_1 - y_2|,$$

with $\mathcal{L}_v = |\theta|$, the drift function $v(y)$ satisfies Lipschitz continuity.

Diffusion term From

$$|g(y_1) - g(y_2)| = \left| -2\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right) \cdot \left(\sqrt{y_1 + \sigma^2 - \frac{2\mu\phi_y}{1 - 2\phi_{vol}}} - \sqrt{y_2 + \sigma^2 - \frac{2\mu\phi_y}{1 - 2\phi_{vol}}}\right)\right|,$$

we apply the mean value theorem: there is a constant ξ for any pair $\{y_1,\ y_2\}$ such that

$$\sqrt{y_1 + \sigma^2 - \frac{2\mu\phi_y}{1 - 2\phi_{vol}}} - \sqrt{y_2 + \sigma^2 - \frac{2\mu\phi_y}{1 - 2\phi_{vol}}} = \frac{1}{2\sqrt{\xi + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}}} \cdot (y_1 - y_2).$$

Therefore, we can express the previous expression as:

$$|g(y_1) - g(y_2)| = \left|\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right) \frac{1}{\sqrt{\xi + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}}}\right| \cdot |y_1 - y_2|, \tag{D.1}$$

The constant ξ defined above is specific to each pair; therefore, we require a constant that uniformly satisfies the Lipschitz condition for any pair. Differentiating with respect to ξ, we find that:

$$\frac{d\left(\frac{1}{\sqrt{\xi + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}}}\right)}{d\xi} < 0.$$

Note from (A.4) that with $\theta > 0$ and $\mu < 0$, we have $\tilde{\sigma}(0) = 0$ and $\tilde{\mu}(0) > 0$. This implies that $\sigma_t^s = 0$ is a natural boundary for the process $\{\sigma_t^s\}$. Consequently, the minimum of y_t occurs at $\sigma_t^s = 0$, yielding $y^{min} = \frac{2\mu\phi_y}{1-2\phi_{vol}}$. Substituting this into the expression for ξ in (D.1) gives:

$$|g(y_1) - g(y_2)| \leq \mathcal{L}_g \cdot |y_1 - y_2| \,,$$

where

$$\mathcal{L}_g = \left| \frac{1}{\sigma} \cdot \left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}} \right) \right|,$$

satisfying Lipschitz continuity for the volatility function $g(y)$.

D.2 Linear Growth Condition

The Linear Growth Condition requires:

$$|v(y)|^2 + |g(y)|^2 \leq \mathcal{K} \cdot \left(1 + |y|^2\right).$$

We prove the above equation by showing that the following conditions separately hold:

$$|v(y)|^2 \leq \mathcal{K}_v \cdot \left(1 + |y|^2\right),$$
$$|g(y)|^2 \leq \mathcal{K}_g \cdot \left(1 + |y|^2\right),$$

which then allow us to prove the Linear Growth Condition with $\mathcal{K} = \mathcal{K}_v + \mathcal{K}_g$.

Drift term It is trivial to see that $|v(y)|^2 = |\theta|^2|y|^2 \leq \mathcal{K}_v \cdot \left(1 + |y|^2\right)$ with $\mathcal{K}_v = \theta^2$.

Diffusion term We see that

$$\begin{aligned} |g(y)|^2 &= \left| 2\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right)\sigma - 2\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right)\sqrt{y + \sigma^2 - \frac{2\mu\phi_y}{1 - 2\phi_{vol}}} \right|^2 \\ &= 4\left(\frac{\theta + \phi_y}{1 - 2\phi_{vol}}\right)^2 \left[\sigma - \sqrt{y + \sigma^2 - \frac{2\mu\phi_y}{1 - 2\phi_{vol}}}\right]^2 . \end{aligned}$$

Since[11]

$$y + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}} \geq \sigma^2,$$

we obtain

$$\begin{aligned}\left[\sigma - \sqrt{y + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}}\right]^2 &< y + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}} \\ &< 1 + |y|^2 + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}} \\ &< \max\left\{1 + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}, 1\right\}\left(1 + |y|^2\right),\end{aligned}$$

leading to

$$|g(y)|^2 < \underbrace{4\left(\frac{\theta + \phi_y}{1-2\phi_{vol}}\right)^2 \max\left\{1 + \sigma^2 - \frac{2\mu\phi_y}{1-2\phi_{vol}}, 1\right\}}_{\equiv \mathcal{K}_g}\left(1 + |y|^2\right),$$

satisfying the Linear Growth Condition for the volatility function $g(y)$.

Note that we defined $y_t = (\sigma + \sigma_t^s)^2 - \sigma^2 + \frac{2\mu\phi_y}{1-2\phi_{vol}}$ and proved that it permits a strong solution. Thus, it is trivial to see that the conditions will also hold for $(\sigma + \sigma_t^s)^2$ and for σ_t^s.

[11] Given that $\sigma_t^s \geq 0$ for all t almost surely, when $\sigma_t^s = 0$, $y_t = y^{min} = \frac{2\mu\phi_y}{1-2\phi_{vol}}$

E Constant Relative Risk Aversion (CRRA) Utility

We modify the baseline model with rigid prices by assuming a CRRA utility function with parameter γ. The representative household solves:[12]

$$\max_{\substack{\{C_s^h, L_s^h\}_{s\geq t} \\ \{B_s^h\}_{s>t}}} \mathbb{E}_t \int_t^\infty e^{-\rho(s-t)} \left[\frac{\left(C_s^h\right)^{1-\gamma} - 1}{1-\gamma} - \frac{\left(L_s^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right] \mathrm{d}s,$$

subject to

$$\mathrm{d}B_t^h = \left[i_t B_t^h - p_t C_t^h + w_t L_t^h + D_t \right] \mathrm{d}t.$$

Since most equilibrium expressions are similar to those in the main text with log-utility, we summarize only the key equations. First, the real and nominal state price densities and stochastic discount factors are given by:

$$\frac{d\xi_t^r}{\xi_t^r} = -\underbrace{\left[\rho + \gamma g_t^C - \frac{\gamma(\gamma+1)}{2} \left(\sigma_t^C\right)^2 \right]}_{\equiv r_t} dt - \gamma \sigma_t^C dZ_t,$$

and

$$dQ_t \equiv \frac{d\xi_t^N}{\xi_t^N} = -\underbrace{\left[r_t + \pi_t - \sigma_t^P(\gamma\sigma_t^C + \sigma_t^P) \right]}_{\equiv i_t} dt - \left[\gamma\sigma_t^C + \sigma_t^P \right] dZ_t.$$

From the Euler equation, the expected consumption growth is given by

$$\mathbb{E}_t \left[\frac{dC_t}{C_t} \right] = \frac{1}{\gamma}(i_t - \rho)dt + \frac{(\gamma+1)}{2} \mathrm{Var}_t \left(\frac{dC_t}{C_t} \right).$$

Let σ_t^s denote the excess volatility of output growth, as defined earlier. Then, output Y_t follows the process:

$$\frac{dY_t}{Y_t} = \frac{1}{\gamma} \left[i_t - \rho + \frac{\gamma(\gamma+1)}{2}(\sigma + \sigma_t^s)^2 \right] dt + (\sigma + \sigma_t^s) dZ_t, \tag{E.1}$$

[12]As in the main text, in equilibrium, we impose $C_t^h = C_t$, $L_t^h = L_t$, $B_t^h = B_t = 0$, $\forall h$, and $C_t = Y_t$, $D_t = \bar{p}Y_t - w_t L_t$.

leading to

$$d\ln Y_t = \frac{1}{\gamma}\left[i_t - \rho + \frac{1}{2}\gamma^2(\sigma+\sigma_t^s)^2\right]dt + (\sigma+\sigma_t^s)dZ_t. \tag{E.2}$$

In a similar way to (E.1), natural output Y_t^n follow

$$\frac{dY_t^n}{Y_t^n} = \frac{1}{\gamma}\left[r^n - \rho + \frac{\gamma(\gamma+1)}{2}\sigma^2\right]dt + \sigma dZ_t, \tag{E.3}$$

where the natural rate of interest r^n is given by

$$r^n = \rho + \gamma g - \frac{\gamma(\gamma+1)}{2}\sigma^2.$$

Equation (E.3) can be written as

$$d\ln Y_t^n = \frac{1}{\gamma}\left[r^n - \rho + \frac{1}{2}\gamma^2\sigma^2\right]dt + \sigma dZ_t. \tag{E.4}$$

From equations (E.2) and (E.4), output gap $\hat{Y}_t$ follows[13]

$$\begin{aligned} d\hat{Y}_t &= \frac{1}{\gamma}\left[i_t - r^n + \gamma^2\frac{1}{2}\left[(\sigma+\sigma_t^s)^2 - \sigma^2\right]\right]dt + \sigma_t^s dZ_t \\ &= \frac{1}{\gamma}\left[i_t - r_t^T\right]dt + \sigma_t^s dZ_t, \end{aligned} \tag{E.5}$$

where the risk-adjusted natural rate r_t^T is similarly defined:

$$r_t^T = r^n - \frac{1}{2}\gamma^2\left[(\sigma+\sigma_t^s)^2 - \sigma^2\right].$$

Precautionary premium Defining the precautionary premium as $pp_t \equiv \gamma^2(\sigma+\sigma_t^s)^2$, we can express r_t^T as[14]

$$r_t^T = r^n - \frac{1}{2}\left(pp_t - pp_t^n\right), \tag{E.6}$$

which has the same form as the log-utility case.

[13]Note that $\frac{1}{\gamma}$ is multiplied to the gap between i_t and r_t^T as it acts as an elasticity of intertemporal substitution.

[14]This definition of pp_t is equivalent to the risk premium in a canonical consumption-based asset pricing model.

Martingale equilibrium We can construct the martingale equilibrium similarly in a similar manner. Plugging policy rule (A.1) into equation (E.5):

$$d\hat{Y}_t = \underbrace{\frac{1}{\gamma}\left[\phi_y \hat{Y}_t + \left[\gamma^2 - 2\phi_{vol}\right]\frac{1}{2}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]\right]}_{=0} dt + \sigma_t^s dZ_t.$$

Under the assumption that $\hat{Y}_t$ is a local martingale, we obtain

$$\hat{Y}_t = -\frac{1}{2\phi_y}\left[\gamma^2 - 2\phi_{vol}\right]\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right].$$

Defining $\phi_{total} \equiv \frac{\phi_y}{\gamma^2 - 2\phi_{vol}}$, the previous expression becomes:

$$\hat{Y}_t = -\frac{1}{2\phi_{total}}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]. \tag{E.7}$$

Then, from equation (E.7), we obtain

$$d(\sigma + \sigma_t^s)^2 = -2\phi_{total}\sigma_t^s dZ_t,$$

and

$$d\sigma_t^s = -\phi_{total}^2 \frac{(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3} dt - \phi_{total}\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right) dZ_t. \tag{E.8}$$

When $\gamma = 1$ (i.e., under log-utility), equation (E.8) reduces to equation (J.5) in Section J.

Ornstein-Uhlenbeck equilibrium If we assume that $\hat{Y}_t$ follows an Ornstein-Uhlenbeck process:

$$d\hat{Y}_t = \theta\left[\mu - \hat{Y}_t\right] dt + \sigma_t^s dZ_t, \tag{E.9}$$

then from

$$d\hat{Y}_t = \underbrace{\frac{1}{\gamma}\left[\phi_y \hat{Y}_t + \left[\gamma^2 - 2\phi_{vol}\right]\frac{1}{2}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]\right]}_{=\theta\left(\mu - \hat{Y}_t\right)} dt + \sigma_t^s dZ_t,$$

we obtain

$$\hat{Y}_t = \left(\frac{\gamma\theta}{\gamma\theta + \phi_y}\right)\mu - \frac{1}{2}\left(\frac{1}{\gamma\theta + \phi_y}\right)\left[\gamma^2 - 2\phi_{vol}\right]\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]. \tag{E.10}$$

If we define $\tilde{\phi}_{total} \equiv \frac{\gamma\theta + \phi_y}{\gamma^2 - 2\phi_{vol}}$,[15] we obtain

$$\hat{Y}_t = \left(\frac{\gamma\theta}{\gamma\theta + \phi_y}\right)\mu - \frac{1}{2\tilde{\phi}_{total}}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right],$$

from which with the help of equations (E.9) and (E.10), we obtain

$$\begin{aligned} d(\sigma + \sigma_t^s)^2 = -2\theta\tilde{\phi}_{total}&\left[\left(\frac{\phi_y}{\gamma\theta + \phi_y}\right)\mu + \frac{1}{2\tilde{\phi}_{total}}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]\right] dt \\ &- 2\tilde{\phi}_{total}\sigma_t^s dZ_t, \end{aligned}$$

and

$$\begin{aligned} d\sigma_t^s = -\theta\tilde{\phi}_{total}\left(\frac{1}{\sigma + \sigma_t^s}\right)&\left[\left(\frac{\phi_y}{\gamma\theta + \phi_y}\right)\mu\right. \\ &\left. + \frac{1}{2\tilde{\phi}_{total}}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right] - \tilde{\phi}_{total}^2 \frac{(\sigma_t^s)^2}{2(\sigma + \sigma_t^s)^3}\right] dt \\ &- \tilde{\phi}_{total}\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right) dZ_t. \end{aligned}$$

Perfect stabilization and growth targeting The policy rule (A.1) with $\phi_{vol} = \gamma^2$ achieves full stabilization and yields $\hat{Y}_t = 0$ for all t as the unique stable equilibrium.

[15]Note that it is just a natural extension of ϕ_{total} constant in (E.7), as $\theta = 0$ corresponds to the martingale equilibrium case.

Or, with precautionary premium as defined in (E.6), we can write:

$$i_t = \underbrace{r^n - \frac{1}{2}\left(pp_t - pp_t^n\right)}_{=r_t^T} + \phi_y \hat{Y}_t, \tag{E.11}$$

which is of the same form as in the log-utility case of Section J.

Finally, we can rewrite (E.11) as a growth mandate rule:

$$\frac{\mathbb{E}_t\left(d\log Y_t\right)}{dt} = \frac{\mathbb{E}_t\left(d\log Y_t^n\right)}{dt} + \frac{\phi_y}{\gamma}\hat{Y}_t. \tag{E.12}$$

The expected growth rate of natural output is given by:

$$\frac{\mathbb{E}_t\left(d\log Y_t^n\right)}{dt} = g - \frac{1}{2\gamma}\sigma^2,$$

so equation (E.12) can be rewritten as:

$$\frac{\mathbb{E}_t\left(d\log Y_t\right)}{dt} = \left(g - \frac{1}{2\gamma}\sigma^2\right) + \frac{\phi_y}{\gamma}\hat{Y}_t.$$

A possible interpretation is that the central bank in practice follows the rule

$$\frac{\mathbb{E}_t\left(d\log Y_t\right)}{dt} = \left(g - \frac{1}{2\gamma}\sigma^2\right) + \tilde{\phi}_y\hat{Y}_t,$$

which results in an implicit output gap target in the Taylor rule for interest rates, with $\phi_y = \gamma\tilde{\phi}_y$.

F Admissible Equilibria

F.1 Martingale Equilibrium

This section shows that the martingale equilibrium with an initial negative excess volatility shock, $\sigma_0^s < 0$, is not allowed in the baseline model, characterized by (19) and $\mathbb{E}_t\left(d\hat{Y}_t\right) = 0$.

As in Appendix I.2, we define $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2$. Then equation (19) can be written as

$$d\mathcal{E}_t = \underbrace{-2\phi_y\left(\sqrt{\mathcal{E}_t + \sigma^2} - \sigma\right)}_{\equiv \sigma^{\mathcal{E}}(\mathcal{E}_t)} dZ_t. \tag{F.1}$$

First, note that $\mathcal{E}_t \geq -\sigma^2$ by definition. At $\mathcal{E}_t = -\sigma^2$, we have

$$\sigma^{\mathcal{E}}\left(-\sigma^2\right) = 2\phi_y\sigma \neq 0. \tag{F.2}$$

Note that with $\mathcal{E}_0 < 0$ (or equivalently, $\sigma_0^s < 0$), $\mathcal{E}_t$ always reaches its lower bound $-\sigma^2$ with positive probability, at which the diffusion term $\sigma^{\mathcal{E}}\left(-\sigma^2\right)$ from equation (F.2) is non-zero. It implies that $\mathcal{E}_t < -\sigma^2$ with some positive probability, which is impossible given that $\mathcal{E}_t \geq -\sigma^2$ for all t almost surely.

Thus, only the martingale equilibrium with $\sigma_0^s > 0$ is feasible.

F.2 Ornstein-Uhlenbeck Equilibria

In this section, we show that given $\theta > 0$, the Ornstein-Uhlenbeck equilibria in the baseline model, characterized by equations (23) and (24), are admitted only for $\mu \leq 0$. Note that $\mu < 0$ and $\mu = 0$ correspond to Property 1 and Property 2, respectively, in Proposition 3.[16]

As in Appendix I.2, we define $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2$. Then equation (24) can be

[16]We appreciate an anonymous referee for pointing this issue out and providing detailed thoughts on the problem.

written as

$$d\mathcal{E}_t = \underbrace{\theta\left(\overline{\mathcal{E}} - \mathcal{E}_t\right)}_{\equiv \mu^{\mathcal{E}}(\mathcal{E}_t)} dt \underbrace{-2\left(\theta + \phi_y\right)\left(\sqrt{\mathcal{E}_t + \sigma^2} - \sigma\right)}_{\equiv \sigma^{\mathcal{E}}(\mathcal{E}_t)} dZ_t. \tag{F.3}$$

where $\overline{\mathcal{E}} = -2\mu\phi_y$.

First, note that $\mathcal{E}_t \geq -\sigma^2$ by definition. At $\mathcal{E}_t = -\sigma^2$, we have

$$\mu^{\mathcal{E}}\left(-\sigma^2\right) = \theta\left(\sigma^2 - 2\mu\phi_y\right) \tag{F.4}$$

$$\sigma^{\mathcal{E}}\left(-\sigma^2\right) = 2(\theta + \phi_y)\sigma. \tag{F.5}$$

Second, note that when $\mathcal{E}_t = 0$, we obtain

$$\mu^{\mathcal{E}}(0) = -\theta\left(2\mu\phi_y\right) \tag{F.6}$$

$$\sigma^{\mathcal{E}}(0) = 0. \tag{F.7}$$

We first note that $\mathcal{E}_0 < 0$ is not admissible. With $\mathcal{E}_0 < 0$, $\mathcal{E}_t$ always reaches its lower bound $-\sigma^2$ with positive probability, at which the diffusion term $\sigma^{\mathcal{E}}\left(-\sigma^2\right)$ from equation (F.5) is non-zero. It implies that $\mathcal{E}_t < -\sigma^2$ with some positive probability, which is impossible given that $\mathcal{E}_t \geq -\sigma^2$. Thus, for any equilibrium to be feasible, it must satisfy $\mathcal{E}_t \geq 0$ for all t almost surely.

If $\mu > 0$, $\mu^{\mathcal{E}}(0) = -\theta\left(2\mu\phi_y\right) < 0$ from (F.6) while $\sigma^{\mathcal{E}}(0) = 0$, implying that the $\mathcal{E}_t$ process will exit the region $(0, \infty)$ with probability 1, which is not allowed because $\mathcal{E}_t \geq 0$ for all t almost surely. Therefore, we impose $\mu \leq 0$ given $\theta > 0$.

We then distinguish between cases of $\mu < 0$ and $\mu = 0$ as follows:

Case 1 If $\mu < 0$, then $\mu^{\mathcal{E}}(0) = -\theta\left(2\mu\phi_y\right) > 0$ while $\sigma^{\mathcal{E}}(0) = 0$, implying that $\mathcal{E}_t \geq 0$ for all t almost surely. Thus, starting from $\mathcal{E}_0 = 0$, the equilibrium exists. In fact, the stationary distribution of σ_t^s in the limit $\sigma \to 0$ given in Property 1 of Proposition 3 (i.e., equation (25)) is feasible only for $\mu < 0$ as well, given that $d > 0$.[17]

Case 2 If $\mu = 0$, then $\mu^{\mathcal{E}}(0) = 0$ and $\sigma^{\mathcal{E}}(0) = 0$. Thus, any $\mathcal{E}_0 > 0$ can be an equilibrium, but the economy eventually gets absorbed at $\mathcal{E}_\infty = 0$, corresponding

[17] From $d = -\frac{2\theta\mu\phi_y}{(\theta+\phi_y)^2} > 0$, it should be that $\mu < 0$.

to Property 2 of Proposition 3.

G Discrete-Time Model

This section shows that our multiplicity result is not specific to the continuous-time specification and also holds in a discrete-time setting, subject to minor adjustments. We consider a discretized version of the model with step size $0 \leq \mathrm{d}t \leq 1$ between periods: as $\mathrm{d}t \to 0$, the model collapses to the continuous-time formulation in the main text, whereas $\mathrm{d}t = 1$ yields a standard discrete-time model. The purpose of this discretization is to establish that the proposed solution to the discrete-time model converges to its continuous-time counterpart as $\mathrm{d}t \to 0$, while remaining well-defined for arbitrary step sizes.[18]

A second-order approximation to the dynamic IS equation with time interval $\mathrm{d}t$ is

$$\hat{Y}_t = \mathbb{E}_t \hat{Y}_{t+dt} - \left(i_t - r^n + \frac{1}{2} \left[\left(\sigma + \sigma^s_{t+dt} \right)^2 - \sigma^2 \right] \right) dt, \tag{G.1}$$

with the approximation becoming exact at $\mathrm{d}t \to 0$. Notice that this expression includes the additional term capturing precautionary savings at the core of our solution construction.[19]

Define excess variance at t as $\mathcal{E}_t \equiv (\sigma + \sigma^s_{t+dt})^2 - \sigma^2$. We conjecture an equilibrium satisfying

$$\hat{Y}_{t+dt} - \hat{Y}_t = -\theta \left[\hat{Y}_t - \mu \right] dt + \sigma^s_{t+dt} \sqrt{dt} \cdot \tilde{\varepsilon}_{t+dt}, \tag{G.2}$$

$$\mathcal{E}_t \geq 0, \quad \forall t, \ a.s., \tag{G.3}$$

where $\mu \leq 0$ and $0 < \theta < 1$. Relative to the continuous-time formulation in the main text, the key difference is that (G.3) must be imposed as an explicit equilibrium restriction. In continuous time, this restriction holds automatically in equilibrium and is therefore omitted from the exposition.

The shock term in equation (G.2) is defined as

$$\tilde{\varepsilon}_{t+dt} = \max\{-\bar{\varepsilon}_{t+dt}, \ \min\{\varepsilon_{t+dt}, \ \bar{\varepsilon}_{t+dt}\}\},$$

[18] Dordal i Carreras and Lee (2025) also offers a discrete-time approximation to the continuous-time New Keynesian framework with endogenous precautionary savings. See their Online Appendix E.

[19] As in the main text, $\left(\sigma^s_{t+dt}\right)^2 dt \equiv \mathrm{Var}_t\left(\hat{Y}_{t+dt}\right)$, so σ^s_{t+dt} is known at t.

where $\varepsilon_t \overset{\text{i.i.d.}}{\sim} N(0,1)$ is the normalized fundamental TFP shock and $\bar{\varepsilon}_{t+dt} \geq 0$ is a (potentially time-varying) truncation bound determined below. Thus, $\tilde{\varepsilon}_{t+dt}$ is a censored version of ε_{t+dt}, symmetrically bounded by $\pm\bar{\varepsilon}_{t+dt}$, and satisfies $\mathbb{E}_t[\tilde{\varepsilon}_{t+dt}] = 0$. Throughout this section, ε_{t+dt} is realized at $t + dt$, after expectations at time t are formed. This truncation scheme prevents $\mathcal{E}_t$ from becoming negative for appropriate choices of the bound $\bar{\varepsilon}_{t+dt}$.

As shown in the main text, the model admits a well-behaved solution on the state space $\mathcal{E}_t \geq 0$ (equivalently, $\sigma_t^s \geq 0$). In continuous time, this boundary is never crossed because the Brownian-motion shocks are continuous. In discrete time, innovations occur in jumps and can push $\mathcal{E}_t$ below zero if the shock support is unbounded, which motivates the truncation of the fundamental shock ε_t as part of the discrete-time solution.

Equation (G.2) can be rewritten as

$$\hat{Y}_{t+dt} = \mathbb{E}_t\hat{Y}_{t+dt} + \sigma^s_{t+dt}\sqrt{dt} \cdot \tilde{\varepsilon}_{t+dt}, \tag{G.4}$$

where $\mathbb{E}_t\hat{Y}_{t+dt} \equiv \hat{Y}_t - \theta\left[\hat{Y}_t - \mu\right]dt$.

Using (G.4), equation (G.1) becomes

$$\hat{Y}_{t+dt} - \hat{Y}_t = \left(\underbrace{i_t - r^n}_{=\phi_y\hat{Y}_t} + \frac{1}{2}\left[\left(\sigma + \sigma^s_{t+dt}\right)^2 - \sigma^2\right]\right)dt + \sigma^s_{t+dt}\sqrt{dt} \cdot \tilde{\varepsilon}_{t+dt}, \tag{G.5}$$

where monetary policy i_t follows the Taylor rule $i_t = r^n + \phi_y\hat{Y}_t$, with the Taylor principle $\phi_y > 0$. Comparing (G.5) and (G.2) yields

$$\hat{Y}_t = \frac{\theta\mu}{\theta + \phi_y} - \frac{1}{2(\theta + \phi_y)}\left[(\sigma + \sigma^s_{t+dt})^2 - \sigma^2\right], \tag{G.6}$$

which matches equation (23) in the continuous-time model, except that σ^s_{t+dt} appears instead of σ^s_t because of discrete-time timing.

Substituting (G.6) into (G.5) and rearranging using the definition of $\mathcal{E}_t$ gives

$$\mathcal{E}_{t+dt} - \mathcal{E}_t = -\theta\left[\mathcal{E}_t - (-2\phi_y\mu)\right]dt - 2(\theta + \phi_y)\left[\sqrt{\mathcal{E}_t + \sigma^2} - \sigma\right]\sqrt{dt} \cdot \tilde{\varepsilon}_{t+dt}. \tag{G.7}$$

Thus, excess variance $\mathcal{E}_t$ follows a mean-reverting process with speed θ and mean level $(-2\phi_y\mu) \geq 0$.

Finally, to satisfy (G.3), equation (G.7) must imply $\mathcal{E}_{t+dt} \geq 0$, which holds whenever

$$\tilde{\varepsilon}_{t+dt} \leq \frac{\mathcal{E}_t - \theta\left[\mathcal{E}_t - (-2\phi_y\mu)\right]dt}{2(\theta+\phi_y)\left[\sqrt{\mathcal{E}_t+\sigma^2}-\sigma\right]\sqrt{dt}} \equiv \bar{\varepsilon}_{t+dt}, \tag{G.8}$$

with $\bar{\varepsilon}_t \geq 0$ whenever $\mathcal{E}_t \geq 0$. Hence $\mathcal{E}_t \geq 0$ for all t if $\mathcal{E}_0 \geq 0$ and $\bar{\varepsilon}_{t+dt}$ is defined as in (G.8). Note that the upper bound $\bar{\varepsilon}_{t+dt}$ is known at t.

It is easy to prove that $\bar{\varepsilon}_{t+dt} \geq 0$ by rewriting (G.8) as

$$\bar{\varepsilon}_{t+dt} = \frac{(1-\theta dt)\mathcal{E}_t + (-2\phi_y\mu)\theta dt}{2(\theta+\phi_y)\left[\sqrt{\mathcal{E}_t+\sigma^2}-\sigma\right]\sqrt{dt}},$$

where $(1-\theta dt) \geq 0$ since $0 \leq \theta$ and $0 \leq \mathrm{d}t \leq 1$, and $-2\phi_y\mu \geq 0$ since $\phi_y > 0$ and $\mu \leq 0$. The denominator is strictly positive whenever $\mathcal{E}_t \geq 0$, so the construction is internally consistent if $\mathcal{E}_0 \geq 0$ (or equivalently, $\sigma^s_{dt} \geq 0$).

Finally, the expression for $\bar{\varepsilon}_{t+dt}$ nests the continuous-time model as $\mathrm{d}t \to 0$:

$$\lim_{\mathrm{d}t\to 0} \bar{\varepsilon}_t = +\infty,$$

which implies

$$\lim_{\mathrm{d}t\to 0} \tilde{\varepsilon}_t = \varepsilon_t.$$

Under this limit, the equilibrium conditions (G.6) and (G.7) coincide with their continuous-time counterparts, as the innovations converge to Brownian increments and the truncation becomes non-binding.

When $\mathrm{d}t = 1$, we recover the discrete-time solution of the model, summarized by

$$
\begin{aligned}
\hat{Y}_t &= \frac{\theta\mu}{\theta+\phi_y} - \frac{1}{2(\theta+\phi_y)}\mathcal{E}_t,\\
\mathcal{E}_{t+1} &= -2\phi_y\mu\theta + (1-\theta)\mathcal{E}_t - 2(\theta+\phi_y)\left[\sqrt{\mathcal{E}_t+\sigma^2}-\sigma\right]\cdot\tilde{\varepsilon}_{t+1},\\
\tilde{\varepsilon}_{t+1} &= \max\{-\bar{\varepsilon}_{t+1},\ \min\{\varepsilon_{t+1},\ \bar{\varepsilon}_{t+1}\}\},\\
\bar{\varepsilon}_{t+1} &= \frac{(1-\theta)\mathcal{E}_t - 2\phi_y\mu\theta}{2(\theta+\phi_y)\left[\sqrt{\mathcal{E}_t+\sigma^2}-\sigma\right]},
\end{aligned}
$$

given $\mathcal{E}_0$.

Figure G.1 plots a representative simulated path of the Ornstein–Uhlenbeck solution in discrete time, with $0<\theta<1$ and $\mu<0$.[20] The dynamics mirror those in Figure 3a: even starting from zero excess volatility (i.e., $\sigma_1^s = 0$), the process $\{\sigma_t^s\}$ converges to a stationary distribution, with $\sigma_t^s \geq 0$ along the sample path and infrequent but recurrent episodes of high excess volatility that generate recessionary downturns.

[20]To make the simulated patterns more visible at quarterly frequency, we increase persistence by lowering θ relative to the baseline calibration. This change has no qualitative implications for our results.

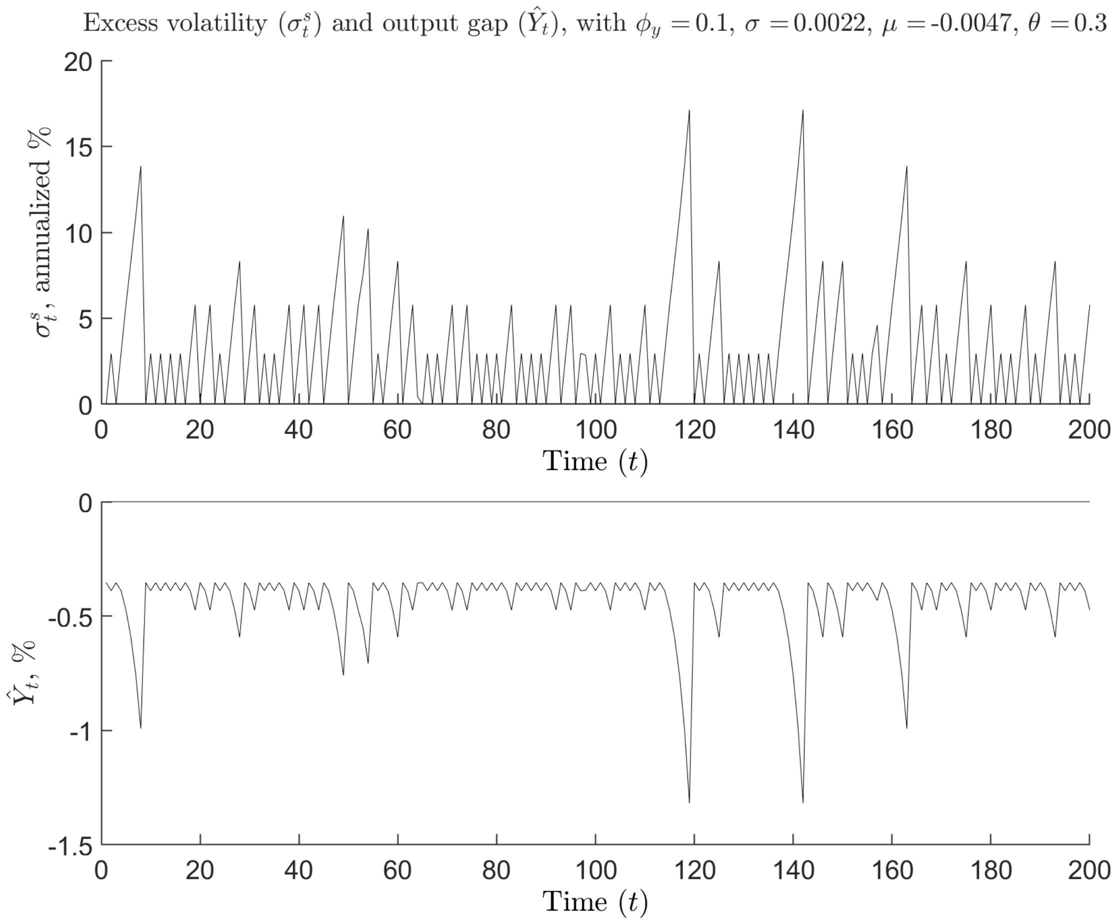

Figure G.1: Representative simulated path in the discrete-time model for the excess volatility $\{\sigma_t^s\}$ and output gap $\{\hat{Y}_t\}$ under the Ornstein–Uhlenbeck solution with $0 < \theta < 1$ and $\mu < 0$. The calibration for the remaining parameters used to construct Figure G.1 is reported in Online Appendix L.

H Fiscal Escape Clause and Equilibrium Selection

This appendix derives the continuous-time perpetual-youth overlapping-generations demand block from the household problem and constructs the fiscal escape clause of Section 4.2 that selects the full-stabilization path. The environment, notation, and Hamilton-Jacobi-Bellman argument follow Section 2 and Online Appendices A and B. The specification of finite lives follows Yaari (1965) and Blanchard (1985).

H.1 Environment

As in the main text, time is continuous. Prices are perfectly rigid, $p_t = \bar{p}$. Technology, firms, and the Taylor rule all follow Section 2. In particular,

$$\frac{dA_t}{A_t} = g\,dt + \sigma\,dZ_t, \tag{H.1}$$

and firm profits rebated to households are $D_t = \bar{p}Y_t - w_t L_t$. There is no government spending, so goods-market clearing remains $C_t = Y_t$.

Households are perpetual-youth overlapping generations. A living household dies with Poisson intensity $\lambda \geq 0$ and is replaced by a newborn of the same index. The measure of households is one at every date. Households do not value the utility of those who replace them. While alive, a household can save in an actuarially fair annuity on the same nominal bond B_t^h. Conditional on survival, bond holdings therefore earn $i_t + \lambda$. When $\lambda = 0$, the demand block collapses to the representative-agent problem of Section 2.

Labor supply is intermediated as in Angeletos, Lian and Wolf (2024), so that in equilibrium $L_t^h = L_t$ for every living household. As in Angeletos, Lian and Wolf (2024), a balanced social fund transfers resources from incumbents to newborns. Newborns receive S^{new} and incumbents pay S^{old}, with $\lambda S^{\text{new}} + S^{\text{old}} = 0$. The fund sets S^{new} equal to steady-state public debt so that every cohort has the same wealth, and hence the same consumption, in steady state. Steady-state public debt is zero in our setting, so $S^{\text{new}} = S^{\text{old}} = 0$. Newborns therefore enter with $B_t^{\text{new}} = 0$.

Let d_t be the debt-to-output ratio, that is, public debt in the hands of households in units of current output,

$$d_t \equiv \frac{B_t}{\bar{p}Y_t}, \qquad B_t \equiv \int B_t^h \, dh, \qquad C_t \equiv \int C_t^h \, dh, \tag{H.2}$$

and let $\mathcal{T}_t$ be the real tax flow in the same units, so that the real tax payment is given by $\mathcal{T}_t Y_t$. Private claims on the government are the only net financial asset.

H.2 Household problem

A living household h chooses consumption and labor to maximize

$$\Gamma_t^h \equiv \max \mathbb{E}_t \int_t^\infty e^{-(\rho+\lambda)(s-t)} \left[\log C_s^h - \frac{(L_s^h)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right] ds \tag{H.3}$$

subject to

$$dB_t^h = \left[(i_t + \lambda) B_t^h - \bar{p} C_t^h + w_t L_t^h + D_t - \bar{p}\mathcal{T}_t Y_t + \bar{p} S_t^h \right] dt. \tag{H.4}$$

The factor $e^{-\lambda(s-t)}$ is the survival probability to date s. The term λB_t^h is the annuity premium received while alive. The real flow S_t^h is the household's transfer from, or contribution to, the social fund.

The household takes the aggregates w_t, D_t, Y_t, $\mathcal{T}_t$, and S_t^h as given and chooses C_t^h, L_t^h, and B_t^h. Equation (H.4) is the absolutely continuous part of the budget and has no diffusion. Jumps in d_t, introduced below, shift B_t^h at intervention dates and do not enter the drift of the Hamilton-Jacobi-Bellman equation. When $\lambda = 0$, $\mathcal{T}_t = 0$, and $S_t^h = 0$, that continuous part collapses to the household budget equation (2) of the main text.

H.3 Deriving the individual Euler equation

As in Online Appendix B, we write $\Gamma^h = \Gamma^h(B_t^h, A_t, \sigma_t^s, t)$ for the value of a living household, taking the same aggregate states. The Hamilton-Jacobi-Bellman equation is

$$(\rho+\lambda)\Gamma^h = \max_{C_t^h, L_t^h}\left\{\log C_t^h - \frac{(L_t^h)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h}\right\}, \tag{H.5}$$

where Ito's formula gives $d\Gamma^h = \mu_t^{\Gamma,h}dt + \sigma_t^{\Gamma,h}dZ_t$ and

$$\begin{aligned}\mu_t^{\Gamma,h} &= \partial_t\Gamma^h + \Gamma_B^h\big[(i_t+\lambda)B_t^h - \bar{p}C_t^h + w_tL_t^h + D_t - \bar{p}\mathcal{T}_tY_t + \bar{p}S_t^h\big] + \Gamma_A^h A_t g + \Gamma_\sigma^h\mu_t^\sigma \\ &\quad + \frac{1}{2}\Gamma_{AA}^h(A_t\sigma)^2 + \frac{1}{2}\Gamma_{\sigma\sigma}^h(\sigma_t^\sigma)^2 + \Gamma_{A\sigma}^h(A_t\sigma)\sigma_t^\sigma. \end{aligned}\tag{H.6}$$

Subscripts on Γ^h denote partial derivatives, and $\partial_t\Gamma^h$ is the time partial. The first-order condition for consumption is thus given by

$$\Gamma_B^h = \frac{1}{\bar{p}C_t^h}. \tag{H.7}$$

The labor first-order condition is the same intratemporal relation as in Section 2.

Differentiate equation (H.5) with respect to B_t^h and apply the envelope theorem. The return on wealth in the budget is $i_t+\lambda$ and the effective discount on the left-hand side is $\rho+\lambda$, so

$$\mu_t^{\Gamma_B,h} = (\rho - i_t)\Gamma_B^h, \tag{H.8}$$

where $\mu_t^{\Gamma_B,h}$ is the drift of $d\Gamma_B^h$. Thus

$$d\Gamma_B^h = (\rho - i_t)\Gamma_B^h\,dt + \sigma_t^{\Gamma_B,h}\,dZ_t, \tag{H.9}$$

which is the same envelope equation as in the representative-agent case of Appendix II. Note that the death intensity and the annuity premium cancel so no λ appears.

Since $\Gamma_B^h = 1/(\bar{p}C_t^h)$ and $\bar{p}$ is constant, Ito's formula yields

$$\frac{d(1/C_t^h)}{1/C_t^h} = -\frac{dC_t^h}{C_t^h} + \mathrm{Var}_t\left(\frac{dC_t^h}{C_t^h}\right). \tag{H.10}$$

Matching equations (H.9) and (H.10) gives the individual Euler equation of a living household:

$$\mathbb{E}_t\left(\frac{dC_t^h}{C_t^h}\right) = (i_t - \rho)\,dt + \mathrm{Var}_t\left(\frac{dC_t^h}{C_t^h}\right), \tag{H.11}$$

which is the same as equation (4) in the main text. Finite lives do not appear in the individual first-order condition: the annuity premium and the death intensity offset each other for individual households.

H.4 Equilibrium labor, the social fund, and the aggregate budget

The profit identity $D_t = \bar{p}Y_t - w_t L_t$ gives

$$w_t L_t + D_t = \bar{p}Y_t. \tag{H.12}$$

Labor intermediation sets $L_t^h = L_t$, so household non-financial income is $w_t L_t^h + D_t = \bar{p}Y_t$. Taxes are the same lump-sum for every household.

With zero steady-state public debt, the incumbent transfer is $S_t^h = 0$. Newborns thus enter with $S^{\text{new}} = 0$, so $B_t^{\text{new}} = 0$, meaning that newborns do not inherit the incumbents' bonds when $d_t \neq 0$.

Substituting equation (H.12), $L_t^h = L_t$, and $S_t^h = 0$ into the household budget equation (H.4) gives

$$dB_t^h = \left[(i_t + \lambda)B_t^h + \bar{p}\big(Y_t(1-\mathcal{T}_t) - C_t^h\big)\right]dt. \tag{H.13}$$

Aggregate bonds do not earn the annuity. Integrating equation (H.13) across living households h, subtracting the estates of the dead, $\lambda B_t\,dt$, and adding newborns with $B_t^{\text{new}} = 0$ yields the absolutely continuous aggregate budget

$$dB_t = \left[i_t B_t + \bar{p}\big(Y_t(1-\mathcal{T}_t) - C_t\big)\right]dt. \tag{H.14}$$

Goods-market clearing $C_t = Y_t$ reduces the continuous part of equation (H.14) to the government flow budget derived below. Note that these laws of motion omit intervention dates; if the government later jumps d_t, real public debt $d_t Y_t$ jumps and aggregate nominal household bonds jump by $\bar{p}\,\Delta(d_t Y_t)$.

H.5 Consumption function

From (H.13), household h's budget in real terms is

$$d\left(\frac{B_t^h}{\bar{p}}\right) = \left[(i_t + \lambda)\frac{B_t^h}{\bar{p}} + Y_t(1 - \mathcal{T}_t) - C_t^h\right]dt. \tag{H.15}$$

Real financial wealth $B_t^h/\bar{p}$ earns $i_t + \lambda$ because of the annuity. The household values after-tax non-financial income with the real stochastic discount factor implied by equation (H.3),

$$m_{t,s}^h \equiv e^{-(\rho+\lambda)(s-t)}\frac{C_t^h}{C_s^h}. \tag{H.16}$$

Note that this discount factor already accounts for survival. From (H.10) and (H.11), we obtain

$$\mathbb{E}_t\left(\frac{d(1/C_t^h)}{1/C_t^h}\right) = -(i_t - \rho)\,dt. \tag{H.17}$$

As a process in s with date t fixed, $m_{t,s}^h$ therefore satisfies

$$\mathbb{E}_t\left(\frac{dm_{t,s}^h}{m_{t,s}^h}\right) = -(i_s + \lambda)\,ds, \tag{H.18}$$

which does not depend on h. Multiply (H.15) by $m_{t,s}^h$, take the expectation, and impose the transversality condition similar to the one in Online Appendix P, which is

$$\lim_{T\to\infty} \mathbb{E}_t\left[m_{t,T}^h \frac{B_T^h}{\bar{p}}\right] = 0, \tag{H.19}$$

we can obtain the following usual intertemporal budget constraint:

$$\frac{B_t^h}{\bar{p}} + H_t^h = \mathbb{E}_t \int_t^\infty m_{t,s}^h C_s^h\,ds, \tag{H.20}$$

where household h's human wealth is the present value of after-tax non-financial income,

$$H_t^h \equiv \mathbb{E}_t \int_t^\infty m_{t,s}^h Y_s(1-\mathcal{T}_s)\, ds. \tag{H.21}$$

Note that in (H.21), the income stream $Y_s(1-\mathcal{T}_s)$ is common, but $m_{t,s}^h$ depends on household h's consumption path, so H_t^h may depend on B_t^h. Total wealth is the sum of real financial wealth and human wealth,

$$W_t^h \equiv \frac{B_t^h}{\bar{p}} + H_t^h. \tag{H.22}$$

Equation (H.20) implies that total wealth equals the present value of consumption.

By equation (H.16),

$$m_{t,s}^h C_s^h = e^{-(\rho+\lambda)(s-t)} C_t^h, \tag{H.23}$$

and therefore

$$\mathbb{E}_t \int_t^\infty m_{t,s}^h C_s^h\, ds = C_t^h \int_t^\infty e^{-(\rho+\lambda)(s-t)}\, ds = \frac{C_t^h}{\rho+\lambda}. \tag{H.24}$$

The C_s^h terms cancel inside the discount factor, due to log utility. Combining equation (H.20) and equation (H.24) gives

$$C_t^h = (\rho+\lambda) W_t^h = (\rho+\lambda)\left(\frac{B_t^h}{\bar{p}} + H_t^h\right). \tag{H.25}$$

Consumption equals a constant fraction $\rho+\lambda$ of total wealth. The variance term in (H.11) does not change that factor. It changes the valuation of H_t^h, because $m_{t,s}^h$ then carries a risk premium. When $\lambda = 0$, (H.25) collapses to the representative-agent consumption function for log utility (Caballero and Simsek, 2020).

Integrate (H.25) across living households and define $H_t \equiv \int H_t^h\, dh$. Then

$$C_t = (\rho+\lambda)\left(\frac{B_t}{\bar{p}} + H_t\right) = (\rho+\lambda)(d_t Y_t + H_t). \tag{H.26}$$

Goods-market clearing $C_t = Y_t$ then gives

$$H_t = Y_t\left(\frac{1}{\rho+\lambda} - d_t\right). \tag{H.27}$$

H.6 Government budget

The government issues the nominal bond B_t and levies the tax flow $\mathcal{T}_t Y_t$. A negative value of $\mathcal{T}_t$ corresponds to a debt-financed flow transfer. Write $\Delta x_t \equiv x_t - x_{t-}$. Real public debt is $B_t/\bar{p} = d_t Y_t$ by (H.2).

Between intervention dates the government budget depends only on interest net of the tax flow. At an intervention date, the government issues a discrete lump of real debt and transfers it directly to living households, in which case the stock that jumps is real public debt. Define Ξ as the cumulative record of those jumps, each divided by pre-jump output, which is

$$\Xi_t \equiv \sum_{0<\tau\le t} \frac{\Delta(d_\tau Y_\tau)}{Y_{\tau-}}, \tag{H.28}$$

which is a pure jump process of locally finite variation. Its increment

$$\chi_t \equiv \Delta\Xi_t \tag{H.29}$$

is the jump in real public debt as a fraction of Y_{t-}. A lump χ_t therefore raises real debt by $\chi_t Y_{t-}$ and nominal bonds by $\bar{p}\,\chi_t Y_{t-}$. The budget is therefore given by

$$d(d_t Y_t) = (i_t d_t - \mathcal{T}_t) Y_t\, dt + Y_{t-}\, d\Xi_t. \tag{H.30}$$

The first term is usual interest net of the tax flow. The second term is the jump in real bonds. Between intervention dates, $d\Xi_t = 0$ and equation (H.30) reduces to

$$d(d_t Y_t) = (i_t d_t - \mathcal{T}_t) Y_t\, dt. \tag{H.31}$$

Write $dY_t/Y_t = \mu_{Y,t}\, dt + \sigma_{Y,t}\, dZ_t$ for the continuous part of output. We then apply Ito's formula to $d_t = (d_t Y_t)/Y_t$ using equation (H.31) for the numerator. Between jumps this gives

$$dd_t = \left[(i_t - \mu_{Y,t} + \sigma^2_{Y,t})d_t - \mathcal{T}_t\right]dt - \sigma_{Y,t} d_t \, dZ_t. \tag{H.32}$$

At an exact intervention date, equations (H.30) and (H.29) give

$$\Delta(d_t Y_t) = \chi_t Y_{t-}. \tag{H.33}$$

Output jumps with the intervention too. Goods-market clearing $C_t = Y_t$ and (H.26) give $Y_t = (\rho + \lambda)(d_t Y_t + H_t)$. With $\lambda > 0$, living households do not capitalize the matching future taxes one-for-one, so H_t falls by less than financial wealth rises, total wealth jumps, and $C_t = Y_t$ jumps. Again, the quotient rule on $d_t = (d_t Y_t)/Y_t$, together with discrete jump equation (H.33), then yields

$$\Delta d_t = (d_{t-} + \chi_t)\frac{Y_{t-}}{Y_t} - d_{t-}. \tag{H.34}$$

The government chooses the jump in the debt-to-output ratio,

$$s_t \equiv \Delta d_t. \tag{H.35}$$

This jump is the government's instrument. Between intervention dates, $s_t = 0$ and $\chi_t = 0$. Substituting equation (H.35) into equation (H.34) and solving for the lump gives

$$\chi_t = (d_{t-} + s_t)\frac{Y_t}{Y_{t-}} - d_{t-}. \tag{H.36}$$

Aggregate household bonds jump by $\bar{p}\,\Delta(d_t Y_t)$. The full law of d_t is the continuous part equation (H.32) together with equation (H.35).

If $d_0 = 0$, $\mathcal{T}_t = 0$, and Ξ is identically zero, then $d_t = 0$ for all t.

H.7 Law of human wealth and aggregate output

We write $dH_t^h = \mu_t^{H,h}\,dt + \sigma_t^{H,h}\,dZ_t$. Differentiating (H.25) and using (H.15) gives

$$\frac{dC_t^h}{C_t^h} = \left[\frac{(i_t+\lambda)B_t^h/\bar{p} + Y_t(1-\mathcal{T}_t) + \mu_t^{H,h}}{W_t^h} - (\rho+\lambda)\right]dt + \frac{\sigma_t^{H,h}}{W_t^h}\,dZ_t. \tag{H.37}$$

The individual Euler equation (H.11) then requires

$$\mu_t^{H,h} = (i_t+\lambda)H_t^h - Y_t(1-\mathcal{T}_t) + \frac{(\sigma_t^{H,h})^2}{W_t^h}. \tag{H.38}$$

We then integrate (H.38) across living households. With $H_t = \int H_t^h\,dh$ and $dH_t = \mu_t^H\,dt + \sigma_t^H\,dZ_t$, which is

$$\mu_t^H = (i_t+\lambda)H_t - Y_t(1-\mathcal{T}_t) + \int \frac{(\sigma_t^{H,h})^2}{W_t^h}\,dh, \tag{H.39}$$

where the first two terms are the generator of the annuity-rate integral. The last term is the cross-sectional integral of the individual risk adjustments in equation (H.21). From (H.37), the diffusion of dC_t^h/C_t^h is $\sigma_t^{H,h}/W_t^h$. Living households load on the same aggregate shock, so that loading is common.

After $C_t = Y_t$, this common loading is the output-growth diffusion $\sigma_{Y,t}$, and therefore $(\sigma_t^{H,h})^2/W_t^h = (\sigma_{Y,t})^2 W_t^h$. Integrating and using $C_t = (\rho+\lambda)\int W_t^h\,dh$ with $C_t = Y_t$ gives

$$\int \frac{(\sigma_t^{H,h})^2}{W_t^h}\,dh = \frac{(\sigma_{Y,t})^2 Y_t}{\rho+\lambda}. \tag{H.40}$$

The first two terms of equation (H.39) are already aggregates. For the last term, integrate the common loading. Aggregate human wealth has diffusion $\sigma_t^H = \int \sigma_t^{H,h}\,dh$, so $\sigma_t^H = \sigma_{Y,t}\int W_t^h\,dh = \sigma_{Y,t}(d_tY_t + H_t)$. Then

$$\frac{(\sigma_t^H)^2}{d_tY_t + H_t} = (\sigma_{Y,t})^2(d_tY_t + H_t), \tag{H.41}$$

which equals the integral in equation (H.40) because $C_t = (\rho+\lambda)(d_tY_t + H_t) = Y_t$.

Substituting into equation (H.39) gives

$$\mu_t^H = (i_t + \lambda)H_t - Y_t(1 - \mathcal{T}_t) + \frac{(\sigma_t^H)^2}{d_t Y_t + H_t}. \tag{H.42}$$

Differentiate $Y_t = (\rho + \lambda)(d_t Y_t + H_t)$, substitute equation (H.31) and equation (H.42), and use equation (H.27). Write $\psi \equiv \lambda(\rho + \lambda)$. Then

$$dY_t = \left[(i_t - \rho)Y_t - \psi d_t Y_t + \sigma_{Y,t}^2 Y_t\right] dt + \sigma_{Y,t} Y_t \, dZ_t, \tag{H.43}$$

where $\sigma_{Y,t} \equiv (\rho + \lambda)\sigma_t^H / Y_t$ is the instantaneous volatility of output growth. Thus

$$\frac{dY_t}{Y_t} = \left(i_t - \rho + \sigma_{Y,t}^2 - \psi d_t\right) dt + \sigma_{Y,t} \, dZ_t. \tag{H.44}$$

When $\lambda = 0$ or $d_t = 0$, this collapses to our baseline representative-agent equation (11). The new term $-\psi d_t$ is the generational-turnover effect of Blanchard (1985). Newborns enter with zero bonds and consume only out of human wealth, while aggregate consumption is $(\rho + \lambda)(d_t Y_t + H_t)$. A positive debt-to-output ratio then lowers aggregate consumption growth.

H.8 IS equation

As in the main text, the flexible-price natural output $\{Y_t^n\}$ is the path with $\sigma_t^s = 0$ and $d_t = \mathcal{T}_t = 0$. That path satisfies

$$\frac{dY_t^n}{Y_t^n} = (r^n - \rho + \sigma^2)\, dt + \sigma \, dZ_t, \qquad r^n = \rho + g - \sigma^2. \tag{H.45}$$

The fiscal term is absent on this path because $d_t = 0$. Define $\hat{Y}_t = \ln(Y_t / Y_t^n)$ and

$$\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2. \tag{H.46}$$

Output remains adapted to the filtration generated by equation (H.1), so $\sigma_{Y,t} = \sigma + \sigma_t^s$ and

$$d \ln Y_t = \left(i_t - \rho + \tfrac{1}{2}(\sigma + \sigma_t^s)^2 - \psi d_t\right) dt + (\sigma + \sigma_t^s)\, dZ_t, \tag{H.47}$$

$$d \ln Y_t^n = \left(r^n - \rho + \tfrac{1}{2}\sigma^2\right) dt + \sigma\, dZ_t. \tag{H.48}$$

Subtracting these processes yields the exact nonlinear IS equation

$$d\hat{Y}_t = \left(i_t - r^n + \frac{1}{2}\mathcal{E}_t - \psi d_t\right) dt + \sigma_t^s\, dZ_t. \tag{H.49}$$

Under the Taylor rule $i_t = r^n + \phi_y \hat{Y}_t$ of Section 2, with $\phi_y > 0$,

$$d\hat{Y}_t = \left(\phi_y \hat{Y}_t + \frac{1}{2}\mathcal{E}_t - \psi d_t\right) dt + \sigma_t^s\, dZ_t. \tag{H.50}$$

If $\lambda = 0$ or $d_t = 0$, equations (H.49) and (H.50) reduce to (12) and (15). Equations (H.43)–(H.50) are the continuous part, between jumps. At an intervention date, d_t jumps by equation (H.35) and $\hat{Y}_t$ jumps through goods-market clearing together with equation (H.26).

By equation (H.35),

$$d_t = d_{t-} + s_t. \tag{H.51}$$

After the jump, equations (H.49) and (H.50) are evaluated at this new value of d_t, so the fiscal term is $-\psi(d_{t-} + s_t)$. If $d_{t-} = 0$, then $d_t = s_t$ and the fiscal term is $-\psi s_t$. A larger debt-to-output ratio raises current consumption out of wealth and therefore lowers expected output-gap growth.

H.9 Robust escape clause

Based on the above model setup, we now construct a robust fiscal escape clause. The escape clause is an off-equilibrium threat, not a standing fiscal rule. If a recessionary volatility path starts with a negative output gap, the government jumps d_t by enough that remaining on that path would violate the boundedness condition in Definition 2. On that path, future cohorts service the resulting debt. Because the threat is credible, that path is not started, implying that the unique remaining bounded equilibrium is the full-stabilization path, along which $\hat{Y}_t = \mathcal{E}_t = s_t = d_t = 0$. The construction requires $\lambda > 0$, so that $\psi \equiv \lambda(\rho + \lambda) > 0$ and a jump in d_t moves aggregate demand.

Default strategy The debt-to-output ratio starts at $d_0 = 0$. If $\hat{Y}_t \geq 0$, the government sets $s_t = 0$. Combined with $d_0 = 0$, $\mathcal{T}_t = 0$, and $\Xi \equiv 0$, (H.32) then keeps $d_t = 0$ along any path that never enters $\hat{Y}_t < 0$.

Threat strategy If $\hat{Y}_t < 0$, the government maintains d_t so that

$$\psi d_t = \lambda_v \mathcal{E}_t + \varepsilon, \qquad \lambda_v \geq \frac{1}{2}, \quad \varepsilon > 0. \tag{H.52}$$

The first term responds at least one-for-one to excess variance. The constant ε is slack. The government implements (H.52) with the jump (H.35), or equivalently with the lump (H.36) when output jumps; as in the martingale equilibrium. If the path starts from $d_{t-} = 0$, then $d_t = s_t$ and equation (H.52) is equivalent to

$$\psi s_t = \lambda_v \mathcal{E}_t + \varepsilon. \tag{H.53}$$

While the output gap remains negative, further jumps reset d_t whenever (H.52) would otherwise fail. Living households spend part of the implied transfer because $\lambda > 0$, which is why d_t appears in the IS equation (H.50). If the economy returns to $\hat{Y}_t \geq 0$, the government can retire residual debt with the tax flow $\mathcal{T}_t$, as in Angeletos, Lian and Wolf (2024). That retirement is an off-equilibrium path.

Lemma H.1 *Under the escape clause, there is no bounded equilibrium in which $\hat{Y}_t < 0$ at any date.*

Proof. Fix a date, labeled 0, at which $\hat{Y}_0 < 0$. After any contemporaneous jump, (H.52) holds, and the continuous part of (H.50) applies while the output gap remains negative. Substitute (H.52) into that continuous part. The residual $\left(\lambda_v - \frac{1}{2}\right)\mathcal{E}_t$ is nonnegative, so

$$d\hat{Y}_t \leq \left(\phi_y \hat{Y}_t - \varepsilon'\right)dt + \sigma_t^s \, dZ_t \tag{H.54}$$

for some $\varepsilon' \geq \varepsilon > 0$. This drift restriction holds only while $\hat{Y}_t < 0$. Let

$$\tau \equiv \inf\{u \geq 0 : \hat{Y}_u \geq 0\}, \tag{H.55}$$

with $\inf \emptyset = \infty$, and set $\check{Y}_t \equiv \hat{Y}_t \, 1_{\{t<\tau\}}$. Then $\check{Y}_t \leq 0$. Integrate the continuous part of (H.54) from date 0 to the stopped time $t \wedge \tau$:

$$\hat{Y}_{t\wedge\tau} = \hat{Y}_0 + \int_0^{t\wedge\tau} \mu_u \, du + \int_0^{t\wedge\tau} \sigma_u^s \, dZ_u, \tag{H.56}$$

where $\mu_u \leq \phi_y \hat{Y}_u - \varepsilon'$ on $u < \tau$. Take $\mathbb{E}_0$. The diffusion term is an Ito integral, hence a martingale, so its expectation is zero, and

$$\mathbb{E}_0\left[\hat{Y}_{t\wedge\tau}\right] \leq \hat{Y}_0 + \phi_y \int_0^t \mathbb{E}_0[\check{Y}_u] \, du - \varepsilon' \, \mathbb{E}_0[t \wedge \tau]. \tag{H.57}$$

The last term is nonpositive. Dropping it raises the upper bound. Split the stopped value as $\hat{Y}_{t\wedge\tau} = \check{Y}_t + \hat{Y}_\tau 1_{\{\tau \leq t\}}$. On $\{\tau \leq t\}$ the process has already left the negative region, so $\hat{Y}_\tau \geq 0$ and $\mathbb{E}_0\left[\hat{Y}_\tau 1_{\{\tau\leq t\}}\right] \geq 0$. Therefore $\mathbb{E}_0[\check{Y}_t] \leq \mathbb{E}_0\left[\hat{Y}_{t\wedge\tau}\right]$, which with (H.57) gives

$$\mathbb{E}_0[\check{Y}_t] \leq \hat{Y}_0 + \phi_y \int_0^t \mathbb{E}_0[\check{Y}_u] \, du. \tag{H.58}$$

Write $N(t) \equiv \int_0^t \mathbb{E}_0[\check{Y}_u] \, du$, so that (H.58) is $\mathbb{E}_0[\check{Y}_t] \leq \hat{Y}_0 + \phi_y N(t)$. Define

$$k(t) \equiv e^{-\phi_y t}\left(\hat{Y}_0 + \phi_y N(t)\right). \tag{H.59}$$

Then $k(0) = \hat{Y}_0$ and

$$k'(t) = e^{-\phi_y t} \phi_y \Big(\mathbb{E}_0[\check{Y}_t] - \left(\hat{Y}_0 + \phi_y N(t)\right)\Big) \leq 0. \tag{H.60}$$

Hence $k(t) \leq \hat{Y}_0$, which is $\hat{Y}_0 + \phi_y N(t) \leq e^{\phi_y t} \hat{Y}_0$. Combined with (H.58),

$$\mathbb{E}_0[\check{Y}_t] \leq e^{\phi_y t} \hat{Y}_0. \tag{H.61}$$

The time t here is calendar time, not the hitting time. Paths that have already reached zero are dropped by $1_{\{t<\tau\}}$ and contribute nothing to $\check{Y}_t$. Since $\hat{Y}_0 < 0$, the right-hand side of equation (H.61) diverges to $-\infty$ as $t \to \infty$. Boundedness in Definition 2 requires $\lim_{t\to\infty} \mathbb{E}_0|\hat{Y}_t| < \infty$. But

$$\mathbb{E}_0|\hat{Y}_t| \geq \mathbb{E}_0|\check{Y}_t| = -\mathbb{E}_0[\check{Y}_t] \geq -e^{\phi_y t}\hat{Y}_0, \tag{H.62}$$

which diverges to $+\infty$, contradicting boundedness. ■

Lemma H.2 *In any equilibrium with $\hat{Y}_t = 0$ for all t, necessarily $\mathcal{E}_t = 0$ and $s_t = 0$ for all t. If also $d_0 = 0$, then $d_t = 0$ for all t.*

Proof. If $\hat{Y}_t = 0$ for all t, the default strategy sets $s_t = 0$. Then, equation (H.32) with $d_0 = 0$, $\mathcal{T}_t = 0$, and $\Xi \equiv 0$ gives $d_t = 0$, yielding $\mathcal{E}_t = 0$. ■

Interpretation The jump s_t is a deficit in the sense of Section 4.2: a debt-financed transfer to living households. In discrete time that object is the period deficit. In continuous time the tax flow $\mathcal{T}_t$ is a rate, so a finite $\mathcal{T}_t$ moves debt only by order dt and cannot deliver a discrete transfer at a date. Thus, the lump χ_t, or equivalently s_t, acts as that transfer. The flow $\mathcal{T}_t$ remains the regular budget residual; the atom is recorded in Ξ. Along the equilibrium path there is no deficit of either kind.

The full-stabilization path $\hat{Y}_t = \mathcal{E}_t = s_t = d_t = 0$ is a bounded equilibrium. It satisfies equation (H.50) because both sides are zero. It satisfies the government budget. The threat is not triggered, so no taxes are raised and no debt is issued. By Lemmas H.1 and H.2, it is the unique bounded equilibrium.

Finally, the escape clause is not a knife-edge rule. Slack $\varepsilon > 0$ is enough. Any $\lambda_v \geq \frac{1}{2}$ is also enough: the residual $\left(\lambda_v - \frac{1}{2}\right)\mathcal{E}_t$ is nonnegative and only tightens equation (H.54).

I Behavioral Frictions

I.1 Fading Memory and Equilibrium Selection

This section shows that introducing a small friction in social memory, following Angeletos and Lian (2023), eliminates alternative equilibria and selects the full-stabilization equilibrium (i.e., $\hat{Y}_t = 0$ for all t) as the unique equilibrium of the economy. These results extend Angeletos and Lian (2023) to our framework with endogenous aggregate volatility and its first-order effects on output.[21]

The intuition is similar: our equilibrium construction with time-varying aggregate excess volatility σ_t^s is sustainable because households coordinate their consumption decisions over time contingent on the realization of past shocks. With even slight loss of memory, this coordination, regardless of whether the resulting output gap paths are non-stationary (e.g., martingale equilibrium) or stationary in the long run (e.g., Ornstein-Uhlenbeck equilibria), becomes non-viable.

Throughout the section, we assume rigid prices and the extended Taylor rule (A.1).

Setup Let the equilibrium stochastic processes for output gap $\hat{Y}_t$ and consumption volatility $(\sigma_t^s + \sigma)^2$ be given by[22]

$$d\hat{Y}_t = \mu_t^{\hat{Y}} dt + \sigma_t^s \, dZ_t,$$
$$d(\sigma_t^s + \sigma)^2 = \mu_t^{\sigma} dt + \tilde{\sigma}_t^{\sigma} \, dZ_t,$$

where $\mu_t^{\hat{Y}}$, μ_t^{σ}, σ_t^s and $\tilde{\sigma}_t^{\sigma}$ are predictable, continuous processes such that $\int_S^T (\sigma_t^s)^2 \, dt < \infty$, $\int_S^T (\tilde{\sigma}_t^{\sigma})^2 \, dt < \infty$, $\int_S^T |\mu_t^{\hat{Y}}| \, dt < \infty$, and $\int_S^T |\mu_t^{\sigma}| \, dt < \infty$ for any $T \geq S$.

[21] Angeletos and Lian (2023) focus on local stationary equilibria in *log-linearized* New Keynesian models, finding that with a small degree of fading social memory, only the so-called minimum-state variable (MSV) equilibrium survives as the unique equilibrium. The MSV solution corresponds to the full-stabilization equilibrium in our framework.

[22] In this section, we assume that $\hat{Y}_t$ and σ_t^s are adapted to the filtration generated by fundamental shocks dZ_t. Even if we allow them to be dependent on a sunspot shock, e.g., another Brownian motion $d\eta_t$ orthogonal to dZ_t (Angeletos and Lian, 2023), our result that only the perfect stabilization path survives as the unique equilibrium under fading memory does not change.

Then, $\hat{Y}_t$ and $(\sigma_t^s + \sigma)^2$ admit an stochastic integral representation of the form:

$$\hat{Y}_t = \psi + \int_{-\infty}^{t} \mu_u^{\hat{Y}}\, du + \int_{-\infty}^{t} \sigma_u^s\, dZ_u, \tag{I.1a}$$

and

$$(\sigma_t^s + \sigma)^2 = \tilde{\delta} + \int_{-\infty}^{t} \mu_u^{\sigma}\, du + \int_{-\infty}^{t} \tilde{\sigma}_u^{\sigma}\, dZ_u, \tag{I.1b}$$

where ψ and $\tilde{\delta}$ are deterministic intercepts in this two-sided representation. If the remote-past levels exist as random variables, they are measurable with respect to the tail σ-algebra generated by $\{Z_t\}$, which is trivial by Kolmogorov's zero-one law (Kallenberg, 2021), so they are almost surely constant. They are not additional processes that can co-move with later Brownian increments. The uniqueness argument below then implies $\psi = 0$ and $\tilde{\delta} = \sigma^2$.

Memory frictions Households have imperfect recall: as the time since a shock increases, the remembered realization of the past Brownian motions decays toward zero, the unconditional mean. Following notations in Angeletos and Lian (2023), let $\bar{\mathbb{E}}_t[\cdot]$ denote time-t average conditional expectations across households under imperfect recall.

We assume exponential fading at rate $\lambda \geq 0$. In particular, we assume that for any adapted process $\{X_t\}$ with $|\bar{\mathbb{E}}_t[X_u]| < \infty$ for all t and any $u \leq t$,

$$\bar{\mathbb{E}}_t\left[X_u dZ_u\right] = e^{-\lambda(t-u)}\bar{\mathbb{E}}_u(X_u)dZ_u,$$

so the recalled Brownian increment $X_u\, dZ_u$ at time t is attenuated by $e^{-\lambda(t-u)}$. When $\lambda = 0$, there is no fading and the formulation nests perfect recall.[23]

[23]The original formulation in Angeletos and Lian (2023) assumes that at each instant t, a fraction $1 - e^{-\lambda(t-u)}$ of households completely forget the past realization of dZ_u, leading to heterogeneous information sets. Our fading formulation is isomorphic in spirit but remains within a representative-household framework, avoiding aggregation issues in a non-linear setting.

Taking expectations of the stochastic integrals in (I.1) we obtain:

$$\bar{\mathbb{E}}_t[\hat{Y}_t] = \psi + \int_{-\infty}^{t} \bar{\mathbb{E}}_t[\mu_u^{\hat{Y}}]\, du + \int_{-\infty}^{t} e^{-\lambda(t-u)} \bar{\mathbb{E}}_u \left[\sigma_u^s\right] dZ_u, \tag{I.2a}$$

$$\bar{\mathbb{E}}_t \left[(\sigma_t^s + \sigma)^2\right] = \tilde{\delta} + \int_{-\infty}^{t} \bar{\mathbb{E}}_t \left[\mu_u^{\sigma}\right] du + \int_{-\infty}^{t} e^{-\lambda(t-u)} \bar{\mathbb{E}}_u \left[\tilde{\sigma}_u^{\sigma}\right] dZ_u, \tag{I.2b}$$

where we use that $\mu_u^{\hat{Y}}$, σ_u^s, μ_u^{σ}, and $\tilde{\sigma}_u^{\sigma}$ are adapted to filtration and integrable, and that $\bar{\mathbb{E}}_t[\cdot]$ is well defined for the past realization of shocks, so the recall operator and the time integral can be interchanged (Whittaker and Watson, 2021).

Dynamic IS equation With the memory friction described above, the dynamic IS equation is given by

$$\hat{Y}_t = \bar{\mathbb{E}}_t \left[\hat{Y}_{t+dt}\right] - \bar{\mathbb{E}}_t \left[(i_t - r^n) + \frac{1}{2}\left(\left(\sigma + \sigma_t^s\right)^2 - \sigma^2\right)\right] dt,$$

where the imperfect-recall expectations operator $\bar{\mathbb{E}}_t[\cdot]$ is applied to the right-hand side to capture both the conditional expectation of future output gap and imperfect observation (or recall) of aggregate variables, following Angeletos and Lian (2023).

Substitute the extended Taylor rule (A.1) in the expression above to obtain:

$$\bar{\mathbb{E}}_t \left[\hat{Y}_{t+dt}\right] - \hat{Y}_t = \bar{\mathbb{E}}_t \left[\phi_y \hat{Y}_t + \left(\frac{1}{2} - \phi_{\text{vol}}\right)\left(\left(\sigma + \sigma_t^s\right)^2 - \sigma^2\right)\right] dt. \tag{I.3}$$

From equations (I.1a) and (I.2a), the expected change in the output gap is given by

$$\begin{aligned}\bar{\mathbb{E}}_t \left[\hat{Y}_{t+dt}\right] - \hat{Y}_t = \bar{\mathbb{E}}_t[\mu_t^{\hat{Y}}]dt &+ \int_{-\infty}^{t} \left(\bar{\mathbb{E}}_t \left(\mu_u^{\hat{Y}}\right) - \mu_u^{\hat{Y}}\right) du \\ &- \int_{-\infty}^{t} \left(\sigma_u^s - e^{-\lambda(t-u)} \bar{\mathbb{E}}_u \left[\sigma_u^s\right]\right) dZ_u.\end{aligned} \tag{I.4}$$

By equating (I.3) and (I.4), and substituting the stochastic-integral representations (I.1a) and (I.1b), together with their expectations (I.2a) and (I.2b), we obtain

$$\begin{aligned} 0 = &\left[\phi_y\psi + \left(\frac{1}{2} - \phi_{vol}\right)(\tilde{\delta} - \sigma^2) - \bar{\mathbb{E}}_t[\mu_t^{\hat{Y}}]\right] dt \\ &+ \left[\int_{-\infty}^t \left[\phi_y \bar{\mathbb{E}}_t[\mu_u^{\hat{Y}}] + \left(\frac{1}{2} - \phi_{vol}\right)\bar{\mathbb{E}}_t\left[\mu_u^{\sigma}\right]\right] du\right] dt \\ &+ \left[\int_{-\infty}^t e^{-\lambda(t-u)}\left[\phi_y \bar{\mathbb{E}}_t\left[\sigma_u^s\right] + \left(\frac{1}{2} - \phi_{vol}\right)\bar{\mathbb{E}}_t\left[\tilde{\sigma}_u^{\sigma}\right]\right] dZ_u\right] dt \\ &+ \int_{-\infty}^t \left(\mu_u^{\hat{Y}} - \bar{\mathbb{E}}_t\left(\mu_u^{\hat{Y}}\right)\right) du + \int_{-\infty}^t \left(\sigma_u^s - e^{-\lambda(t-u)}\bar{\mathbb{E}}_u\left[\sigma_u^s\right]\right) dZ_u. \end{aligned} \tag{I.5}$$

Taking the limit when $dt \longrightarrow 0$ of equation (I.5), we obtain the following condition:

$$0 = \int_{-\infty}^t \left(\mu_u^{\hat{Y}} - \bar{\mathbb{E}}_t\left(\mu_u^{\hat{Y}}\right)\right) du + \int_{-\infty}^t \left(\sigma_u^s - e^{-\lambda(t-u)}\bar{\mathbb{E}}_u[\sigma_u^s]\right) dZ_u,$$

which by an application of Lemma I.3 implies

$$0 = \sigma_u^s - e^{-\lambda(t-u)}\bar{\mathbb{E}}_u\left[\sigma_u^s\right], \quad \text{for all } u \leq t. \tag{I.6}$$

For any positive recall friction $\lambda > 0$, equation (I.6) can hold for all $u \leq t$ only if

$$\sigma_t^s = 0, \quad \text{for all } t.^{24} \tag{I.7}$$

Plugging equation (I.7) into (I.1b), and making use again of Lemma I.3, we obtain the following component-wise conditions:

$$\tilde{\delta} = \sigma^2, \tag{I.8a}$$

$$0 = \mu_t^{\sigma} = \tilde{\sigma}_t^{\sigma}, \quad \text{for all } t, \tag{I.8b}$$

which hold for any positive degree of memory fading $\lambda > 0$, even when $\lambda \longrightarrow 0^+$.

Since $\hat{Y}_t$ and σ_t^s become deterministic by the results in (I.7) and (I.8b), no memory loss occurs in equilibrium and $\bar{\mathbb{E}}_t(X_t) = X_t$ for any aggregate variable X_t. Hence, using equations (I.7) and (I.8), equation (I.5) reduces to

$$\mu_t^{\hat{Y}} = \phi_y\psi + \phi_y \int_{-\infty}^t \mu_u^{\hat{Y}}\, du, \quad \text{for all } t. \tag{I.9}$$

[24] Taking $t \longrightarrow \infty$ in equation (I.6) yields $\sigma_u^s = 0$ for all u.

When $\phi_y \neq 0$, using equation (I.9) to compute $\int_{-\infty}^{t} \mu_u^{\hat{Y}}\, du$ and rearranging yields

$$0 = \int_{-\infty}^{t} \left[\psi + \left(t - u - \frac{1}{\phi_y}\right) \mu_u^{\hat{Y}}\right] du.$$

By Lemma I.3, the integrand must be zero for all $u \leq t$, which leads to $\psi = 0$ and $\mu_t^{\hat{Y}} = 0$ for all t. Thus the unique equilibrium exhibits perfect stabilization: $\hat{Y}_t = 0$ and $\sigma_t^s = 0$ for all t.

Finally, in the knife-edge case $\phi_y = 0$, equation (I.9) implies zero drift, $\mu_t^{\hat{Y}} = 0$ for all t, and a constant output gap equal to the initial condition, $\hat{Y}_t = \psi$. Multiple steady states are therefore possible in this case, which is not specific to our model, and without loss of generality we can set $\psi = 0$ (Galí, 2015; Angeletos and Lian, 2023).

Lemma I.3 (Uniqueness of stochastic–integral representation for Brownian motion) *Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfy the usual conditions and $\{Z_t\}$ be a Brownian motion with $Z_S = 0$ for some $S \in \mathbb{R}$. Let X, Y be predictable, continuous processes with $\int_S^T X_t^2\, dt < \infty$ and $\int_S^T Y_t^2\, dt < \infty$ a.s. for every $T \geq S$. Let V, K be predictable, continuous processes with $\int_S^T |V_t|\, dt < \infty$ and $\int_S^T |K_t|\, dt < \infty$ a.s. for every $T \geq S$. If there exist constants $C, D \in \mathbb{R}$ such that*

$$C + \int_S^T V_t\, dt + \int_S^T X_t\, dZ_t = D + \int_S^T K_t\, dt + \int_S^T Y_t\, dZ_t \qquad \text{a.s. for all } T \geq S,$$

then $C = D$, $V = K$ $dt \times \mathbb{P}$-a.e., and $X = Y$ $dt \times \mathbb{P}$-a.e. If, in addition, V, K, X, Y are continuous, then these equalities hold indistinguishably.

Proof. Setting $T = S$ yields $C = D$. Define $G := V - K$ and $H := X - Y$, and for $t \geq S$ set

$$A_t := \int_S^t G_u\, du, \qquad N_t := \int_S^t H_u\, dZ_u.$$

By construction,

$$A_S = \int_S^S G_u\, du = 0 \quad \text{and} \quad N_S = \int_S^S H_u\, dZ_u = 0,$$

i.e., both processes are initialized at zero at time S. The stated identity implies

$$A_t + N_t = 0 \qquad \text{a.s. for all } t \geq S.$$

Here A is continuous with finite variation and N is a continuous local martingale. Hence $N = -A$ is a continuous local martingale of finite variation, and therefore constant. Using $N_S = 0$ gives $N \equiv 0$, and then $A \equiv 0$.

From $N \equiv 0$ we obtain its quadratic variation:

$$\langle N \rangle_t = \int_S^t H_u^2\, du = 0 \quad \text{for all } t \geq S,$$

which implies $H = 0$ $dt \times \mathbb{P}$-a.e., i.e., $X = Y$ $dt \times \mathbb{P}$-a.e. From $A \equiv 0$ we have $\int_S^t G_u du = 0$ for all $t \geq S$, hence $G = 0$ $dt \times \mathbb{P}$-a.e., i.e., $V = K$ $dt \times \mathbb{P}$-a.e. If V, K, X, Y are continuous, vanishing almost everywhere in time forces pointwise equality by continuity, yielding indistinguishability. ■

I.2 Level-k Thinking

We formally present a mathematical treatment of the level-k reasoning (see, e.g., Farhi and Werning (2019)) that may eliminate the alternative equilibria. To see this, note first that the IS equation is given by

$$d\hat{Y}_t = \left(i_t - r^n + \frac{1}{2}\mathcal{E}_t\right) dt + \sigma_t^s dZ_t,$$

where $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2$ is the excess variance. It can be written as

$$\begin{aligned} \hat{Y}_t &= \mathbb{E}_t\left[\hat{Y}_{t+dt}\right] - \left(i_t - r^n + \frac{1}{2}\mathcal{E}_t\right) dt \\ &= \lim_{s\to\infty} \mathbb{E}_t\left[\hat{Y}_s\right] - \int_t^\infty \mathbb{E}_t\left(i_s - r^n + \frac{1}{2}\mathcal{E}_s\right) ds. \end{aligned} \tag{I.10}$$

Given that i_t follows the Taylor rule ($i_s = r^n + \phi_y \hat{Y}_s$), (I.10) can be written in the following general form:

$$\hat{Y}_t = Y^* \left(\hat{Y}_t, \mathcal{E}_t, \left\{ \hat{Y}_{t+h}, \mathcal{E}_{t+h} \right\}_{h>0} \right), \tag{I.11}$$

for some functional Y^*. Thus, bounded equilibria are fixed-point solutions of (I.11), while satisfying the boundedness condition (2).

In contrast, under level-k thinking, output gap is recursively determined as a fixed-point solution to

$$\hat{Y}_t^k = Y^* \left(\hat{Y}_t^k, \mathcal{E}_t^k, \left\{ \hat{Y}_{t+h}^{k-1}, \mathcal{E}_{t+h}^{k-1} \right\}_{h>0} \right), \tag{I.12}$$

where $\hat{Y}_t^k, \mathcal{E}_t^k$ are the output gap and excess variance up to a k-level reasoning. When the economy is currently at the flexible price equilibrium, i.e., for all t, $\hat{Y}_t^0 = \mathcal{E}_t^0 = 0$, we obtain $\hat{Y}_t^1 = \mathcal{E}_t^1 = 0$ from (I.12), which recursively leads to $\hat{Y}_t^k = \mathcal{E}_t^k = 0$. Consequently, households are unable to transition into alternative equilibria through self-fulfilling expectations, as level-k thinking substantially undermines their ability to coordinate intertemporal consumption decisions.

J A Knife-Edge Monetary Policy Benchmark

This section studies a monetary rule that selects the full-stabilization path by exactly offsetting the precautionary-saving feedback from aggregate volatility. The targeted allocation is the constrained-efficient flexible-price equilibrium characterized by equation (9) and Appendix II.2.2.[25] The exercise is a theoretical implementation benchmark rather than a practical policy proposal. Under log utility, uniqueness requires a response ϕ_{vol} by the extended Taylor rule of exactly one half to the volatility gap in order to cancel the term in the drift of the IS equation (12). As we will see below, coefficient misspecification and multiplicative measurement error in that gap both prevent this exact cancellation and restore the possibility of alternative bounded equilibria. These fragilities motivate the behavioral and fiscal selection mechanisms presented in the main text as better alternatives to policy rules for restoring uniqueness, but we include the exercise here for completeness.

J.1 Exact Volatility-Gap Targeting

Instead of the conventional Taylor rule in (14), consider

$$i_t = r^n + \phi_y \hat{Y}_t - \underbrace{\frac{1}{2}\left((\sigma + \sigma_t^s)^2 - \sigma^2\right)}_{\text{Aggregate volatility targeting}}, \quad \text{with} \quad \phi_y > 0. \tag{J.1}$$

This rule targets the output gap $\hat{Y}_t$ and the aggregate volatility gap with an exact coefficient of $\frac{1}{2}$. Substituting (J.1) into the non-linear IS equation (12) cancels the volatility feedback term from the drift. The dynamics then reduce to equation (18), making the benchmark equilibrium of Proposition 1, with $\hat{Y}_t = \sigma_t^s = 0$ for all t, the unique bounded rational expectations equilibrium under the Taylor principle $\phi_y > 0$.

The same rule can be written as $i_t = r_t^T + \phi_y \hat{Y}_t$, where r_t^T is the risk-adjusted natural rate defined in (13). This representation makes the exact cancellation transparent: the policy intercept must move one-for-one with the endogenous effect of aggregate volatility on the effective reference rate.[26]

[25] With a proper production subsidy eliminating the real distortion generated by monopolistic competition, the flexible-price allocation becomes the first best. See Woodford (2003).

[26] Acharya and Benhabib (2024) similarly account for endogenous changes in the natural interest

Along the selected equilibrium path, excess volatility becomes zero, so rule (J.1) is observationally equivalent to the conventional Taylor rule (14). The distinction lies entirely off the equilibrium path: rule (J.1) commits the monetary authority to offset excess volatility if it emerges. The same condition can be expressed in terms of expected output growth:[27]

$$\underbrace{\frac{\mathbb{E}_t\left(d\log Y_t\right)}{dt}}_{\text{Growth rate}} = \underbrace{\frac{\mathbb{E}_t\left(d\log Y_t^n\right)}{dt}}_{\substack{\text{Benchmark}\\ \text{growth rate}}} + \underbrace{\phi_y \hat{Y}_t}_{\substack{\text{Business cycle}\\ \text{targeting}}} = \left(g - \frac{1}{2}\sigma^2\right) + \phi_y \hat{Y}_t. \tag{J.2}$$

Equation (J.2) restates the exact rule using an expected growth-rate target rather than the policy instrument i_t.

J.2 Implementation

The uniqueness result requires the monetary authority to offset the volatility gap with exactly the coefficient $1/2$ in (J.1). If the response is either stronger or weaker because of policy misspecification or measurement error, the rule no longer cancels the precautionary-saving feedback in the non-linear IS equation (12).[28]

To examine coefficient misspecification, consider the extended Taylor rule (A.1) in the martingale environment of Section 3.1, with $\phi_{\text{vol}} \neq \frac{1}{2}$.[29] Substituting (A.1) into the IS equation gives

$$d\hat{Y}_t = \left[\phi_y \hat{Y}_t + \left(\frac{1}{2} - \phi_{\text{vol}}\right)\left(\left(\sigma + \sigma_t^s\right)^2 - \sigma^2\right)\right] dt + \sigma_t^s dZ_t. \tag{J.3}$$

rate when eliminating fluctuations in precautionary saving in a heterogeneous-agent New Keynesian model with countercyclical idiosyncratic risk. Here, the risk-adjusted natural rate reflects endogenous aggregate volatility.

[27] This is only an algebraic representation of the same exact-cancellation condition, not a separate policy proposal.

[28] Online Appendix E extends the model to constant relative risk aversion utility and derives

$$r_t^T = r^n - \frac{\gamma^2}{2}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right],$$

where γ is relative risk aversion. The exact volatility-targeting coefficient becomes $\frac{\gamma^2}{2}$, so the required response varies with preferences.

[29] The rule in equation (A.1) with $\phi_{vol} \neq \frac{1}{2}$ also permits the Ornstein-Uhlenbeck equilibrium. Online Appendix A gives the corresponding equilibrium conditions.

When $\phi_{\text{vol}} \neq \frac{1}{2}$, the martingale equilibrium with $\sigma_0^s > 0$ reappears and satisfies[30]

$$\hat{Y}_t = -\frac{\left(\sigma + \sigma_t^s\right)^2}{2\phi_{\text{total}}} + \frac{\sigma^2}{2\phi_{\text{total}}}, \quad \text{with} \quad \phi_{\text{total}} \equiv \frac{\phi_y}{1 - 2\phi_{\text{vol}}}, \tag{J.4}$$

where the process for σ_t^s after the initial volatility shock is

$$d\sigma_t^s = -\frac{\phi_{\text{total}}^2 \left(\sigma_t^s\right)^2}{2\left(\sigma + \sigma_t^s\right)^3} dt - \phi_{\text{total}} \frac{\sigma_t^s}{\sigma + \sigma_t^s} dZ_t. \tag{J.5}$$

For $\phi_y > 0$, taking $\phi_{\text{vol}} \to \frac{1}{2}$ is equivalent to taking $\phi_y \to \infty$ while holding $\phi_{\text{vol}} = 0$: both make $\phi_{\text{total}} \to \infty$ and recover determinacy. The latter limit corresponds to an impractical infinitely aggressive off-equilibrium response to output-gap deviations (Cochrane, 2007, 2011). Parameter combinations that raise ϕ_{total} speed average convergence to stabilization, but also increase the likelihood of a more severe crisis path over a fixed horizon, as illustrated in Figure 2.

Implementation also requires observing the fundamental and excess-volatility components, or equivalently the risk-adjusted natural rate. Let the observed volatility gap contain a multiplicative measurement error φ_t:

$$\mathcal{E}_t^{\text{obs}} = \varphi_t \cdot \mathcal{E}_t, \quad \varphi_t \neq 1,$$

where $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2$. Even when the authority responds to the observed gap with coefficient $\frac{1}{2}$, the effective coefficient on the true gap differs from $\frac{1}{2}$. Additive measurement errors instead enter as conventional monetary policy shocks and do not alter the exact coefficient on the true gap.

J.3 Comparison with Related Policy Rules

Woodford (2001b, 2003) and Galí (2015) study Taylor rules in the following log-linearized New Keynesian model, abstracting from inflation for comparison:[31]

$$\begin{aligned} \mathbb{E}_t(d\hat{Y}_{t+1}) &= (i_t - r^n)\, dt, \\ i_t &= i_t^* + \phi_y \hat{Y}_t, \quad \text{with} \quad \phi_y > 0, \end{aligned} \tag{J.6}$$

[30] Equations (J.4) and (J.5) follow by the same argument as Proposition 2.

[31] We thank an anonymous referee for suggesting this comparison.

where i_t^* is the monetary authority's target for the policy rate. The standard result is

B.1 If $i_t^* = r^n$, the rule delivers $\hat{Y}_t = 0$ for all t as the unique equilibrium, consistent with Section 2.4. If $i_t^* \neq r^n$, equilibrium remains unique but the output gap is nonzero.[32]

Cúrdia and Woodford (2010, 2016) study a modified log-linearized model in which households face a rate i_t^m that differs from the policy rate:

$$\begin{aligned} \mathbb{E}_t(d\hat{Y}_{t+1}) &= \left(i_t^m - r^n\right) dt, \\ i_t &= i_t^* + \phi_y \hat{Y}_t, \quad \text{with} \quad \phi_y > 0 \\ i_t^m &= i_t + \omega_t, \end{aligned} \tag{J.7}$$

where ω_t is the gap between the policy rate and the rate entering the IS relation. In this literature, ω_t is exogenous or endogenous only to first order.[33]

C.1 The linear system (J.7) has a unique equilibrium under the Taylor principle.

C.2 Setting $i_t^* = r^n - \omega_t$ yields $\hat{Y}_t = 0$ for all t as the unique equilibrium.[34] If $i_t^* = r^n - \phi_w \omega_t$ with $\phi_w < 1$, the equilibrium output gap remains nonzero even as stronger responses reduce its reaction to a shock in ω_t.

Our global setting differs because $i_t^m - i_t$ depends on the endogenous volatility of the output-gap process. With $r_t^T \equiv r^n + (i_t - i_t^m)$ as in equation (13), write

$$\begin{aligned} \mathbb{E}_t(d\hat{Y}_{t+1}) &= \left(i_t^m - r^n\right) dt, \\ i_t &= i_t^* + \phi_y \hat{Y}_t, \quad \text{with} \quad \phi_y > 0, \\ \omega_t &= i_t^m - i_t = \frac{1}{2}\left[(\sigma + \sigma_t^s)^2 - \sigma^2\right]. \end{aligned} \tag{J.8}$$

Here ω_t is an endogenous higher-order moment, not an exogenous or first-order wedge.[35] The comparison yields

[32] As usual, we assume that $i_t^* - r^n$ is not itself a multiple of $\hat{Y}_t$.

[33] In Cúrdia and Woodford (2010, 2016), ω_t is proportional to credit spreads generated by financial intermediation frictions.

[34] The corresponding policy rule is $i_t = r^n + \phi_y \hat{Y}_t - \omega_t$.

[35] The wedge plays a role similar to a financial friction in reducing aggregate demand (Cúrdia and Woodford, 2010, 2016), but it is determined globally as part of the equilibrium.

D.1 If $i_t^* = r_t^T = r^n - \omega_t$, the full-stabilization path is the unique equilibrium. This parallels C.2 but requires the exact rule in (J.1).

D.2 If $i_t^* \neq r_t^T$, the rule does not generally ensure uniqueness, and the martingale or Ornstein-Uhlenbeck equilibria described in Sections 3.1 and 3.2 may arise.

D.3 Because ω_t depends only on the volatility gap, a knife-edge specification in which $i_t^* + \omega_t$ is independent of excess volatility can still yield uniqueness away from full stabilization.

On- and off-equilibrium objects Aggregate volatility has two distinct components. While fundamental volatility σ is exogenous and comes from the technology process, excess volatilit σ_t^s arises when households coordinate on intertemporal contingent consumption plans. Both objects are rational expectations equilibrium quantities.

The literature has also considered perceived volatility arising from sentiment (Caballero and Simsek, 2020). Perceived risk can change even when fundamentals remain unchanged, with the policy rate responding within a given equilibrium. We abstract from this channel.

Fundamental and perceived volatility are on-equilibrium-path objects as they belong to primitives or beliefs, and the interest rate responds to them within the equilibrium. Self-fulfilling excess volatility is instead deterred by off-equilibrium commitments. It can be ruled out by the exact threat in rule (J.1) or by the behavioral and fiscal selection mechanisms in Section 4. Without those devices, a conventional Taylor rule continues to admit the additional bounded equilibria.

K Model with Sticky Prices à la Rotemberg (1982)

In this section, we derive a New Keynesian Phillips curve based on nominal rigidities à la Rotemberg (1982). First, we present the equilibrium conditions when firms are monopolistically competitive as in a canonical New Keynesian model (Woodford, 2003).

K.1 Equilibrium Conditions

Firm i setting its price p_t^i when p_t is the price aggregator faces the demand given by

$$D(p_t^i, p_t) = \left(\frac{p_t^i}{p_t}\right)^{-\epsilon} Y_t. \tag{K.1}$$

Other equilibrium conditions do not change. The optimality conditions for households are given by

$$\frac{1}{p_t C_t} = \frac{L_t^{1/\eta}}{w_t}, \tag{K.2}$$

and

$$\frac{dY_t}{Y_t} = \underbrace{\left(i_t + (\sigma + \sigma_t^s)^2 - \pi_t - \rho\right)}_{\equiv \mu_t^Y} dt + (\sigma + \sigma_t^s) dZ_t. \tag{K.3}$$

In equilibrium, $Y_t = C_t = A_t L_t$ hold. Natural output Y_t^n is given by

$$Y_t^n = A_t \left(\frac{\epsilon - 1}{\epsilon}\right)^{\frac{\eta}{\eta+1}}, \tag{K.4}$$

and output gap $\hat{Y}_t$ is similarly defined as follows:

$$\hat{Y}_t = \log\left(\frac{Y_t}{Y_t^n}\right). \tag{K.5}$$

Combining equations (K.2) to (K.5), we obtain:

$$\frac{w_t}{p_t A_t} = \left(\frac{\epsilon - 1}{\epsilon}\right) e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}. \tag{K.6}$$

Using equations (K.3)–(K.5) and the technology process (8) we can obtain an expression for the Is equation with inflation as:

$$d\hat{Y}_t = \left[i_t - \pi_t - r_t^T\right] dt + \sigma_t^s dZ_t. \tag{K.7}$$

K.2 Firm Price Setting

The price p_t^i of an individual firm i evolves according to:

$$dp_t^i = \pi_t^i p_t^i dt. \tag{K.8}$$

From firm i's perspective, the current price p_t^i is taken as given and acts as a state variable when solving for π_t^i, which the firm controls at each point t.

Firms face price nominal rigidities à la Rotemberg (1982), i.e., they need to pay convex adjustment costs whenever $\pi_t^i \neq 0$, given by:

$$\Theta(\pi_t^i) = \frac{\tau}{2}(\pi_t^i - \bar{\pi})^2 p_t Y_t.$$

where $\bar{\pi}$ denotes the trend inflation rate, or equivalently, the long-run inflation target.[36] For simplicity, we assume that this costs are rebated to the representative household in a lump-sum fashion, so no output ends up "missing" in the final equilibrium.

Therefore, nominal firm profits Ψ_t^i at time t are given by:

$$\begin{aligned}
\Psi_t^i &= \left[p_t^i - \frac{w_t}{A_t}\right] D(p_t^i, p_t) - \Theta(\pi_t^i) \\
&= p_t Y_t \left[\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} - \frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-\epsilon} - \frac{\tau}{2}(\pi_t^i - \bar{\pi})^2\right].
\end{aligned}$$

Let S_t^i be a vector that contains the state variables of the firm's dynamic optimization problem. Therefore, it contains the current price of the firm, $p_t^i \in S_t^i$. We can express this vector of states as following an Ito process of the form:

$$dS_t^i = \mu_t^{S,i} dt + G_t^{S,i} dZ_t,$$

[36] Section 5.1 assumes $\bar{\pi} = 0$ while Section 5.2 explicitly incorporates $\bar{\pi}$.

where $\mu_t^{S,i}$ and $G_t^{S,i}$ are vectors containing the drift and stochastic components of the states. For example, $\pi_t^i p_t^i$ is a component of vector $\mu_t^{S,i}$.

We define the value function of firm i, denoted $V(S_t^i)$, as the present discounted sum of the real profits of the firm on the optimal path, formally:

$$\begin{aligned} V(S_t^i) &= \max_{\{\pi_s^i\}_{s\geq t}} \mathbb{E}_t \int_t^\infty \frac{\xi_s^r}{\xi_t^r}\frac{\Psi_s^i}{p_s} ds \\ &= \max_{\{\pi_s^i\}_{s\geq t}} \mathbb{E}_t \int_t^\infty e^{-\rho(s-t)}\frac{C_t}{C_s} Y_s \left[\left(\frac{p_s^i}{p_s}\right)^{1-\epsilon} - \frac{w_s}{p_s A_s}\left(\frac{p_s^i}{p_s}\right)^{-\epsilon} - \frac{\tau}{2}(\pi_s^i - \bar{\pi})^2\right] ds, \end{aligned}$$

where $\xi_t^r = e^{-\rho t}\frac{1}{C_t}$ is the (real) state price density. We write the Hamilton-Jacobi-Bellman (HJB) equation of the firm's problem as:

$$\rho V(S_t^i) = \max_{\{\pi_s^i\}_{s\geq t}} \left\{ Y_t \left[\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} - \frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-\epsilon} - \frac{\tau}{2}(\pi_t^i - \bar{\pi})^2\right] + \frac{E_t\left[dV(S_t^i)\right]}{dt} \right\}. \quad \text{(K.9)}$$

Based on Ito's lemma, we can express the process for $V(S_t^i)$ as:

$$dV(S_t^i) = \left[(\nabla_S V)^T \mu_t^{S,i} + \frac{1}{2} Tr\left[\left(G_t^{S,i}\right)^T (H_S V)\, G_t^{S,i}\right]\right] dt + (\nabla_S V)^T G_t^{S,i} dZ_t,$$

where $\nabla_S V$ and $H_S V$ are the gradient and Hessian of the value function V with respect to the vector of states S_t^i, respectively.

This allows to alternatively write the value function as:

$$\begin{aligned} \rho V(S_t^i) = \max_{\{\pi_s^i\}_{s\geq t}} Y_t &\left[\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} - \frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-\epsilon} - \frac{\tau}{2}(\pi_t^i - \bar{\pi})^2\right] + (\nabla_S V)^T \mu_t^{S,i} \\ &+ \frac{1}{2} Tr\left[\left(G_t^{S,i}\right)^T (H_S V)\, G_t^{S,i}\right]. \end{aligned} \quad \text{(K.10)}$$

Computing the first-order condition of equation (K.10) with respect to π_s^i, we obtain:

$$\frac{\partial V(S_t^i)}{\partial p_t^i} = \tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}. \quad \text{(K.11)}$$

Taking derivative of the HJB equation in (K.9) evaluated at the optimum with respect to the price of the individual firm, we obtain:

$$\rho \frac{\partial V_t^i}{\partial p_t^i} = \left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\left(\frac{1}{\pi_t^i - \bar{\pi}}\right)\left(\frac{\epsilon - 1}{\tau}\right) \times \left[\left(\frac{\epsilon}{\epsilon - 1}\right)\frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-1} - 1\right]\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} + \frac{E_t\left[d\left(\frac{\partial V_t^i}{\partial p_t^i}\right)\right]}{dt}. \tag{K.12}$$

Plugging equation (K.11) into (K.12), we obtain:

$$\rho \left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right) = \left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\left(\frac{1}{\pi_t^i - \bar{\pi}}\right)\left(\frac{\epsilon - 1}{\tau}\right) \times \left[\left(\frac{\epsilon}{\epsilon - 1}\right)\frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-1} - 1\right]\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} + \frac{E_t\left[d\left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\right]}{dt}. \tag{K.13}$$

Let us assume that individual firm inflation π_t^i follows a process of the form:

$$d\pi_t^i = \mu_t^{\pi,i} dt + \sigma_t^{\pi,i} dZ_t. \tag{K.14}$$

Based on Ito's lemma, together with equations (K.3), (K.8) and (K.14), we obtain:

$$d\left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right) = \left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\left[\mu_t^Y + \frac{\mu_t^{\pi,i}}{\pi_t^i - \bar{\pi}} - \pi_t^i + \frac{\sigma_t^{\pi,i}}{\pi_t^i - \bar{\pi}}(\sigma + \sigma_t^s)\right] dt + \left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\left(\sigma + \sigma_t^s + \frac{\sigma_t^{\pi,i}}{\pi_t^i - \bar{\pi}}\right) dZ_t.$$

Taking expectations and using the fact that $\mu_t^{\pi,i} dt = \mathbb{E}_t\left[d\pi_t^i\right]$, we obtain:

$$\frac{\mathbb{E}_t\left[d\left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\right]}{dt} = \left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right) \times \left[\frac{\mathbb{E}_t\left[d\pi_t^i\right]}{(\pi_t^i - \bar{\pi})dt} + \mu_t^Y - \pi_t^i + \frac{\sigma_t^{\pi,i}}{\pi_t^i - \bar{\pi}}(\sigma + \sigma_t^s)\right]. \tag{K.15}$$

Plugging equation (K.15) into equation (K.13) and rearranging, we obtain:

$$\begin{aligned}\rho\left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right) = &\left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\left(\frac{1}{\pi_t^i - \bar{\pi}}\right)\left(\frac{\epsilon - 1}{\tau}\right) \\ &\times \left[\left(\frac{\epsilon}{\epsilon - 1}\right)\frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-1} - 1\right]\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} \\ &+ \left(\tau Y_t \frac{\pi_t^i - \bar{\pi}}{p_t^i}\right)\left[\frac{\mathbb{E}_t\left[d\pi_t^i\right]}{(\pi_t^i - \bar{\pi})dt} + \mu_t^Y - \pi_t^i + \frac{\sigma_t^{\pi,i}}{\pi_t^i - \bar{\pi}}(\sigma + \sigma_t^s)\right],\end{aligned}$$

which becomes

$$\begin{aligned}\mathbb{E}_t\left[d\pi_t^i\right] = \Bigg[&\left[(\rho + \bar{\pi}) + (\pi_t^i - \bar{\pi}) - \mu_t^Y\right](\pi_t^i - \bar{\pi}) \\ &- \sigma_t^{\pi,i}(\sigma + \sigma_t^s) \\ &- \left(\frac{\epsilon - 1}{\tau}\right)\left[\left(\frac{\epsilon}{\epsilon - 1}\right)\frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-1} - 1\right]\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon}\Bigg]dt. \end{aligned} \tag{K.16}$$

Under the symmetric equilibrium, i.e., $p_t^i = p_t, \forall i$, $\pi_t^i = \pi_t, \forall i$, and $\sigma_t^{\pi,i} = \sigma_t^{\pi}, \forall i$, equation (K.16) becomes:

$$\begin{aligned}\mathbb{E}_t\left[d\pi_t\right] = \Big[&\left[(\rho + \bar{\pi}) + (\pi_t - \bar{\pi}) - \mu_t^Y\right](\pi_t - \bar{\pi}) - \sigma_t^{\pi}(\sigma + \sigma_t^s) \\ &- \left(\frac{\epsilon - 1}{\tau}\right)\left[\left(\frac{\epsilon}{\epsilon - 1}\right)\frac{w_t}{p_t A_t} - 1\right]\Big] dt. \end{aligned} \tag{K.17}$$

Finally, we plug equation (K.6) into (K.17) and obtain

$$\mathbb{E}_t\left[d\pi_t\right] = \Big[\left[(\rho+\bar{\pi}) + (\pi_t - \bar{\pi}) - \mu_t^Y\right](\pi_t - \bar{\pi}) - \sigma_t^\pi(\sigma + \sigma_t^s) + \left(\frac{\epsilon - 1}{\tau}\right)\left[1 - e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}\right]\Big]dt. \tag{K.18}$$

which is the New Keynesian Phillips curve in our environment. We can further replace the expression for μ_t^Y in (K.3) into (K.18), and use equation (K.14) to obtain:

$$\begin{aligned} d\pi_t = \Bigg[&\left[2(\rho+\bar{\pi}) + 2(\pi_t - \bar{\pi}) - i_t - (\sigma+\sigma_t^s)^2\right](\pi_t - \bar{\pi}) \\ &- \sigma_t^\pi(\sigma+\sigma_t^s) + \left(\frac{\epsilon-1}{\tau}\right)\left[1 - e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}\right]\Bigg]dt \\ &+ \sigma_t^\pi dZ_t. \end{aligned} \tag{K.19}$$

Interpretation Equation (K.18) can be interpreted as follows. An increase in μ_t^Y prompts households to borrow against the future and raise current consumption, leading to higher aggregate demand and higher current inflation while reducing $d\pi_t$ on average. Similarly, a rise in current income, $\hat{Y}_t$, increases inflation now and lowers $d\pi_t$ on average.

Moreover, a higher $\sigma_t^\pi(\sigma + \sigma_t^s)$ implies that inflation is typically higher when aggregate output is higher too. It raises the expected marginal price-adjustment cost (see (K.11)) in the future (after dt period), inducing firms to raise inflation now and leading to higher $d\pi_t$ on average.

K.3 Linearization

Assume the extended Taylor rule

$$i_t = r^n + \phi_y \hat{Y}_t + \phi_\pi \pi_t - \phi_{vol}\left[(\sigma+\sigma_t^s)^2 - \sigma^2\right]. \tag{K.20}$$

This is the sticky-price counterpart of the rigid-price rule (A.1). Plugging (K.20) into the Phillips curve (K.19), we obtain:

$$d\pi_t = \Bigg[\left[\rho - g + (2-\phi_\pi)\pi_t - \phi_y \hat{Y}_t - (1-\phi_{vol})\left[(\sigma+\sigma_t^s)^2 - \sigma^2\right]\right](\pi_t - \bar{\pi}) \\ - \sigma_t^\pi(\sigma+\sigma_t^s) + \left(\frac{\epsilon-1}{\tau}\right)\left[1 - e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}\right]\Bigg] dt + \sigma_t^\pi dZ_t, \tag{K.21}$$

We assume $\bar{\pi} = 0$ and linearly approximate the drift in (K.21) around the zero-inflation and zero-output gap steady state, i.e., $\pi_t \simeq 0$ and $\hat{Y}_t \simeq 0$. In this approximation, we ignore deviations in terms involving higher-order moments, such as inflation volatility σ_t^π and excess volatility σ_t^s. These terms must nevertheless be evaluated at an expansion point. We can therefore set $\sigma_t^\pi \simeq 0$ and $\sigma_t^s \simeq 0$, which ensures that the inflation drift in (K.21) is zero when $(\pi_t, \hat{Y}_t, \sigma_t^s, \sigma_t^\pi) = (0,0,0,0)$. This leads to the continuous-time New Keynesian Phillips curve in (31):

$$\mathbb{E}_t(d\pi_t) = \left(\kappa^\pi \pi_t - \kappa^y \hat{Y}_t\right) dt, \tag{K.22}$$

where the constants are defined as

$$\kappa^\pi = \rho - g, \quad \text{and} \quad \kappa^y = \left(\frac{\epsilon-1}{\tau}\right)\left(\frac{\eta+1}{\eta}\right).$$

This expression is consistent with the continuous-time literature[37] (e.g., Werning (2012)) and is equivalent to a discrete-time formulation (e.g., Galí (2015)).

The constructions of Section 5.1 extend under (K.20). Define

$$\phi_{total} \equiv \frac{\phi}{1-2\phi_{vol}} = \frac{1}{1-2\phi_{vol}}\left(\phi_y + \frac{\kappa^y}{\kappa^\pi}(\phi_\pi - 1)\right) \tag{K.23}$$

and

$$\phi_2 \equiv \frac{\phi_y + \theta + (\phi_\pi - 1)\frac{\kappa^y}{\kappa^\pi+\theta}}{1-2\phi_{vol}}, \tag{K.24}$$

[37] Most textbook treatments assume $g = 0$ for fundamental shocks, which explains the presence of g in our New Keynesian Phillips curve.

which satisfy $\phi_{total} > 0$ and $\phi_2 > 0$ when $\phi_{vol} < \frac{1}{2}$ and the Taylor principle (34) holds with $\phi_\pi > 1$. The associated output gap and inflation in the Ornstein-Uhlenbeck equilibrium are

$$\hat{Y}_t = \mu - \frac{1}{2\phi_2}\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right] \tag{K.25}$$

and

$$\pi_t = \frac{\kappa^y}{\kappa^\pi}\mu - \frac{1}{2\phi_2}\left(\frac{\kappa^y}{\kappa^\pi + \theta}\right)\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]. \tag{K.26}$$

The long-run expected excess variance is $-2\mu\phi_{total}$. When $\phi_{vol} = 0$, ϕ_{total} reduces to ϕ in (34) and ϕ_2 reduces to (43). At $\phi_{vol} = \frac{1}{2}$, the extended rule offsets the precautionary-saving feedback from excess variance and can select the full-stabilization path as the unique bounded equilibrium; any departure from that exact half response restores multiplicity. Section J studies this knife-edge case. Away from it, the qualitative results remain the same.

K.4 Linearization with long-run targets

We consider an alternative monetary policy rule with explicit long-run targets. Let aggregate output variance be defined as $x_t \equiv (\sigma + \sigma_t^s)^2$. The policy rule is

$$i_t = r^T + \bar{\pi} + \phi_\pi(\pi_t - \bar{\pi}) + \phi_y(\hat{Y}_t - \mu) - \phi_{vol}(x_t - \bar{x}), \tag{K.27}$$

where r^n is the natural rate and r^T is the risk-adjusted steady-state real rate:

$$r^T = r^n - \frac{1}{2}(\bar{x} - \sigma^2) \quad \text{and} \quad r^n = \rho + g - \sigma^2.$$

Here, $\bar{\pi}$ and $\bar{x}$ denote the long-run (steady-state) values of inflation and aggregate output variance and can be treated as free parameters. Under the policy rule (K.27), the long-run nominal rate becomes $r^T + \bar{\pi}$, and policy responses are defined in terms of deviations from corresponding long-run means.

Substituting the monetary policy rule (K.27) into (K.19) results in

$$d\pi_t = \Bigg[\Big[2(\rho + \bar{\pi}) - (r^T + \bar{\pi} + \bar{x}) + (2 - \phi_\pi)(\pi_t - \bar{\pi}) \\ - \phi_y(\hat{Y}_t - \mu) - (1 - \phi_{vol})(x_t - \bar{x}) \Big] (\pi_t - \bar{\pi}) \\ - \sigma_t^\pi \sqrt{x_t} + \left(\frac{\epsilon - 1}{\tau} \right) \left[1 - e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t} \right] \Bigg] dt + \sigma_t^\pi dZ_t, \tag{K.28}$$

We linearly approximate the drift in (K.28) around the long-run targets for inflation and the output gap, $\bar{\pi}$ and μ, respectively. In doing so, we ignore deviations in terms involving higher-order moments, such as inflation volatility σ_t^π and aggregate output variance x_t. Nevertheless, these terms must be evaluated at an expansion point. We therefore expand x_t around $\bar{x}$ and σ_t^π around $\overline{\sigma^\pi}$. We set $\overline{\sigma^\pi}$ so that the inflation drift in (K.28) is equal to zero when π_t, $\hat{Y}_t$, x_t, and σ_t^π are evaluated at their expansion points $(\bar{\pi}, \mu, \bar{x}, \overline{\sigma^\pi})$:

$$\overline{\sigma^\pi} = \left(\frac{\epsilon - 1}{\tau\sqrt{\bar{x}}} \right) \left[1 - e^{\left(\frac{\eta+1}{\eta}\right)\mu} \right].$$

This results in a linearized Phillips curve given by

$$\mathbb{E}_t d(\pi_t - \bar{\pi}) = \Big(\tilde{\kappa}^\pi (\pi_t - \bar{\pi}) - \tilde{\kappa}^y (\hat{Y}_t - \mu) \Big) dt, \tag{K.29}$$

where the constants are defined as

$$\tilde{\kappa}^\pi = \rho + \bar{\pi} - g - \frac{1}{2}(\bar{x} - \sigma^2),$$
$$\tilde{\kappa}^y = \left(\frac{\epsilon - 1}{\tau} \right) \left(\frac{\eta + 1}{\eta} \right) e^{\left(\frac{\eta+1}{\eta}\right)\mu}.$$

Despite differences in the expansion points and in the constants multiplying inflation and the output gap, equations (K.22) and (K.29) share the same functional form, and they coincide when $\bar{\pi} = 0$, $\mu = 0$, and $\bar{x} = \sigma^2$.

K.5 Inflation Dynamics with Non-linear New Keynesian Phillips Curve

This appendix characterizes equilibria in which inflation follows the exact nonlinear New Keynesian Phillips curve (K.19). We use a policy rule expressed around the long-run targets $(\bar{\pi}, \bar{x})$, as described in Online Appendix (K.4), to match data moments more precisely during numerical simulation. We focus on equilibria in which $\sigma_t^s \neq 0$ along the equilibrium path.

K.5.1 Definitions and state dynamics

We define *aggregate output variance* as $x_t \equiv (\sigma + \sigma_t^s)^2$, where $\sigma > 0$ is the baseline volatility parameter and σ_t^s is the *excess volatility* component. By construction, $x_t \in [\sigma^2, +\infty)$.

Let the process for x_t be

$$\mathrm{d}x_t = \mu_t^x \mathrm{d}t + \sigma_t^x \mathrm{d}Z_t, \tag{K.30}$$

where Z_t is a Brownian motion and $\mu_t^x \equiv \mu^x(x_t)$ and $\sigma_t^x \equiv \sigma^x(x_t)$ are endogenous functions of the state.

Excess volatility is recovered from x_t via

$$\sigma_t^s = \sqrt{x_t} - \sigma.$$

Stationary density Assuming the diffusion defined above admits an invariant (i.e., stationary) density, it is given up to normalization by

$$n(x) \propto \frac{1}{\left(\sigma^x(x)\right)^2} \exp\left\{ \int_{\sigma^2}^{x} \frac{2\mu^x(y)}{\left(\sigma^x(y)\right)^2} \mathrm{d}y \right\}.$$

Define the normalized stationary density $\tilde{n}(x)$, satisfying $\int_{\sigma^2}^{+\infty} \tilde{n}(x)\mathrm{d}x = 1$, by

$$\tilde{n}(x) = \frac{n(x)}{\int_{\sigma^2}^{+\infty} n(y)\mathrm{d}y}.$$

Once a candidate policy function $f(\cdot)$ is obtained, the implied functions $\mu^x(\cdot)$

and $\sigma^x(\cdot)$ can be derived. The stationary density is then used to evaluate the unconditional moments that appear in the boundary conditions (Online Appendix K.6.1).

K.5.2 Model Derivations

Substituting the monetary policy rule with long-run targets (K.27) into the IS equation (K.7) gives

$$\begin{aligned} d\hat{Y}_t = &\left[\phi_y(\hat{Y}_t - \mu) + (\phi_\pi - 1)(\pi_t - \bar{\pi}) + \left(\frac{1}{2} - \phi_{vol}\right)(x_t - \bar{x})\right] dt \\ &+ (\sqrt{x_t} - \sigma) dZ_t. \end{aligned} \tag{K.31}$$

As in the baseline case with fully rigid prices, we conjecture a solution to the system of equations given by (K.28) and (K.31), and then verify numerically that the conjecture is internally consistent. Because the state vector includes inflation, we impose an additional restriction linking inflation to total variance. Formally, we conjecture:

$$d\hat{Y}_t = -\theta\left[\hat{Y}_t - \mu\right] dt + (\sqrt{x_t} - \sigma) dZ_t, \quad \theta \geq 0 \text{ and } \mu \leq 0, \tag{K.32}$$

$$\pi_t = f(x_t), \tag{K.33}$$

where equation (K.32) is an Ornstein–Uhlenbeck specification for the output gap and equation (K.33) posits that inflation is a smooth function of total variance. The goal is to find a smooth function $f(\cdot)$ such that (K.32) and (K.33) jointly satisfy the equilibrium conditions.

Combining equations (K.31), (K.32), and (K.33) yields an expression for the output gap:

$$\hat{Y}_t = \mu - \left(\frac{1}{\theta + \phi_y}\right)\left[(\phi_\pi - 1)(f(x_t) - \bar{\pi}) + \left(\frac{1}{2} - \phi_{vol}\right)(x_t - \bar{x})\right]. \tag{K.34}$$

Substituting equations (K.34) and (K.33) into (K.28) and (K.32) yields:

$$\begin{aligned} d\pi_t &= \mu_t^\pi\, dt + \sigma_t^\pi dZ_t, \\ \mu_t^\pi &\equiv \big[\psi_1 + \psi_2(f(x_t) - \bar{\pi}) - \psi_3(x_t - \bar{x})\big](f(x_t) - \bar{\pi}) \\ &\qquad - \sigma_t^\pi \sqrt{x_t} \\ &\qquad + \psi_4\big[1 - e^{\psi_5 - \psi_6(f(x_t) - \bar{\pi}) - \psi_7(x_t - \bar{x})}\big]. \end{aligned} \tag{K.35}$$

and

$$\begin{aligned} d\hat{Y}_t &= \left(\frac{\theta}{\theta + \phi_y}\right)\left[(\phi_\pi - 1)(f(x_t) - \bar{\pi}) + \left(\frac{1}{2} - \phi_{vol}\right)(x_t - \bar{x})\right] dt \\ &\quad + (\sqrt{x_t} - \sigma) dZ_t, \end{aligned} \tag{K.36}$$

where the constants are:

$$\begin{aligned} \psi_1 &= 2(\rho + \bar{\pi}) - (r^T + \bar{\pi} + \bar{x}), & \psi_5 &= \left(\frac{\eta + 1}{\eta}\right)\mu, \\ \psi_2 &= 1 - \left(\frac{\theta}{\theta + \phi_y}\right)(\phi_\pi - 1), & \psi_6 &= \left(\frac{\eta + 1}{\eta}\right)\left(\frac{\phi_\pi - 1}{\theta + \phi_y}\right), \\ \psi_3 &= \left(\frac{\theta}{\theta + \phi_y}\right)(1 - \phi_{vol}) + \left(\frac{\phi_y}{\theta + \phi_y}\right)\frac{1}{2}, & \psi_7 &= \left(\frac{\eta + 1}{\eta}\right)\left(\frac{1}{\theta + \phi_y}\right)\left(\frac{1}{2} - \phi_{vol}\right). \\ \psi_4 &= \left(\frac{\epsilon - 1}{\tau}\right), \end{aligned}$$

Equation (K.34) can be written as $N(\hat{Y}_t, x_t) = 0$, where:

$$N(\hat{Y}_t, x_t) \equiv \hat{Y}_t - \mu + \left(\frac{1}{\theta + \phi_y}\right)\left[(\phi_\pi - 1)(f(x_t) - \bar{\pi}) + \left(\frac{1}{2} - \phi_{vol}\right)(x_t - \bar{x})\right],$$

which implicitly writes x_t as a function of $\hat{Y}_t$. To compute the implicit derivatives, note that $N_{\hat{Y}} = 1, N_{\hat{Y}\hat{Y}} = 0, N_{\hat{Y}x} = 0$, and:

$$N_x = \left(\frac{1}{\theta + \phi_y}\right)\left[(\phi_\pi - 1)f'(x_t) + \left(\frac{1}{2} - \phi_{vol}\right)\right] \quad \text{and} \quad N_{xx} = \left(\frac{\phi_\pi - 1}{\theta + \phi_y}\right) f''(x_t),$$

which implies the first implicit derivative of x_t with respect to $\hat{Y}_t$:

$$\frac{\partial x_t}{\partial \hat{Y}_t} = -\left[\frac{\theta + \phi_y}{(\phi_\pi - 1)f'(x_t) + \left(\frac{1}{2} - \phi_{vol}\right)}\right], \tag{K.37}$$

and the second-order implicit derivative as:

$$\frac{\partial^2 x_t}{\partial^2 \hat{Y}_t} = -\left[\frac{(\theta + \phi_y)^2(\phi_\pi - 1)f''(x_t)}{\left[(\phi_\pi - 1)f'(x_t) + \left(\frac{1}{2} - \phi_{vol}\right)\right]^3}\right]. \tag{K.38}$$

Using Itô's lemma together with (K.36), (K.37), and (K.38), we obtain the implied drift and diffusion terms for the law of motion for x_t in (K.30):

$$\begin{aligned}
\mu_t^x &= -\left[\frac{\psi_8(f(x_t) - \bar{\pi}) + \psi_9(x_t - \bar{x})}{\psi_{12}f'(x_t) + \psi_{13}}\right. \\
&\qquad \left. + \psi_{10}\frac{f''(x_t)(\sqrt{x_t} - \sigma)^2}{\left[\psi_{12}f'(x_t) + \psi_{13}\right]^3}\right], \\
\sigma_t^x &= -\left[\frac{\psi_{11}(\sqrt{x_t} - \sigma)}{\psi_{12}f'(x_t) + \psi_{13}}\right],
\end{aligned}$$

with constants defined as

$$\begin{aligned}
\psi_8 &= \theta(\phi_\pi - 1), & \psi_{11} &= \theta + \phi_y, \\
\psi_9 &= \theta\left(\tfrac{1}{2} - \phi_{vol}\right), & \psi_{12} &= \phi_\pi - 1, \\
\psi_{10} &= \tfrac{(\theta+\phi_y)^2(\phi_\pi - 1)}{2}, & \psi_{13} &= \tfrac{1}{2} - \phi_{vol}.
\end{aligned}$$

Applying Itô's lemma to equation (K.33) and using equation (K.30) yields:

$$d\pi_t = \left[\mu_t^x f'(x_t) + \frac{(\sigma_t^x)^2}{2}f''(x_t)\right]dt + \sigma_t^x f'(x_t)dZ_t \tag{K.39}$$

Matching the diffusion terms in equation (K.35) and equation (K.39) gives:

$$\sigma_t^\pi = -\left[\frac{\psi_{11}f'(x_t)(\sqrt{x_t} - \sigma)}{\psi_{12}f'(x_t) + \psi_{13}}\right], \tag{K.40}$$

which can be used in the drift of (K.35) to obtain:

$$\mu_t^{\pi} = \left[\psi_1 + \psi_2(f(x_t) - \bar{\pi}) - \psi_3(x_t - \bar{x})\right](f(x_t) - \bar{\pi}) + \sqrt{x_t}\left[\frac{\psi_{11} f'(x_t)(\sqrt{x_t} - \sigma)}{\psi_{12} f'(x_t) + \psi_{13}}\right] \\ + \psi_4\left[1 - e^{\psi_5 - \psi_6(f(x_t) - \bar{\pi}) - \psi_7(x_t - \bar{x})}\right], \tag{K.41}$$

In equilibrium, the drift terms in (K.35) and (K.39) must coincide. Equating the drift in equation (K.39) with equation (K.41) yields the following non-linear second-order ordinary differential equation (ODE):

$$\mu_t^{\pi} = \mu_t^{x} f'(x_t) + \frac{(\sigma_t^{x})^2}{2} f''(x_t). \tag{K.42}$$

Solving the second-order ODE equation (K.42) requires two additional boundary conditions to pin down a solution. In practice, multiple solutions may exist; the numerical procedure in Online Appendix K.6.1 selects among them by imposing moment restrictions and feasibility constraints.

K.6 Numerical Solution and Simulation

This appendix describes the numerical solution algorithm, numerical robustness tests used to validate the solution routines, and the simulation algorithm.

K.6.1 Numerical solution algorithm

The ODE equation (K.42) has two free boundary conditions. We parameterize them by the level and slope of the policy function at the steady state, $(f(\bar{x}), f'(\bar{x}))$, and choose this pair to minimize the moment loss:

$$\begin{aligned} m_1 &= \left[\mathbb{E}\left[f(x_t)\right] - \bar{\pi}\right]^2, \\ m_2 &= \left[\mathbb{E}\left[f'(x_t)\right] - \left[-\frac{1}{2\check{\phi}_2}\left(\frac{\tilde{\kappa}^{y}}{\tilde{\kappa}^{\pi} + \theta}\right)\right]\right]^2, \end{aligned} \tag{K.43}$$

where $\mathbb{E}[\cdot]$ denotes an unconditional expectation operator under the stationary distribution of x_t, numerically approximated using the density in Online Appendix K.5.1. The targets $\bar{\pi}$ and $-\frac{1}{2\check{\phi}_2}\left(\frac{\tilde{\kappa}^{y}}{\tilde{\kappa}^{\pi}+\theta}\right)$ are the steady-state inflation level and slope implied by the linearized model of Section O. Matching these moments enforces local consistency of the non-linear solution around $\bar{x}$.

For a wide range of candidate boundary conditions, the implied diffusion coefficient, $\sigma^x(x)$, approaches zero at finite points on both sides of $\bar{x}$. Let $x^{\min}$ and $x^{\max}$ denote the corresponding lower and upper endpoints. We use these points as endpoints of the numerical domain for both the solution and the simulation.

Informally, a boundary is called *natural* (i.e., unattainable) if the process cannot reach it in finite time from an interior point. Operationally, we use a practical criterion consistent with this intuition: we keep domains where drift points inward at both endpoints. Since the diffusion coefficient approaches zero at the boundaries, trajectories are pushed back toward the interior and remain within the numerical domain.

Accordingly, we impose the following endpoint restrictions on the drift $\mu^x(x)$ over the numerical domain:

$$\mu^x_t(x^{\min}) \geq \mu^{x,\min} \quad \text{and} \quad \mu^x_t(x^{\max}) \leq \mu^{x,\max}, \tag{K.44}$$

with user-chosen thresholds $\mu^{x,\min} > 0$ and $\mu^{x,\max} < 0$ (typically close to zero). These inequalities enforce inward drift at $x^{\min}$ and $x^{\max}$. We treat equation (K.44) as hard feasibility constraints: they must hold even when equation (K.43) cannot be matched exactly.

The algorithm proceeds in two steps. First, define the reference pair

$$b_1^{ini} = \bar{\pi} \quad \text{and} \quad b_2^{ini} = -\frac{1}{2\check{\phi}_2}\left(\frac{\tilde{\kappa}^y}{\tilde{\kappa}^\pi + \theta}\right),$$

and construct a coarse grid of candidate boundary conditions $(f(\bar{x}), f'(\bar{x}))$ centered around (b_1^{ini}, b_2^{ini}). At each grid point, we solve the non-linear ODE, compute the implied moments in equation (K.43), and test feasibility via equation (K.44).

Second, among feasible grid points, we choose the pair with the smallest loss and use it as the initial guess for a local constrained optimizer over $(f(\bar{x}), f'(\bar{x}))$. Any candidate that violates equation (K.44) is treated as infeasible (equivalently, assigned infinite loss). Thus, feasibility is enforced strictly, while equation (K.43) is matched in a least-squares sense within the feasible set.

Figure 5 reports the resulting drift and diffusion functions of the total-variance process x_t in equation (K.30), evaluated over the numerical range used in the solution and simulation steps. The figure shows finite lower and upper boundaries for x_t, with diffusion approaching zero at both endpoints. In addition, the drift is weakly positive near the lower boundary and strongly negative near the upper boundary, implying inward mean reversion over the relevant region, consistent with the natural-boundary interpretation discussed above.

Figure 4 plots the mapping from aggregate volatility x_t to inflation π_t and the output gap $\hat{Y}_t$. Note that higher x_t is associated with lower π_t and a lower $\hat{Y}_t$, consistent with the model's precautionary channel under heightened uncertainty.

K.6.2 Numerical simulation

Simulation. Given that the policy function $\pi = f(x)$ on the numerical grid is solved as above, we simulate the scalar state x_t using the model-implied drift and diffusion under an Euler–Maruyama discretization, with parameters calibrated as in Online Appendix L. Starting from $x_0 = \bar{x}$, we iterate

$$x_{t+\Delta t} = x_t + \mu^x(x_t)\Delta t + \sigma^x(x_t)\Delta W_t,$$

and recover $\pi_t = f(x_t)$, the output gap $\hat{Y}_t$, and the excess volatility $\sigma_t^s = \sqrt{x_t} - \sigma$ along the simulated path.

Figure 6 shows a representative path for these key variables: the dynamics are mostly centered around the steady state, with infrequent high-volatility episodes accompanied by strongly negative output-gap realizations and deflation.

Trimmed-moments robustness check. As a robustness exercise, we recompute simulation moments after asymmetric tail trimming to verify that results are not driven by a small number of extreme realizations. Concretely, letting $p \in (0, 1/2)$ denote the trimming share, we retain only observations that satisfy $\pi_t \geq q_p(\pi)$, $\hat{Y}_t \geq q_p(\hat{Y})$, and $\sigma_t^s \leq q_{1-p}(\sigma^s)$, with $p = 0.10$ in our baseline trimming test.

Figure K.2 shows that the qualitative comovement pattern is preserved after trimming, while the largest volatility realization is attenuated. Typical realizations are therefore more moderate still, and the main simulation implications are not driven solely by a small number of tail observations.

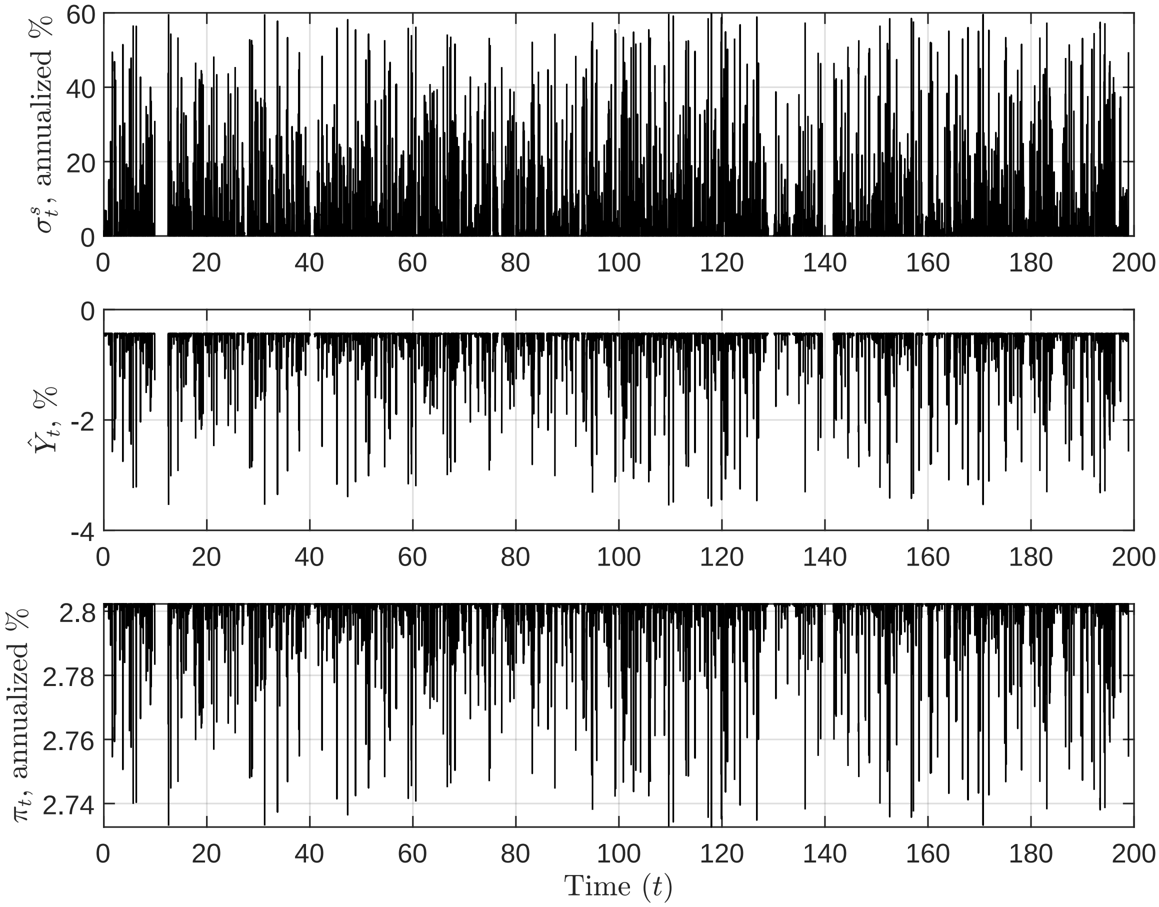

Figure K.2: Trimmed-moments robustness check of the non-linear Rotemberg model. Trimming removes the lower 10% tails of inflation and the output gap and the upper 10% tail of total variance.

K.6.3 Numerical robustness tests

This section summarizes three numerical robustness checks used to validate the numerical solution and simulation:

- *Discretization and boundary-handling sensitivity.* Simulated moments should be stable with respect to the simulation time step Δt. We therefore run Euler–Maruyama simulations over a fixed horizon for multiple step sizes (multiple N, with $\Delta t = 1/N$) and compare summary statistics across discretizations. Because the state x_t process occasionally overshoots the effective upper endpoint when $\sigma^x(x)$ becomes small, we also compare two numerical boundary treatments for such overshoots (reflection back into the domain vs. capping at the endpoint).

- *Right-end boundary vs. numerical stiffness diagnostics.* The ODE solver can stop early on the right (high-x) side either because the continuous-time system approaches a genuine boundary (e.g., $\sigma^x(x) \to 0$ together with inward drift) or because the ODE becomes numerically stiff due to near-singular denominator terms. We diagnose this by repeating the ODE solution under multiple solvers and tolerances and by reporting tail diagnostics for the baseline solution, including endpoint and tail behavior of $\mu^x(x)$ and $\sigma^x(x)$ and explicit checks of denominator expressions that may blow up as the solver approaches the stopping point.

- *A posteriori residual (defect) validation of the ODE solution.* After solving for $\pi = f(x)$ on a grid, we verify that the stored solution is internally consistent with the first-order system used by the ODE solver. Specifically, we compute finite-difference "defects" for both $f'(x)$ and $f''(x)$ using adjacent grid points (with scaled versions that normalize by $1 + |\cdot|$), and construct summary statistics over the full grid and over a right-tail window where numerical issues are most likely. This check detects insufficient grid resolution or loss of accuracy near the boundary or stopping region.

L Parameter calibration

This section summarizes our calibration for the model with sticky prices à la Rotemberg (1982). Table L.2 reports the baseline parameter values and sources.

Standard values are used for common structural parameters, including the household discount rate ρ, the Frisch labor-supply elasticity η, the elasticity of substitution across varieties ϵ, and the monetary policy-rule coefficients ϕ_y and ϕ_π. In the baseline specification, the policy response to volatility is set to $\phi_{vol} = 0$; alternative values are used in exercises that study volatility targeting.

Trend inflation $\bar{\pi}$ is estimated as the average quarterly growth rate of the Personal Consumption Expenditures (PCE) price index over 1959Q1–2019Q1. For nominal rigidities, we calibrate the Rotemberg adjustment-cost parameter τ to be first-order consistent with a Calvo specification. We first choose the Calvo parameter δ based on standard Calvo calibrations in the literature. Using the linearized Phillips-curve expressions under Calvo and Rotemberg pricing in equation (M.29) and equation (K.29), we set the Rotemberg adjustment cost to match the relative contribution of inflation and the output gap in the Phillips curve under the two microfoundations, i.e., $\frac{\tilde{\kappa}^y_{\text{Rotemberg}}}{\tilde{\kappa}^\pi_{\text{Rotemberg}}} = \frac{\tilde{\kappa}^y_{\text{Calvo}}}{\tilde{\kappa}^\pi_{\text{Calvo}}}$. This yields the mapping from δ to the Rotemberg parameter τ as:

$$\tau = \left(\frac{\tilde{\kappa}^\pi_{\text{Calvo}}}{\tilde{\kappa}^y_{\text{Calvo}}}\right)\left(\frac{\epsilon - 1}{\tilde{\kappa}^\pi_{\text{Rotemberg}}}\right)\left(\frac{\eta+1}{\eta}\right)e^{\left(\frac{\eta+1}{\eta}\right)\mu}.$$

TFP growth g is estimated as the average quarterly growth rate of real GDP per capita over 1959Q1–2019Q1. Volatility σ is calibrated as the standard deviation of CBO potential-output growth, consistent with the natural-output growth expression in equation (9). Over the same sample, we measure the output gap as the log deviation of real GDP from CBO potential output. We then infer the steady-state total-variance term, $\bar{x} \equiv (\sigma + \overline{\sigma^s})^2$, from output-gap volatility. Using equation (12), this identifies average excess volatility $\overline{\sigma^s}$ and therefore $\bar{x}$ given σ. As a robustness check, calibrating $\bar{x}$ from real GDP growth volatility (equation equation (10)) yields a similar value (not reported).

Finally, we estimate the output-gap autoregressive component θ and steady-state component μ by OLS regression of the discretized counterpart of equation

equation (22) over the same sample period. The estimated coefficients are reported in Table L.1:

	Coefficient	Standard Error
θ	0.064755***	0.022938
μ	-0.004716	0.007763

Table L.1: Estimated output-gap autoregressive and steady-state components. The sample period is 1959Q1–2019Q1. *** indicates significance at the 1% level.

Two points are worth noting. First, the estimate of the steady-state component of output gap μ is negative (about -0.5%), which is consistent with the model specification, but it is not statistically significant. We use this estimate as the baseline value for μ in the comparative statics to illustrate the properties of the Ornstein–Uhlenbeck process for the output gap. Second, the estimate of θ is statistically significant at the 1% level but quantitatively small, implying dynamics close to a martingale. In practice, such a low value generates very large fluctuations in the output gap, inflation, and excess volatility, likely reflecting the parsimonious structure of the model, which does not include the frictions and persistence mechanisms of a medium-scale DSGE model.[38] Therefore, we set $\theta = 0.95$, which implies a more autoregressive output-gap process and produces more realistic dynamics for the variables of interest in Online Appendix K.6.

	Parameter Description	Value	Reference
ρ	Household discount rate	0.01	Standard value (Iacoviello, 2005); implies a 4% annualized rate.
η	Frisch labor supply elasticity	1	Standard value; see Peterman (2016).
g	TFP growth	0.004960	Estimated from 1959Q1–2019Q1 real GDP per capita quarterly growth.

[38]Alternatively, one could argue that the OLS estimates in Table L.1 reflect the modeling assumptions embedded in the CBO procedure used to construct potential output, which may not be fully consistent with the model considered here.

σ	TFP volatility	0.002179	Estimated from 1959Q1–2019Q1 CBO potential-output growth volatility; see equation (9).
$\overline{\sigma^s}$	Steady-state excess volatility	0.007771	Estimated from 1959Q1–2019Q1 output-gap volatility; see equation (12).
ϵ	Elasticity of substitution across CES varieties	7	Targets a 16.7% steady-state markup; see Galí (2015).
δ	Calvo price non-adjustment probability (quarterly)	0.45	See Coibion et al. (2012).
τ	Rotemberg price adjustment cost	744.9327	Implied by the Calvo–Rotemberg mapping discussed above.
ϕ_y	Monetary policy output-gap response	0.5	Taylor-rule benchmark; see Taylor (1993).
ϕ_π	Monetary policy inflation response	1.5	Taylor-rule benchmark; see Taylor (1993).
ϕ_{vol}	Monetary policy volatility response	0	Baseline calibration choice. Alternative values are considered in comparative statics.
θ	Output-gap autoregressive component	0.95	Baseline calibration choice; see discussion above.
μ	Output-gap steady-state component	-0.004716	See OLS estimation in Table L.1, 1959Q1–2019Q1.
$\bar{\pi}$	Trend inflation component	0.008025	Estimated from 1959Q1–2019Q1 PCE inflation (quarterly growth rate).
$\bar{x}$	Steady-state total variance	$(\sigma + \overline{\sigma^s})^2$	Implied by σ and $\overline{\sigma^s}$ using 1959Q1–2019Q1 volatility moments.

Table L.2: Calibrated parameters.

M Alternative Pricing and the Supply Block

This section explores different models of sticky prices in a non-linear environment: pricing à la Calvo (1983), pricing à la Taylor (1980), and pricing based on menu costs (Gertler and Leahy, 2008), and derives the model's supply block under each pricing. We demonstrate that these alternative sticky-price models deliver comparable implications for aggregate supply even without relying on linear approximations, ultimately producing the same New Keynesian Phillips curve to a first-order approximation.

M.1 Model with Sticky Prices à la Calvo (1983)

We first derive a New Keynesian Phillips curve based on nominal rigidities à la Calvo (1983). First, we present the equilibrium conditions when firms are monopolistically competitive as in a canonical New Keynesian model (Woodford, 2003).

M.1.1 Equilibrium Conditions

Firm i setting its price p_t^i when p_t is the price aggregator faces the demand given by

$$D(p_t^i, p_t) = \left(\frac{p_t^i}{p_t}\right)^{-\epsilon} Y_t.$$

and the following production function:

$$Y_t^i = A_t L_t^i, \tag{M.1}$$

with demand equal to supply in equilibrium, $D(p_t^i, p_t) = Y_t^i$. The price aggregatior is given by

$$p_t = \left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)^{\frac{1}{1-\epsilon}}. \tag{M.2}$$

Other equilibrium conditions remain the same. The optimality conditions for households are given by

$$\frac{1}{p_t C_t} = \frac{L_t^{1/\eta}}{w_t}, \tag{M.3}$$

and

$$\frac{dY_t}{Y_t} = \underbrace{\left(i_t + (\sigma + \sigma_t^s)^2 - \pi_t - \rho\right)}_{\equiv \mu_t} dt + (\sigma + \sigma_t^s) dZ_t. \tag{M.4}$$

In equilibrium, $Y_t = C_t$ holds. Natural output Y_t^n is given by

$$Y_t^n = A_t \left(\frac{\epsilon - 1}{\epsilon}\right)^{\frac{\eta}{\eta+1}},$$

and output gap $\hat{Y}_t$ is similarly defined as follows:

$$\hat{Y}_t = \log\left(\frac{Y_t}{Y_t^n}\right). \tag{M.5}$$

Combining equations (M.3) to (M.5), we obtain:

$$d\hat{Y}_t = \left[i_t - \pi_t - \left(r^n - \frac{1}{2}(\sigma + \sigma_t^s)^2 + \frac{1}{2}\sigma^2\right)\right] dt + \sigma_t^s dZ_t, \tag{M.6}$$

$$\frac{w_t}{p_t A_t} = \left(\frac{\epsilon - 1}{\epsilon}\right) e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}.$$

We define price dispersion as:

$$\Delta_t = \int_0^1 \left(\frac{p_t^i}{p_t}\right)^{-\epsilon} di, \tag{M.7}$$

and (M.1) and the linear aggregation of labor, i.e.,$L_t = \int_0^1 L_t^i di$, we obtain:

$$Y_t = \frac{A_t L_t}{\Delta_t}.$$

Finally, we conjecture that the aggregate price follows a diffusion process of the form:

$$dp_t = \pi_t p_t dt + \sigma_t^p p_t dZ_t \tag{M.8}$$

where π_t is inflation, and σ_t^p is a potentially endogenous and unknown price volatility.

M.1.2 Firms Problem

Firms set prices subject to Calvo (1983) price-setting friction, with the framework adapted to continuous time. Over a dt interval, from t to $t+dt$, an individual firm i adjusts its price with probability $\delta\, dt$. From the perspective of time 0, the probability that a firm resets its price for the first time at time t is given by:

$$\delta e^{-\delta t} dt = \underbrace{\delta dt}_{\text{change now}} \cdot \underbrace{e^{-\delta t}}_{\text{No change until } t} .$$

Formally, we can describe the evolution of individual firm prices as jump processes:

$$dp_t^i = \left(p_t^{i,*} - p_{t-}^i\right) d\Lambda_t^i, \quad \text{where: } d\Lambda_t^i = \begin{cases} 1, & \text{with probability } \delta dt \\ 0, & \text{with probability } 1-\delta dt \end{cases}, \tag{M.9}$$

where $d\Lambda_t^i$ is an i.i.d Poisson random variable, with rate parameter $\delta \geq 0$, p_{t-}^i stands for the individual price of the firm just before t, and $p_t^{i,*}$ stands for the optimal reset price for firm i at time t.

Nominal firm profits Ψ_t^i at time t are given by:

$$\begin{aligned}\Psi_t^i &= \left[p_t^i - \frac{w_t}{A_t}\right] D(p_t^i, p_t) \\ &= p_t Y_t \left[\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} - \frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-\epsilon}\right].\end{aligned}$$

Define $\Psi_{s|t}^i$ as the nominal profits at time $s \geq t$ of an individual firm i that last reset its prices in time t, formally:

$$\Psi_{s|t}^i = p_s Y_s \left[\left(\frac{p_t^i}{p_s}\right)^{1-\epsilon} - \frac{w_s}{p_s A_s}\left(\frac{p_t^i}{p_s}\right)^{-\epsilon}\right].$$

At time t, a price-changing firm i chooses $p_t^{i,*}$ to solve

$$\max_{p_t^{i,*}} \mathbb{E}_t \int_t^\infty e^{-\delta(s-t)} \frac{\xi_s^r}{\xi_t^r} \frac{\Psi_{s|t}^i}{p_s} ds$$

$$= \mathbb{E}_t \int_t^\infty e^{-(\rho+\delta)(s-t)} \frac{C_t}{C_s} Y_s \\ \times \left[\left(\frac{p_t^{i,*}}{p_s} \right)^{1-\epsilon} - \frac{w_s}{p_s A_s} \left(\frac{p_t^{i,*}}{p_s} \right)^{-\epsilon} \right] ds,$$

where $\xi_t^r = e^{-\rho t} \frac{1}{C_t}$ is the (real) state price density as defined in the main text.

Computing the first-order condition with respect to $p_t^{i,*}$ and rearranging, we obtain:

$$\frac{p_t^*}{p_t} \equiv \frac{p_t^{i,*}}{p_t} = \frac{F_t}{G_t}, \tag{M.10}$$
$$F_t \equiv \mathbb{E}_t \int_t^\infty e^{-(\rho+\delta)(s-t)} \left(\frac{p_s}{p_t} \right)^{\epsilon} \times e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_s} ds,$$
$$G_t \equiv \mathbb{E}_t \int_t^\infty e^{-(\rho+\delta)(s-t)} \left(\frac{p_s}{p_t} \right)^{\epsilon-1} ds.$$

where it follows that the optimal reset price is the same for all firms, $p_t^{i,*} = p_t^*$ for all i.

Based on the Hamilton-Jacobi-Bellman (HJB) method, we can find a recursive expression for F_t and G_t as:

$$(\rho+\delta)F_t = e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_s} + \frac{E_t\left[dF_t\right]}{dt}, \tag{M.11}$$
$$(\rho+\delta)G_t = 1 + \frac{E_t\left[dG_t\right]}{dt}. \tag{M.12}$$

This can be rewritten as:

$$dF_t = \underbrace{\left[(\rho+\delta) - \frac{1}{F_t} e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_s} \right]}_{\equiv \mu_t^F} F_t dt + \sigma_t^F F_t dZ_t, \tag{M.13}$$

$$dG_t = \underbrace{\left[(\rho+\delta) - \frac{1}{G_t} \right]}_{\equiv \mu_t^G} G_t dt + \sigma_t^G G_t dZ_t, \tag{M.14}$$

where σ_t^F and σ_t^G are endogenous unknown variables.

M.1.3 Price Process

Using equation (M.2), we can find an expression for the following derivatives:

$$\frac{dp_t}{d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)} = -\left(\frac{1}{\epsilon-1}\right) p_t^{\epsilon},$$
$$\frac{d^2 p_t}{d^2\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)} = \frac{\epsilon}{(\epsilon-1)^2} p_t^{2\epsilon-1},$$

which we can use to obtain an expression for the price process as:

$$\begin{aligned} dp_t = & -\left(\frac{1}{\epsilon-1}\right) p_t^{\epsilon} d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right) \\ & + \frac{1}{2}\frac{\epsilon}{(\epsilon-1)^2} p_t^{2\epsilon-1}\left[d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)\right]^2. \end{aligned} \tag{M.15}$$

Now notice that by the individual price process in (M.9) and $p_t^{i,*} = p^*$, we have that:

$$d(p_t^i)^{1-\epsilon} = \left[(p_t^*)^{1-\epsilon} - (p_{t-}^i)^{1-\epsilon}\right] d\Lambda_t^i$$

Then, we can compute:

$$\begin{aligned} d\int_0^1 (p_t^i)^{1-\epsilon} di = \mathbb{E}_{i,t}\left[d(p_t^i)^{1-\epsilon}\right] &= \delta\left[(p_t^*)^{1-\epsilon} - (p_t)^{1-\epsilon}\right] dt \\ &= \delta (p_t)^{1-\epsilon}\left[\left(\frac{p_t^*}{p_t}\right)^{1-\epsilon} - 1\right] dt. \end{aligned} \tag{M.16}$$

Plugging equations (M.10) and (M.16) into equation (M.15) and eliminating all terms of order higher than dt, we obtain

$$dp_t = \left(\frac{\delta}{\epsilon-1}\right)\left[1 - \left(\frac{F_t}{G_t}\right)^{1-\epsilon}\right] p_t dt.$$

which validates the conjecture in equation (M.8) with $\sigma_t^p = 0$ and inflation given by:

$$\pi_t = \left(\frac{\delta}{\epsilon - 1}\right)\left[1 - \left(\frac{F_t}{G_t}\right)^{1-\epsilon}\right]. \tag{M.17}$$

For later use, we can rearrange the previous expression as:

$$\frac{F_t}{G_t} = \left[1 - \left(\frac{\epsilon - 1}{\delta}\right)\pi_t\right]^{\frac{1}{1-\epsilon}}. \tag{M.18}$$

M.1.4 Price Dispersion

From equation (M.7), we observe that price dispersion can be alternatively interpreted as a cross-sectional expectation on price dispersion

$$\Delta_t = \mathbb{E}_{i,t}\left[\left(\frac{p_t^i}{p_t}\right)^{-\epsilon}\right],$$

where $\mathbb{E}_{i,t}$ stands for the expectations operator over the cross section i. As reset prices $p_t^{i,*}$ are the same for firms resetting on the same instant, i.e., $p_t^{i,*} = p_t^*$,[39] we obtain

$$\Delta_t = \int_{-\infty}^{t} \delta e^{-\delta(t-s)}\left(\frac{p_s^*}{p_t}\right)^{-\epsilon} ds. \tag{M.19}$$

We can now differentiate equation (M.19) with respect to time to obtain:

$$\begin{aligned}\frac{d\Delta}{dt} &= \delta\left(\frac{p_t^*}{p_t}\right)^{-\epsilon} + \left[\epsilon\left(\frac{1}{p_t}\right)\frac{dp_t}{dt} - \delta\right]\int_{-\infty}^{t}\delta e^{-\delta(t-s)}\left(\frac{p_s^*}{p_t}\right)^{-\epsilon} ds \\ &= \delta\left(\frac{p_t^*}{p_t}\right)^{-\epsilon} + \left[\epsilon\pi_t - \delta\right]\Delta_t,\end{aligned} \tag{M.20}$$

where the last line follows from equations (M.19) and (M.8). We can rearrange (M.20) as:[40]

$$d\Delta_t = \left[\delta\left(\left(\frac{F_t}{G_t}\right)^{-\epsilon} - \Delta_t\right) + \epsilon\pi_t\Delta_t\right]dt. \tag{M.21}$$

[39] See Woodford (2003) for the derivation of (M.19).

[40] Yun (2005) derives (M.21) in a discrete-time version of the New Keynesian model.

M.1.5 Inflation Process

We now compute the following derivatives from (M.17) and (M.18):

$$\begin{aligned}
\frac{\partial \pi_t}{\partial F_t} &= \delta \left(\frac{F_t}{G_t}\right)^{1-\epsilon} \left(\frac{1}{F_t}\right) = [\delta-(\epsilon-1)\pi_t] \left(\frac{1}{F_t}\right),\\
\frac{\partial \pi_t}{\partial G_t} &= -\delta \left(\frac{F_t}{G_t}\right)^{1-\epsilon} \left(\frac{1}{G_t}\right) = -[\delta-(\epsilon-1)\pi_t] \left(\frac{1}{G_t}\right),\\
\frac{\partial^2 \pi_t}{\partial F_t^2} &= -\epsilon\delta \left(\frac{F_t}{G_t}\right)^{1-\epsilon} \left(\frac{1}{F_t}\right)^2 = -\epsilon\,[\delta-(\epsilon-1)\pi_t] \left(\frac{1}{F_t}\right)^2,\\
\frac{\partial^2 \pi_t}{\partial G_t^2} &= -(\epsilon-2)\delta \left(\frac{F_t}{G_t}\right)^{1-\epsilon} \left(\frac{1}{G_t}\right)^2 = -(\epsilon-2)\,[\delta-(\epsilon-1)\pi_t] \left(\frac{1}{G_t}\right)^2,\\
\frac{\partial^2 \pi_t}{\partial F_t \partial G_t} &= (\epsilon-1)\delta \left(\frac{F_t}{G_t}\right)^{1-\epsilon} \left(\frac{1}{F_t G_t}\right) = (\epsilon-1)\,[\delta-(\epsilon-1)\pi_t] \left(\frac{1}{F_t G_t}\right).
\end{aligned}$$

Using equations (M.13), (M.14), (M.17) and the derivatives above together with Ito's Lemma to find an expression for the inflation process:

$$\begin{aligned}
d\pi_t = {}& [\delta-(\epsilon-1)\pi_t] \left[(\mu_t^F - \mu_t^G) - \frac{1}{2}\left[\epsilon(\sigma^F)^2 + (\epsilon-2)(\sigma_t^G)^2\right] + (\epsilon-1)\sigma_t^F\sigma_t^G\right] dt\\
&+ [\delta-(\epsilon-1)\pi_t]\,(\sigma_t^F - \sigma_t^G)dZ_t,
\end{aligned}$$

which, after substituting the expressions for μ_t^F and μ_t^G in (M.13) and (M.14), becomes

$$\begin{aligned}
d\pi_t = {}& [\delta-(\epsilon-1)\pi_t] \left[\frac{1}{G_t}\left[1 - \frac{1}{\delta^{\frac{1}{\epsilon-1}}}[\delta-(\epsilon-1)\pi_t]^{\frac{1}{\epsilon-1}} \times e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}\right]\right.\\
&\left. - \frac{1}{2}\left[(\epsilon-1)(\sigma_t^F-\sigma_t^G)^2 + (\sigma_t^F+\sigma_t^G)(\sigma_t^F-\sigma_t^G)\right]\right]dt\\
&+ [\delta-(\epsilon-1)\pi_t]\,(\sigma_t^F - \sigma_t^G)dZ_t,
\end{aligned} \tag{M.22}$$

Let the stochastic process for inflation be given by

$$d\pi_t = \mu_t^\pi dt + \sigma_t^\pi dZ_t, \tag{M.23}$$

where μ_t^π and σ_t^π are the drift and diffusion terms of the inflation process to be determined in equilibrium. Because π_t is a deterministic function of the ratio $R_t \equiv F_t/G_t$ (see equation (M.17)), only the drift and diffusion of R_t matter for the induced coefficients $(\mu_t^\pi, \sigma_t^\pi)$; hence two components of the joint (F_t, G_t) specification in (M.13) and (M.14) are redundant and can be fixed by normalization, e.g. by setting:

$$G_t = \frac{1}{\rho + \delta} \quad \text{and} \quad \sigma_t^G = 0, \tag{M.24}$$

which implies $dG_t = 0$ and solves the SDE for G_t in (M.14), leaving π_t fully determined by the process for F_t.

Combining equations (M.18) and (M.24), we obtain

$$F_t = \left(\frac{1}{\delta^{\frac{1}{1-\epsilon}}(\rho + \delta)}\right) [\delta - (\epsilon - 1)\pi_t]^{\frac{1}{1-\epsilon}} . \tag{M.25}$$

Computing the derivatives of F_t with respect to π_t:

$$\frac{\partial F_t}{\partial \pi_t} = \left(\frac{1}{\delta^{\frac{1}{1-\epsilon}}(\rho + \delta)}\right) [\delta - (\epsilon - 1)\pi_t]^{\frac{\epsilon}{1-\epsilon}} ,$$

$$\frac{\partial^2 F_t}{\partial^2 \pi_t} = \left(\frac{\epsilon}{\delta^{\frac{1}{1-\epsilon}}(\rho + \delta)}\right) [\delta - (\epsilon - 1)\pi_t]^{\frac{2\epsilon-1}{1-\epsilon}} .$$

Now we apply Ito's Lemma using equations (M.23), (M.25) and the derivatives above to obtain:

$$\begin{aligned} dF_t = &\left(\frac{1}{\delta^{\frac{1}{1-\epsilon}}(\rho + \delta)}\right) [\delta - (\epsilon - 1)\pi_t]^{\frac{\epsilon}{1-\epsilon}} \\ &\times \left[\mu_t^\pi + \frac{\epsilon}{2} \frac{(\sigma_t^\pi)^2}{\delta - (\epsilon - 1)\pi_t}\right] dt \\ &+ \left(\frac{1}{\delta^{\frac{1}{1-\epsilon}}(\rho + \delta)}\right) [\delta - (\epsilon - 1)\pi_t]^{\frac{\epsilon}{1-\epsilon}} \sigma_t^\pi dZ_t. \end{aligned} \tag{M.26}$$

Equating the diffusion terms in (M.13) and (M.26), and using (M.25), we obtain:

$$\sigma_t^F = \frac{\sigma_t^\pi}{\delta - (\epsilon - 1)\pi_t}. \tag{M.27}$$

Plugging equations (M.24) and (M.27) into (M.22) and rearranging, we obtain:

$$
\begin{aligned}
d\pi_t &= \mu_t^\pi \, dt + \sigma_t^\pi dZ_t, \\
\mu_t^\pi &\equiv [\delta - (\epsilon - 1)\pi_t] \left[(\rho + \delta) \left[1 - \frac{1}{\delta^{\frac{1}{\epsilon-1}}} [\delta - (\epsilon - 1)\pi_t]^{\frac{1}{\epsilon-1}} \times e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t} \right] \right. \\
&\qquad \left. - \frac{\epsilon}{2} \left(\frac{\sigma_t^\pi}{\delta - (\epsilon - 1)\pi_t} \right)^2 \right].
\end{aligned} \tag{M.28}
$$

Linearization We linearize the drift μ_t^π in (M.28) around the steady-state values of inflation $\bar{\pi}$ and the output gap μ. In the linearization we ignore deviations in terms involving higher-order moments, such as inflation volatility σ_t^π. Nevertheless, we must choose an expansion point $\overline{\sigma^\pi}$ for σ_t^π to evaluate the rest of the terms involved. We set $\overline{\sigma^\pi}$ so that the inflation drift satisfies $\mu^\pi = 0$ when π_t, $\hat{Y}_t$, and σ_t^π are evaluated at their expansion points $(\bar{\pi}, \mu, \overline{\sigma^\pi})$:

$$
\begin{aligned}
\overline{\sigma^\pi} &= [\delta - (\epsilon - 1)\bar{\pi}] \\
&\quad \times \sqrt{\frac{2}{\epsilon}(\rho + \delta) \left[1 - \frac{1}{\delta^{\frac{1}{\epsilon-1}}} [\delta - (\epsilon - 1)\bar{\pi}]^{\frac{1}{\epsilon-1}} e^{\left(\frac{\eta+1}{\eta}\right)\mu} \right]}.
\end{aligned}
$$

This results in a linearized Phillips curve given by

$$
\mathbb{E}_t d(\pi_t - \bar{\pi}) = \left(\tilde{\kappa}^\pi (\pi_t - \bar{\pi}) - \tilde{\kappa}^y (\hat{Y}_t - \mu) \right) dt, \tag{M.29}
$$

where the constants are defined as

$$
\begin{aligned}
\tilde{\kappa}^\pi &= (\rho + \delta) \left[1 - \epsilon \left[1 - \frac{1}{\delta^{\frac{1}{\epsilon-1}}} [\delta - (\epsilon - 1)\bar{\pi}]^{\frac{1}{\epsilon-1}} e^{\left(\frac{\eta+1}{\eta}\right)\mu} \right] \right] \\
&\quad - \frac{\epsilon(\epsilon - 1)}{2} \left(\frac{\overline{\sigma^\pi}}{\delta - (\epsilon - 1)\bar{\pi}} \right)^2 \\
\tilde{\kappa}^y &= \left(\frac{\eta + 1}{\eta} \right) \left(\frac{\rho + \delta}{\delta^{\frac{1}{\epsilon-1}}} \right) [\delta - (\epsilon - 1)\bar{\pi}]^{\frac{\epsilon}{\epsilon-1}} e^{\left(\frac{\eta+1}{\eta}\right)\mu}.
\end{aligned}
$$

This leads to a standard continuous-time New Keynesian Phillips curve (Werning, 2012).[41] Note that (M.29) is mathematically isomorphic to the log-linearized Phillips curve (5) of Section 5, which is based on pricing à la Rotemberg (1982).

M.2 Model with Sticky Prices à la Taylor (1980)

This section adapts the staggered price-setting mechanism of Taylor (1980) to a continuous-time environment with fixed contract length T by assuming a uniform distribution of contract ages on $[0, T)$. The resulting aggregate price index and optimal price-resetting condition are the continuous-time analogues of the standard discrete-time Taylor contracts (Taylor, 1980; Whelan, 2007; Dixon and Kara, 2011).

We will show that, under Assumption M.1 and Assumption M.2 (to be described) on low and stable inflation, the inflation and price dispersion processes under Taylor (1980) pricing are well approximated by those of the Calvo pricing model derived in Section M.1.

M.2.1 Firms Problem

Firms set prices à la Taylor (1980), which is a standard way of generating price stickiness in New Keynesian models, except that we have to adapt the framework to continuous time.

Each firm i can change its price only at exogenous times that are spaced by a fixed duration $T > 0$ ("Taylor contract"). Between two consecutive reset dates, the firm keeps its price constant. In cross-section, firms are uniformly distributed over contract ages in $[0, T)$, so in each instant a measure $1/T$ of firms resets their prices.

Formally, under Taylor pricing the price of firm i evolves as a pure-jump process:

$$dp_t^i = \left(p_t^{i,*} - p_{t-}^i\right) d\Lambda_t^i,$$
$$\Lambda_t^i := \sum_{n=0}^{\infty} \mathbf{1}_{\{t \geq \tau_0^i + nT\}}. \tag{M.30}$$

where $\tau_0^i \in [0, T)$ is the (idiosyncratic) initial reset time of firm i, so that $d\Lambda_t^i$ has a unit jump at each deterministic reset date $\tau_0^i + nT, \forall n \in \mathbb{N}$ and is zero otherwise, p_{t-}^i stands for the individual price of the firm just before t, and $p_t^{i,*}$ stands for the

[41] Note from (M.29) that we have $\rho + \delta$ instead of ρ as the coefficient on π_t under $\bar{\pi} = \mu = 0$. Gabaix (2020) obtains the same result for the case of rational agents.

optimal reset price for firm i at time t.

Nominal firm profits Ψ_t^i at t are given by:

$$\begin{aligned}\Psi_t^i &= \left[p_t^i - \frac{w_t}{A_t}\right] D(p_t^i, p_t)\\ &= p_t Y_t \left[\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} - \frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-\epsilon}\right].\end{aligned}$$

We define $\Psi_{s|t}^i$ as the nominal profits at instant $s \in [t, t+T)$ of an individual firm i that last reset its price p_t^i at time t, formally:

$$\Psi_{s|t}^i = p_s Y_s \left[\left(\frac{p_t^i}{p_s}\right)^{1-\epsilon} - \frac{w_s}{p_s A_s}\left(\frac{p_t^i}{p_s}\right)^{-\epsilon}\right].$$

At time t, a price-changing firm i chooses optimal $p_t^{i,*}$ to solve

$$\begin{aligned}&\max_{p_t^{i,*}} \mathbb{E}_t \int_t^{t+T} \frac{\xi_s^r}{\xi_t^r}\frac{\Psi_{s|t}^i}{p_s} ds\\ &= \max_{p_t^{i,*}} \mathbb{E}_t \int_t^{t+T} e^{-\rho(s-t)}\frac{C_t}{C_s} Y_s\\ &\qquad \times \left[\left(\frac{p_t^{i,*}}{p_s}\right)^{1-\epsilon} - \frac{w_s}{p_s A_s}\left(\frac{p_t^{i,*}}{p_s}\right)^{-\epsilon}\right] ds,\end{aligned}$$

where $\xi_t^r = e^{-\rho t}\frac{1}{C_t}$ is the real state price density. From the first-order condition with respect to $p_t^{i,*}$, we obtain:

$$\begin{aligned}\frac{p_t^*}{p_t} &\equiv \frac{p_t^{i,*}}{p_t} = \frac{F_t}{G_t}, \\ F_t &\equiv \mathbb{E}_t \int_t^{t+T} e^{-\rho(s-t)}\left(\frac{p_s}{p_t}\right)^{\epsilon} \times e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_s} ds,\\ G_t &\equiv \mathbb{E}_t \int_t^{t+T} e^{-\rho(s-t)}\left(\frac{p_s}{p_t}\right)^{\epsilon-1} ds.\end{aligned} \tag{M.31}$$

where it follows that the optimal reset price is the same for all firms, $p_t^{i,*} = p_t^*$ for all

i.

Based on the Hamilton–Jacobi–Bellman (HJB) method for finite-horizon functionals, we can find recursive expressions for F_t and G_t of the form

$$\rho F_t = e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t} - e^{-\rho T}\,\mathbb{E}_t\left[\left(\frac{p_{t+T}}{p_t}\right)^{\epsilon} e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_{t+T}}\right] + \frac{\mathbb{E}_t\left[dF_t\right]}{dt}, \tag{M.32}$$

$$\rho G_t = 1 - e^{-\rho T}\,\mathbb{E}_t\left[\left(\frac{p_{t+T}}{p_t}\right)^{\epsilon-1}\right] + \frac{\mathbb{E}_t\left[dG_t\right]}{dt}, \tag{M.33}$$

which can be rewritten as

$$dF_t = \mu_t^F F_t dt + \sigma_t^F F_t dZ_t, \tag{M.34}$$

$$dG_t = \mu_t^G G_t dt + \sigma_t^G G_t dZ_t, \tag{M.35}$$

where μ_t^F and μ_t^G are the drifts implied by (M.32) and (M.33). Note that σ_t^F and σ_t^G are endogenous variables capturing the exposure of F_t and G_t to the underlying shocks Z_t.

M.2.2 Price Process

Using equation (M.2), we can find an expression for the following derivatives:

$$\frac{dp_t}{d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)} = -\left(\frac{1}{\epsilon-1}\right)(p_t)^{\epsilon},$$

$$\frac{d^2 p_t}{d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)^2} = \frac{\epsilon}{(\epsilon-1)^2}(p_t)^{2\epsilon-1},$$

which we can use to obtain an expression for the price process as:

$$dp_t = -\left(\frac{1}{\epsilon-1}\right) p_t^{\epsilon} d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right) + \frac{1}{2}\frac{\epsilon}{(\epsilon-1)^2} p_t^{2\epsilon-1}\left[d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)\right]^2. \tag{M.36}$$

Under the individual price process (M.30), at each instant a fraction dt/T of firms resets its price. The firms that reset at time t are exactly those that last reset at time $t-T$, so their pre-reset price is p^*_{t-T}. Therefore,

$$\begin{aligned} d\int_0^1 (p^i_t)^{1-\epsilon} di &= \int_0^1 d(p^i_t)^{1-\epsilon} di \\ &= \frac{1}{T}\left[(p^*_t)^{1-\epsilon} - (p^*_{t-T})^{1-\epsilon}\right] dt. \end{aligned} \tag{M.37}$$

Plugging equation (M.37) into equation (M.36) and eliminating all terms of order higher than dt,[42] we obtain

$$\begin{aligned} dp_t &= -\left(\frac{1}{\epsilon-1}\right) p^\epsilon_t \cdot \frac{1}{T}\left[(p^*_t)^{1-\epsilon} - (p^*_{t-T})^{1-\epsilon}\right] dt \\ &= \frac{1}{(\epsilon-1)T} p_t \left[\left(\frac{p^*_{t-T}}{p_t}\right)^{1-\epsilon} - \left(\frac{p^*_t}{p_t}\right)^{1-\epsilon}\right] dt. \end{aligned} \tag{M.38}$$

which validates $\sigma^p_t = 0$ in equation (M.8). Thus, inflation is given by:

$$\pi_t = \frac{1}{(\epsilon-1)T}\left[\left(\frac{p^*_{t-T}}{p_t}\right)^{1-\epsilon} - \left(\frac{p^*_t}{p_t}\right)^{1-\epsilon}\right]. \tag{M.39}$$

M.2.3 Price Dispersion

From (M.7), price dispersion can be alternatively interpreted as a cross-sectional expectation on relative prices:

$$\Delta_t = \mathbb{E}_{i,t}\left[\left(\frac{p^i_t}{p_t}\right)^{-\epsilon}\right],$$

where $\mathbb{E}_{i,t}$ denotes the expectations on the cross section i. Under Taylor pricing, firms are uniformly distributed over contract ages $[0, T)$, and a firm that last reset at time $s \in [t-T, t]$ charges $p^i_t = p^*_s$ at time t. Thus, similar to Section M.1, we obtain[43]

[42]Since $d\left(\int_0^1 (p^i_t)^{1-\epsilon} di\right) = O(dt)$ under Taylor pricing, the quadratic term becomes $O(dt^2)$ and can be dropped.

[43]See Woodford (2003) for the derivation of (M.40).

$$\Delta_t = \frac{1}{T}\int_{t-T}^{t}\left(\frac{p_s^*}{p_t}\right)^{-\epsilon} ds, \tag{M.40}$$

from which we obtain

$$\begin{aligned}\frac{d\Delta_t}{dt} &= \underbrace{\frac{1}{T}\left[\left(\frac{p_t^*}{p_t}\right)^{-\epsilon} - \left(\frac{p_{t-T}^*}{p_t}\right)^{-\epsilon}\right]}_{\text{boundary term}} \\ &\quad + \frac{1}{T}\int_{t-T}^{t}\frac{\partial}{\partial t}\left\{\left(\frac{p_s^*}{p_t}\right)^{-\epsilon}\right\} ds \\ &= \frac{1}{T}\left[\left(\frac{F_t}{G_t}\right)^{-\epsilon} - \left(\frac{p_{t-T}^*}{p_t}\right)^{-\epsilon}\right] \\ &\quad + \frac{1}{T}\int_{t-T}^{t}\left[\epsilon\left(\frac{1}{p_t}\right)\frac{dp_t}{dt}\right]\left(\frac{p_s^*}{p_t}\right)^{-\epsilon} ds \\ &= \frac{1}{T}\left[\left(\frac{F_t}{G_t}\right)^{-\epsilon} - \left(\frac{p_{t-T}^*}{p_t}\right)^{-\epsilon}\right] + \epsilon\pi_t\,\Delta_t, \end{aligned} \tag{M.41}$$

where the last equality follows from equations (M.8) and (M.40).[44] Equation (M.41) can be rewritten as

$$d\Delta_t = \left[\frac{1}{T}\left(\left(\frac{F_t}{G_t}\right)^{-\epsilon} - \left(\frac{p_{t-T}^*}{p_t}\right)^{-\epsilon}\right) + \epsilon\pi_t\Delta_t\right] dt. \tag{M.42}$$

M.2.4 Approximate Price and Dispersion Dynamics under Taylor Pricing

To obtain closed-form expressions analogous to the Calvo pricing case of Section M.1, we introduce the following approximations. These assumptions are related to a stable, low inflation environment, e.g., close-to-flat Phillips curve (Del Negro et al., 2020; Beaudry et al., 2024).

[44] In the case of Taylor pricing, we observe from (M.38) that $\sigma_t^p = 0$, implying $dp_t = \pi_t p_t dt$.

Assumption M.1 (Small effective price change over one contract duration)

Over a single contract duration T, inflation and relative price dispersion are small enough that the price of the cohort that is just expiring at time t, p^*_{t-T}, is close to the current aggregate price. Formally,

$$\left(\frac{p^*_{t-T}}{p_t}\right)^{1-\epsilon} \simeq 1.$$

Assumption M.2 (Flat within-window price profile)

Relative prices inside the Taylor window $[t-T, t]$ are approximately constant, so that the price dispersion term at the back of the window can be expressed in terms of Δ_t:

$$\left(\frac{p^*_{t-T}}{p_t}\right)^{-\epsilon} \simeq \Delta_t.$$

Note that Assumptions M.1 and M.2 usually hold under a low, stable inflation environment.

Approximate price dynamics and inflation Starting from the exact Taylor inflation expression in (M.39), substituting Assumption M.1 leads to $\left(\frac{p^*_{t-T}}{p_t}\right)^{1-\epsilon} \simeq 1$, so that (M.39) together with the optimal reset price condition (M.31) simplifies to

$$\pi_t \simeq \frac{1}{(\epsilon-1)T}\left[1-\left(\frac{F_t}{G_t}\right)^{1-\epsilon}\right]. \tag{M.43}$$

Approximate price dispersion dynamics Substituting Assumption M.2 into equation (M.42) yields

$$d\Delta_t \simeq \left[\frac{1}{T}\left(\left(\frac{F_t}{G_t}\right)^{-\epsilon} - \Delta_t\right) + \epsilon\pi_t\,\Delta_t\right]dt. \tag{M.44}$$

Equations (M.43) and (M.44), together with Assumptions M.1 and M.2, provide a tractable Taylor-pricing counterpart to the Calvo expressions. When we set the contract duration T equal to the inverse of the Calvo Poisson price reset rate, $T = 1/\delta$, the approximate inflation and price dispersion processes under the Taylor pricing (i.e., equations (M.43) and (M.44)) coincide with those under the Calvo pricing (i.e., equations (M.17) and (M.21)).

Under the long-contract limit, $T \to +\infty$ (equivalently, $\delta \to 0$) that leads to Assumptions M.1 and M.2, generating an economy approximated around rigid prices, then the reset-price conditions (M.32) and (M.33) under Taylor pricing reduce to the same HJB equations as in the Calvo case, given by (M.11) and (M.12).

Since the Taylor pricing case becomes isomorphic to the Calvo pricing case under these conditions, the log-linearized Phillips curve under Taylor pricing will coincide as well with (M.29) under Calvo pricing.

M.3 Model with Menu Costs

This section develops a continuous-time version of the supply block based on menu costs. For modeling firm-level price adjustments, we introduce a fixed cost of changing prices, in line with the (S, s) and menu-cost literature (see e.g., Gertler and Leahy, 2008; Alvarez et al., 2017; Auclert et al., 2024; Blanco et al., 2024).

At the aggregate level, the optimal adjustment policy can be summarized by an endogenous adjustment hazard λ_t and the associated survival function $H_t(s)$ for prices last reset at time t, which together determine the menu-cost analogues of the reset-price condition and the New Keynesian Phillips curve derived under Calvo pricing in Section M.1.

We will show that, under Assumptions M.3 and M.4 (which will be described) on the adjustment hazard and the cross-sectional distribution of prices, the inflation and price dispersion processes implied by the menu-cost model can be well-approximated by those of the Calvo pricing model of Section M.1. In this sense, our continuous-time menu-cost specification provides an explicit state-dependent microfoundation that nests the Calvo block as a limiting case, echoing the equivalence results in Gertler and Leahy (2008) and the broader state- vs. time-dependent pricing comparisons in Alvarez et al. (2017), Auclert et al. (2024), and Blanco et al. (2024).

M.3.1 Firms Problem

Changing price p_t^i requires firm i to pay a fixed cost $\varrho > 0$ expressed in real units (units of the final good). Let $\{\tau_n^i\}_{n \geq 1}$ denote the (endogenous) sequence of stopping times at which firm i chooses to reset its price. Between two adjustment times, p_t^i remains constant.

Formally, we can describe the evolution of individual firm prices as a pure-jump process:

$$dp_t^i = \left(p_t^{i,*} - p_{t-}^i\right) dJ_t^i, \tag{M.45}$$

where dJ_t^i is the increment of a counting process that jumps from 0 to 1 precisely at the (endogenous) adjustment times $\{\tau_n^i\}$, and $p_t^{i,*}$ stands for the optimal reset price for firm i at time t. Note that in contrast with the Calvo or Taylor case, the arrival of dJ_t^i is not governed by an exogenous Poisson process, but by the firm's optimal menu-cost policy.

Nominal period-t profits of firm i are:

$$\begin{aligned}\Psi_t^i &= \left[p_t^i - \frac{w_t}{A_t}\right] D(p_t^i, p_t) \\ &= p_t Y_t \left[\left(\frac{p_t^i}{p_t}\right)^{1-\epsilon} - \frac{w_t}{p_t A_t}\left(\frac{p_t^i}{p_t}\right)^{-\epsilon}\right].\end{aligned}$$

Let $\Psi_{s|t}^i$ denote nominal profits at time $s \geq t$ of a firm that last reset its price p_t^i at time t and keeps that price in place:

$$\Psi_{s|t}^i = p_s Y_s \left[\left(\frac{p_t^i}{p_s}\right)^{1-\epsilon} - \frac{w_s}{p_s A_s}\left(\frac{p_t^i}{p_s}\right)^{-\epsilon}\right].$$

Let $\xi_t^r = e^{-\rho t}\frac{1}{C_t}$ denote the real state price density defined in the main text. The value at time t for firm i of a given price path $\{p_s^i\}_{s\geq t}$ and an adjustment policy $\{\tau_n^i\}_{n\geq 1}$ is:

$$V_t^i = \mathbb{E}_t\left[\int_t^\infty \frac{\xi_s^r}{\xi_t^r}\frac{\Psi_s^i}{p_s} ds \; - \sum_{n\geq 1} \frac{\xi_{\tau_n^i}^r}{\xi_t^r}\varrho\right].$$

At each adjustment time $t = \tau_n^i$, the firm i chooses a new price $p_t^{i,*}$ and commits to the new price until the next adjustment. Let

$$H_t(s) \equiv \Pr\left(\tau_{n+1}^i > s \,\middle|\, \mathcal{F}_t,\, \tau_n^i = t\right), \qquad s \geq t, \tag{M.46}$$

be the endogenous survival probability that the new price set at t is still in place at time s under the optimal menu-cost policy.[45] Conditional on adjusting at time t and choosing $p_t^{i,*}$, the continuation value is

$$
\begin{aligned}
V_t^{\text{adj}}(p_t^{i,*}) &= -\varrho + \mathbb{E}_t\left[\int_t^\infty H_t(s)\frac{\xi_s^r}{\xi_t^r}\frac{\Psi_{s|t}^i}{p_s}ds\right] \\
&= -\varrho + \mathbb{E}_t\Bigg[\int_t^\infty H_t(s)e^{-\rho(s-t)}\frac{C_t}{C_s}Y_s \\
&\qquad \times \left(\left(\frac{p_t^{i,*}}{p_s}\right)^{1-\epsilon} - \frac{w_s}{p_s A_s}\left(\frac{p_t^{i,*}}{p_s}\right)^{-\epsilon}\right)ds\Bigg]
\end{aligned}
$$

At time t, a price-changing firm i chooses $p_t^{i,*}$ to solve[46]

$$
\max_{p_t^{i,*}} V_t^{\text{adj}}(p_t^{i,*}). \tag{M.47}
$$

Computing the first-order condition with respect to $p_t^{i,*}$ and rearranging, we obtain:

$$
\begin{aligned}
\frac{p_t^*}{p_t} &\equiv \frac{p_t^{i,*}}{p_t} = \frac{F_t}{G_t}, \qquad\qquad (\text{M.48})\\
F_t &\equiv \mathbb{E}_t\int_t^\infty e^{-\rho(s-t)}H_t(s)\left(\frac{p_s}{p_t}\right)^{\epsilon} \times e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_s}ds, \\
G_t &\equiv \mathbb{E}_t\int_t^\infty e^{-\rho(s-t)}H_t(s)\left(\frac{p_s}{p_t}\right)^{\epsilon-1}ds.
\end{aligned}
$$

where it follows that the optimal reset price is the same for all firms, $p_t^{i,*} = p_t^*$ for all i.

Based on the Hamilton-Jacobi-Bellman (HJB) method, we find recursive expressions for F_t and G_t that are analogous to the Calvo pricing case in Section M.1 and

[45] In an explicit (S,s) specification, $H_t(s)$ is the survival probability associated with the first-exit time of the relevant state (e.g. the firm-specific price gap) from the inaction region.

[46] Firm i changes its price at time t only if

$$
\max_{p_t^{i,*}} V_t^{\text{adj}}(p_t^{i,*}) \geq V_t^{\text{adj}}(p_{t^-}^i) + \varrho.
$$

the Taylor pricing case in Section M.2. Let δ_t denote the endogenous instantaneous adjustment hazard implied by the optimal menu-cost policy, defined by[47]

$$\frac{dH_t(s)}{H_t(s)} = -\delta_s \, ds, \qquad H_t(t) = 1. \tag{M.49}$$

Under this definition, the HJB conditions imply:

$$\begin{aligned}
(\rho + \delta_t)F_t &= e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t} + \frac{E_t\left[dF_t\right]}{dt}, \\
(\rho + \delta_t)G_t &= 1 + \frac{E_t\left[dG_t\right]}{dt}.
\end{aligned}$$

This can be rewritten as:

$$\begin{aligned}
dF_t &= \underbrace{\left[(\rho + \delta_t) - \frac{1}{F_t} e^{\left(\frac{\eta+1}{\eta}\right)\hat{Y}_t}\right]}_{\equiv \mu_t^F} F_t dt \\
&\quad + \sigma_t^F F_t dZ_t, && \text{(M.50)} \\
dG_t &= \underbrace{\left[(\rho + \delta_t) - \frac{1}{G_t}\right]}_{\equiv \mu_t^G} G_t dt + \sigma_t^G G_t dZ_t, && \text{(M.51)}
\end{aligned}$$

where σ_t^F and σ_t^G are endogenous unknown variables.

M.3.2 Price Process

Using equation (M.2), we can find an expression for the following derivatives:

$$\begin{aligned}
\frac{dp_t}{d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)} &= -\left(\frac{1}{\epsilon - 1}\right) p_t^{\epsilon}, \\
\frac{d^2 p_t}{d^2\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)} &= \frac{\epsilon}{(\epsilon - 1)^2} p_t^{2\epsilon - 1},
\end{aligned}$$

[47] For tractability, we assume that there is an initial distribution of firm prices, which leads to a distribution of prices at each time t consistent with the hazard function (M.46), i.e., for each dt, $\delta_t dt$ measure of firms change their prices to p_t^* solving (M.47).

which leads to an expression for the price process as:

$$
\begin{aligned}
dp_t =& -\left(\frac{1}{\epsilon-1}\right) p_t^{\epsilon} d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right) \\
&+ \frac{1}{2}\frac{\epsilon}{(\epsilon-1)^2} p_t^{2\epsilon-1}\left[d\left(\int_0^1 (p_t^i)^{1-\epsilon} di\right)\right]^2 .
\end{aligned} \tag{M.52}
$$

Under menu costs, the individual price process is given by equation (M.45), and $p_t^{i,*} = p_t^*$ for all i in equilibrium. Therefore,

$$
d(p_t^i)^{1-\epsilon} = \left[(p_t^*)^{1-\epsilon} - (p_{t-}^i)^{1-\epsilon}\right] dJ_t^i .
$$

Let

$$
\Theta_t \equiv \mathbb{E}_t\left[\left(\frac{p_{t-}^i}{p_t}\right)^{1-\epsilon} \middle| \text{firm } i \text{ adjusts at time } t\right]
$$

denote the average relative price (raised by $1-\epsilon$) among adjusting firms at time t. Using the law of large numbers over the continuum of firms, we then obtain

$$
\begin{aligned}
d\int_0^1 (p_t^i)^{1-\epsilon} di =& \int_0^1 d(p_t^i)^{1-\epsilon} di \\
&= \delta_t dt\, \mathbb{E}_t\left[(p_t^*)^{1-\epsilon} - (p_{t-}^i)^{1-\epsilon} \middle| \text{firm } i \text{ adjusts at } t\right] \\
&= \delta_t p_t^{1-\epsilon}\left[\left(\frac{p_t^*}{p_t}\right)^{1-\epsilon} - \Theta_t\right] dt .
\end{aligned} \tag{M.53}
$$

Plugging equations (M.48) and (M.53) into equation (M.52) and eliminating all terms of order higher than dt, we obtain

$$
dp_t = \left(\frac{\delta_t}{\epsilon-1}\right)\left[\Theta_t - \left(\frac{F_t}{G_t}\right)^{1-\epsilon}\right] p_t dt .
$$

This validates $\sigma_t^p = 0$ in equation (M.8). Thus, inflation is given by:

$$
\pi_t = \left(\frac{\delta_t}{\epsilon-1}\right)\left[\Theta_t - \left(\frac{F_t}{G_t}\right)^{1-\epsilon}\right] . \tag{M.54}
$$

M.3.3 Price Dispersion

Due to the absence of firm-specific idiosyncratic shocks in our model, if firms start at the same initial price, there would be no price dispersion ($\Delta_t = 1$). To compare with the Calvo pricing case, we assume that there is an initial distribution of firm prices, which leads to a distribution of prices at each time t consistent with the hazard function (M.46).

From equation (M.7), price dispersion can be alternatively interpreted as a cross-sectional expectation on price dispersion:

$$\Delta_t = \mathbb{E}_{i,t}\left[\left(\frac{p_t^i}{p_t}\right)^{-\epsilon}\right],$$

where $\mathbb{E}_{i,t}$ stands for the expectations operator over the cross section i. Under menu costs, all firms that reset at a given time s choose the same reset price p_s^*.[48] Then, $\delta_s H_s(t)\, ds$ is the measure of firms at time t whose last adjustment took place in the interval $[s, s+ds)$. Their price at time t is p_s^*, so that $\left(p_t^i/p_t\right)^{-\epsilon} = \left(p_s^*/p_t\right)^{-\epsilon}$ for those firms. Hence we can write

$$\Delta_t = \int_{-\infty}^{t} \delta_s H_s(t) \left(\frac{p_s^*}{p_t}\right)^{-\epsilon} ds. \tag{M.55}$$

We can now differentiate equation (M.55) with respect to time to obtain:

$$\begin{aligned}
\frac{d\Delta_t}{dt} &= \underbrace{\delta_t H_t(t)\left(\frac{p_t^*}{p_t}\right)^{-\epsilon}}_{\text{boundary term}} + \int_{-\infty}^{t} \frac{d}{dt}\left\{\delta_s H_s(t)\left(\frac{p_s^*}{p_t}\right)^{-\epsilon}\right\} ds \\
&= \delta_t \left(\frac{p_t^*}{p_t}\right)^{-\epsilon} + \int_{-\infty}^{t} \delta_s \left[\frac{dH_s(t)}{dt}\left(\frac{p_s^*}{p_t}\right)^{-\epsilon} + H_s(t)\frac{d}{dt}\left(\frac{p_s^*}{p_t}\right)^{-\epsilon}\right] ds \\
&= \delta_t \left(\frac{p_t^*}{p_t}\right)^{-\epsilon} + (\epsilon\pi_t - \delta_t)\Delta_t,
\end{aligned} \tag{M.56}$$

where the last equality follows from (M.8), (M.49) and (M.55). We can equivalently write

[48] The argument is similar to Woodford (2003).

$$d\Delta_t = \left[\delta_t\left(\left(\frac{F_t}{G_t}\right)^{-\epsilon} - \Delta_t\right) + \epsilon\pi_t\Delta_t\right]dt. \tag{M.57}$$

M.3.4 Approximate Price and Dispersion Dynamics under Menu Costs

To obtain closed-form expressions similar to the Calvo pricing case, we introduce the following assumptions on the endogenous adjustment hazard and cross-sectional price dispersion.

Assumption M.3 (*State-independent Poisson hazard.*) *The optimal menu-cost policy generates an approximately constant and state-independent adjustment hazard. Formally, there exists a constant $\delta > 0$ such that*

$$\delta_t \simeq \delta$$

for all t. Thus, the times of price adjustments of each firm are approximately governed by a Poisson process with intensity δ.[49]

Assumption M.4 (*Random adjustment and small dispersion.*) *At each time t, the set of firms that adjust is approximately a random draw from the cross section, and relative price dispersion is small. In particular, in a neighborhood of the steady state we have*

$$\Theta_t = \mathbb{E}_t\left[\left(\frac{p^i_{t-}}{p_t}\right)^{1-\epsilon} \middle| \textit{firm } i \textit{ adjusts at } t\right] \simeq 1,$$

Approximate price dynamics and inflation Starting from the exact menu-cost inflation expression in (M.54), and substituting Assumption M.4 implies $\Theta_t \approx 1$, so that

$$\pi_t \approx \frac{\delta_t}{\epsilon - 1}\left[1 - \left(\frac{F_t}{G_t}\right)^{1-\epsilon}\right]. \tag{M.58}$$

Approximate price dispersion dynamics Starting from the exact menu-cost price dispersion dynamics in (M.57), note that equation (M.57) becomes its counterpart under Calvo pricing, equation (M.21).

[49]This assumption effectively minimizes the state dependency of price-resetting, making it closer to time-dependent pricing.

Equations (M.57) and (M.58), together with Assumptions M.3 and M.4, provide a tractable menu-cost counterpart to the expressions under Calvo pricing.

Under Assumption M.3, when the hazard rate is approximately constant, i.e., $\delta_t \simeq \delta$, the inflation and price dispersion processes under menu costs coincide with those under Calvo pricing, given by equations (M.17) and (M.21).

Since the model with menu costs becomes isomorphic to the model with Calvo pricing under these conditions, the log-linearized New Keynesian Phillips curve under menu costs will coincide as well with (M.29), which is derived under Calvo pricing.

N Equilibrium Value Function for the Representative Household and the Transversality Condition with the Linearized Phillips Curve

In this section, we characterize the equilibrium value function of households with the linearized New Keynesian Phillips curve (31), and prove the transversality condition for individual households in this case. It turns out that the proof is very similar to Online Appendix B. First, we summarize the key equations.

Equilibrium interest rates We rewrite the equilibrium nominal interest rate from equation (K.20), with ϕ_{total} and ϕ_2 given by (K.23) and (K.24), as:

$$i_t = r^n - \frac{1}{2}\left[\left[\sigma^2 - 2\mu\phi_{total}\right] - \sigma^2\right] + \frac{\kappa^y}{\kappa^\pi}\mu + \phi_y\left(\hat{Y}_t - \mu\right) + \phi_\pi\left(\pi_t - \frac{\kappa^y}{\kappa^\pi}\mu\right) \tag{N.1}$$

$$- \phi_{vol}\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]. \tag{N.2}$$

With equations (K.25), (K.26) and (N.1), the real interest rate $r_t = i_t - \pi_t$ is given by

$$r_t = r^n - \frac{1}{2}\left[\left[\sigma^2 - 2\mu\phi_{total}\right] - \sigma^2\right] - \frac{1}{2}\left(1 - \frac{\theta}{\phi_2}\right)\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]. \tag{N.3}$$

Formulation The Hamilton-Jacobi-Bellman (HJB) equation of household h is given by:

$$\rho \cdot \Gamma^h = \max_{C_t^h, L_t^h}\left\{\log C_t^h - \frac{\left(L_t^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h}\right\}. \tag{N.4}$$

where we assume B_t^h as the only endogenous state variable,[50] and $\{p_t, A_t, \sigma_t^s\}$ as exogenous state variables from individual household's perspective. Note that p_t follows (28) while π_t is a function of $\hat{Y}_t$, which is in turn a function of σ_t^s. We

[50]Eventually, we will impose the bond market equilibrium condition, i.e., $B_t = 0$ as in Online Appendix B.

calculate $\mu_t^{\Gamma,h}$ as follows:

$$\mu_t^{\Gamma,h} = \Gamma_t^h + \Gamma_B^h \cdot \left(i_t B_t^h - p_t \cdot C_t^h + w_t L_t^h + D_t \right) + \Gamma_A^h \cdot A_t g + \Gamma_\sigma^h \cdot \mu_t^\sigma$$
$$+ \frac{1}{2}\Gamma_{AA}^h \cdot (A_t\sigma)^2 + \frac{1}{2}\Gamma_{\sigma\sigma}^h \cdot (\sigma_t^\sigma)^2 + \Gamma_{A\sigma}^h \cdot (A_t\sigma)(\sigma_t^\sigma) + \Gamma_p^h \cdot \pi_t p_t,$$

where subscripts denote differentiation with respect to the corresponding variable, yielding the following first-order conditions:

$$\Gamma_B^h = \frac{1}{p_t C_t} = \frac{L_t^{\frac{1}{\eta}}}{w_t},$$

where we impose equilibrium conditions $C_t^h = C_t$ and $L_t^h = L_t$.

Under the general Ornstein-Uhlenbeck equilibrium described above, we can characterize the exact functional form of $\Gamma^h(B_t^h, p_t, A_t, \sigma_t^s, t)$. Again, using the fact that $w_t L_t - p_t \cdot C_t + D_t = 0$, together with $\log(C_t) = \left(\frac{\eta}{\eta+1}\right) \log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \log(A_t) + \hat{Y}_t$ and equation (A.2) to substitute for labor L_t, we can express the value function Γ^h of the household evaluated at the optimum as:[51]

$$\begin{aligned}
\Gamma^h =& \frac{1}{\rho} \cdot \left(\frac{\eta}{\eta+1}\right) \log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{\rho} \cdot \Gamma_B^h \cdot i_t B_t^h \\
&+ \frac{1}{\rho} \cdot \log A_t + \frac{1}{\rho} \cdot \Gamma_A^h \cdot A_t g + \frac{1}{2\rho}\Gamma_{AA}^h \cdot (A_t\sigma)^2 \\
&- \frac{1}{\rho}\Gamma_{A\sigma}^h \cdot (A_t\sigma) \cdot \phi_2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right) + \frac{1}{\rho} \cdot \Gamma_p^h \cdot \pi_t p_t + \frac{1}{\rho} \cdot \Gamma_t^h \\
&+ \frac{1}{\rho} \cdot \left[\mu - \frac{1}{2\phi_2}\left[\left(\sigma + \sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]\right] \\
&- \frac{1}{\rho} \cdot \left(\frac{\varepsilon-1}{\varepsilon}\right) \frac{e^{\left(\frac{\eta+1}{\eta}\right)\left[\mu - \frac{1}{2\phi_2}\left[\left(\sigma+\sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]\right]}}{1 + \frac{1}{\eta}} \\
&- \frac{1}{2\rho}\Gamma_\sigma^h \cdot \left(\frac{1}{\sigma + \sigma_t^s}\right)\left[\theta\left[(\sigma + \sigma_t^s)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right] + \left(\phi_2\right)^2\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)^2\right] \\
&+ \frac{1}{2\rho}\Gamma_{\sigma\sigma}^h \cdot \left(\phi_2\right)^2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)^2.
\end{aligned} \tag{N.5}$$

[51] Note that $i_t - \pi_t$ is a sole function of σ_t^s in equilibrium, as shown in (N.3).

As in Online Appendix B, we adopt a guess-and-verify approach by assuming that the value function Γ^h takes the following form:

$$\Gamma^{guess} = \frac{1}{\rho} \cdot \log(A_t) + h\left(\sigma_t^s\right) \cdot \frac{B_t^h}{p_t A_t} + \tilde{\Gamma}\left(\sigma_t^s\right). \tag{N.6}$$

Therefore, we assume that the value function depends on $\log(A_t)$; on a term that is a multiple of a function of σ_t^s and the ratio $\frac{B_t^h}{p_t A_t}$; and on a function of σ_t^s. Differentiating yields:

$$\begin{aligned}
\Gamma_B^{guess} &= \frac{h\left(\sigma_t^s\right)}{p_t A_t}, \ \Gamma_A^{guess} = \frac{1}{\rho A_t} - h\left(\sigma_t^s\right) \cdot \frac{B_t^h}{p_t A_t^2}, \ \Gamma_p^{guess} = -h(\sigma_t^s) \cdot \frac{B_t^h}{p_t^2 A_t} \\
\Gamma_{AA}^{guess} &= -\frac{1}{\rho\left(A_t\right)^2} + 2h\left(\sigma_t^s\right) \cdot \frac{B_t^h}{p_t A_t^3}, \ \Gamma_{A\sigma}^{guess} = -h'\left(\sigma_t^s\right) \cdot \frac{B_t^h}{p_t A_t^2}, \\
\Gamma_\sigma^{guess} &= h'\left(\sigma_t^s\right) \cdot \frac{B_t^h}{p_t A_t} + \tilde{\Gamma}'\left(\sigma_t^s\right), \ \Gamma_{\sigma\sigma}^{guess} = h''\left(\sigma_t^s\right) \cdot \frac{B_t^h}{p_t A_t} + \tilde{\Gamma}''\left(\sigma_t^s\right).
\end{aligned}$$

Finding $h(\sigma_t^s)$ First, collecting all the terms that have $\frac{B_t^h}{A_t}$, we see that $h(\sigma_t^s)$ satisfies

$$\begin{aligned}
h(\sigma_t^s) =& \frac{1}{\rho} \underbrace{\left(i_t - \pi_t\right)}_{=r_t} h(\sigma_t^s) - \frac{1}{\rho} g h(\sigma_t^s) + \frac{1}{\cancel{2}\rho}\sigma^2 \cdot \cancel{2} h(\sigma_t^s) \\
& - \frac{1}{\rho} h'(\sigma_t^s)\sigma\left[-\phi_2 \cdot \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)\right] \\
& - \frac{1}{2\rho} h'(\sigma_t^s)\left(\frac{1}{\sigma + \sigma_t^s}\right)\left[\theta\left[(\sigma + \sigma_t^s)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right] + \left(\phi_2\right)^2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)^2\right] \\
& + \frac{1}{2\rho} h''(\sigma_t^s)\left(\phi_2\right)^2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)^2.
\end{aligned} \tag{N.7}$$

We conjecture

$$h(\sigma_t^s) \propto \exp\left[\left(\frac{1}{2\phi_2}\right)\left[\left(\sigma + \sigma_t^s\right)^2\right]\right]. \tag{N.8}$$

If conjecture (N.8) is true, we obtain

$$h'(\sigma_t^s) = h(\sigma_t^s) \cdot \frac{1}{\phi_2}(\sigma + \sigma_t^s),$$
$$h''(\sigma_t^s) = h(\sigma_t^s) \cdot \left[\frac{1}{\phi_2} + \left(\frac{1}{\phi_2}\right)^2 (\sigma + \sigma_t^s)^2\right],$$

which can be plugged into (N.7) and lead to

$$\begin{aligned}\rho =&(r_t - g + \sigma^2) + \sigma\sigma_t^s \\ &- \frac{1}{2}\frac{1}{\phi_2}\cancel{(\sigma+\sigma_t^s)}\left(\frac{1}{\cancel{\sigma+\sigma_t^s}}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right] + (\phi_2)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\ &+ \frac{1}{2}\left[\frac{1}{\phi_2} + \left(\frac{1}{\phi_2}\right)^2(\sigma+\sigma_t^s)^2\right](\phi_2)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2,\end{aligned}$$

leading to the equilibrium real interest rate (N.3), thereby confirming our conjectural functional form (N.8). Finally, from the optimality condition

$$\Gamma_B^h = \Gamma_B = h(\sigma_t^s)\frac{1}{p_t A_t} = \frac{1}{p_t C_t} = \frac{1}{p_t Y_t}, \tag{N.9}$$

and with the help of (A.2), we obtain

$$h\left(\sigma_t^s\right) = \frac{1}{\left(\frac{\varepsilon-1}{\varepsilon}\right)^{\left(\frac{\eta}{\eta+1}\right)}} \cdot e^{-\left[\mu - \frac{1}{2\phi_2}\left[\left(\sigma+\sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]\right]}.$$

Finding $\tilde{\Gamma}(\sigma_t^s)$ The final term $\tilde{\Gamma}(\sigma_t^s)$ satisfies the following ordinary differential equation (ODE):

$$\begin{aligned}\tilde{\Gamma}\left(\sigma_t^s\right) =& \frac{1}{\rho}\cdot\left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{\rho^2}\left(g - \frac{1}{2}\sigma^2\right) \\ &+ \frac{1}{\rho}\cdot\left[\mu - \frac{1}{2\phi_2}\left[\left(\sigma+\sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]\right] \\ &- \frac{1}{\rho}\left(\frac{\eta}{\eta+1}\right)\left(\frac{\varepsilon-1}{\varepsilon}\right)\cdot e^{\left(\frac{\eta+1}{\eta}\right)\left[\mu - \frac{1}{2\phi_2}\left[\left(\sigma+\sigma_t^s\right)^2 - \left[\sigma^2 - 2\mu\phi_{total}\right]\right]\right]}\end{aligned}$$

$$
\begin{aligned}
&- \frac{1}{2\rho}\tilde{\Gamma}'\left(\sigma_t^s\right)\cdot\left(\frac{1}{\sigma+\sigma_t^s}\right)\Bigg[\theta\left[(\sigma+\sigma_t^s)^2-\left[\sigma^2-2\mu\phi_{total}\right]\right] \\
&\quad + \left(\phi_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\Bigg] \\
&+ \frac{1}{2\rho}\tilde{\Gamma}''\left(\sigma_t^s\right)\cdot\left(\phi_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2.
\end{aligned}
$$

We can rearrange the previous equation as

$$
\tilde{\Gamma}\left(\sigma_t^s\right)+\mathcal{A}\left(\sigma_t^s\right)\tilde{\Gamma}'\left(\sigma_t^s\right)+\mathcal{B}\left(\sigma_t^s\right)\tilde{\Gamma}''\left(\sigma_t^s\right)=\mathcal{R}\left(\sigma_t^s\right) \tag{N.10}
$$

where:

$$
\begin{aligned}
\mathcal{A}\left(\sigma_t^s\right) &= \frac{1}{2\rho}\left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2-\left[\sigma^2-2\mu\phi_{total}\right]\right]+\left(\phi_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
\mathcal{B}\left(\sigma_t^s\right) &= -\frac{1}{2\rho}\left(\phi_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2 \\
\mathcal{R}\left(\sigma_t^s\right) &= \frac{1}{\rho}\cdot\left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right)+\frac{1}{\rho^2}\left(g-\frac{1}{2}\sigma^2\right) \\
&\quad + \frac{1}{\rho}\cdot\left[\mu-\frac{1}{2\phi_2}\left[\left(\sigma+\sigma_t^s\right)^2-\left[\sigma^2-2\mu\phi_{total}\right]\right]\right] \\
&\quad - \frac{1}{\rho}\left(\frac{\eta}{\eta+1}\right)\left(\frac{\varepsilon-1}{\varepsilon}\right)\cdot e^{\left(\frac{\eta+1}{\eta}\right)\left[\mu-\frac{1}{2\phi_2}\left[\left(\sigma+\sigma_t^s\right)^2-\left[\sigma^2-2\mu\phi_{total}\right]\right]\right]}.
\end{aligned}
$$

Then, by Peano's Theorem (Walter, 1973), a function $\tilde{\Gamma}\left(\sigma_t^s\right)$ solving the ODE in (N.10) locally exists. It becomes global since all three functions $\mathcal{A}\left(\sigma_t^s\right)$, $\mathcal{B}\left(\sigma_t^s\right)$, and $\mathcal{R}\left(\sigma_t^s\right)$ are globally Lipschitz in $\mathcal{E}_t \equiv (\sigma+\sigma_t^s)^2-\sigma^2$,[52] and σ_t^s converges to a stationary distribution from Proposition 5, covering the full support up to ∞ on the equilibrium paths (Murray and Miller, 2007).

Several points are worth noting. First, the initial assumption that Γ depends on $\{B_t^h, p_t, A_t, \sigma_t^s, t\}$, i.e., $\Gamma=\Gamma(B_t^h, p_t, A_t, \sigma_t^s, t)$, can be simplified to $\Gamma(B_t, p_t, A_t, \sigma_t^s)$ by dropping the explicit time dependence (and hence eliminating Γ_t from the derivations). Second, with aggregate bonds in zero net supply, $B_t = 0$ for all t, the

[52]Note that $\mathcal{E}_t \geq 0$ almost surely for $\theta \geq 0$ and $\mu \leq 0$.

representative household's value function simplifies to:

$$\Gamma\left(A_t, \sigma_t^s\right) = \frac{1}{\rho} \cdot \log(A_t) + \tilde{\Gamma}\left(\sigma_t^s\right). \tag{N.11}$$

Transversality condition With the equilibrium value function given by (N.11) having the same form as (B.7), we can prove the transversality condition in the same way as in Online Appendix B.

O A Policy Rule Specification with Long-Run Targets

In this section, we solve the linearized model under the alternative monetary policy rule in (K.29), which specifies long-run targets for the output gap $\hat{Y}_t$, aggregate output variance $x_t \equiv (\sigma + \sigma_t^s)^2$, and inflation π_t. We treat these three long-run targets as free parameters. The equilibrium construction, however, remains largely the same as in Section 5.

Here, we repeat for convenience the expressions for the IS equation (K.7) and the linearized Phillips curve (K.29), given by:

$$d\hat{Y}_t = \left[i_t - \pi_t - r_t^T\right] dt + \left[\sqrt{x_t} - \sigma\right] dZ_t,$$
$$\mathbb{E}_t d(\pi_t - \bar{\pi}) = \left(\tilde{\kappa}^\pi(\pi_t - \bar{\pi}) - \tilde{\kappa}^y(\hat{Y}_t - \mu)\right) dt,$$

where $r_t^T = r^n - \frac{1}{2}(\sigma + \sigma_t^s)^2 + \frac{1}{2}\sigma^2$.

We conjecture, as usual, that the output gap follows an Ornstein-Uhlenbeck process of the form

$$d\hat{Y}_t = -\theta(\hat{Y}_t - \mu)dt + \left[\sqrt{x_t} - \sigma\right] dZ_t. \tag{O.1}$$

We also correctly conjecture that inflation satisfies

$$\pi_t - \bar{\pi} = \left(\frac{\tilde{\kappa}^y}{\theta + \tilde{\kappa}^\pi}\right)(\hat{Y}_t - \mu), \tag{O.2}$$

which satisfies (K.29) given our conjecture (O.1).

Combining the IS equation (K.7) with the Taylor rule (K.27) and the inflation relation (O.2), we obtain

$$d\hat{Y}_t = \left[\check{\phi}\left(\hat{Y}_t - \mu\right) + \left(\frac{1}{2} - \phi_{vol}\right)[x_t - \bar{x}]\right] dt + \left[\sqrt{x_t} - \sigma\right] dZ_t, \tag{O.3}$$

where $\check{\phi} \equiv \phi_y + (\phi_\pi - 1)\left(\frac{\tilde{\kappa}^y}{\theta + \tilde{\kappa}^\pi}\right)$.

Comparing the drift terms in (O.1) and (O.3), it follows that the output gap and volatility processes are determined as

$$\hat{Y}_t - \mu = -\frac{1}{2\check{\phi}_2}[x_t - \bar{x}], \tag{O.4}$$

where

$$\check{\phi}_2 = \frac{\theta + \check{\phi}}{1 - 2\phi_{vol}}. \tag{O.5}$$

Plugging (O.4) into (O.2), we obtain:

$$\pi_t - \bar{\pi} = -\frac{1}{2\check{\phi}_2}\left(\frac{\tilde{\kappa}^y}{\theta + \tilde{\kappa}^\pi}\right)[x_t - \bar{x}], \tag{O.6}$$

Using (O.1) and (O.4), we obtain the process for volatility:

$$dx_t = -\theta\,[x_t - \bar{x}]\,dt - 2\check{\phi}_2\left[\sqrt{x_t} - \sigma\right]dZ_t, \tag{O.7}$$

or equivalently,

$$\begin{aligned} d\sigma_t^s =& -\frac{1}{2}\left(\frac{1}{\sigma + \sigma_t^s}\right)\left[\theta\left[(\sigma + \sigma_t^s)^2 - \bar{x}\right] + (\check{\phi}_2)^2\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)^2\right]dt \\ & - \check{\phi}_2\left(\frac{\sigma_t^s}{\sigma + \sigma_t^s}\right)dZ_t. \end{aligned} \tag{O.8}$$

With equations (K.27), (O.4) and (O.6), the real interest rate $r_t = i_t - \pi_t$ is given by

$$r_t = r^T - \frac{1}{2}\left(1 - \frac{\theta}{\check{\phi}_2}\right)[x_t - \bar{x}] \tag{O.9}$$

Due to the isomorphic mathematical structure between (45) and (O.8), we can similarly prove the following proposition.

Proposition O.1 (Ornstein-Uhlenbeck Equilibrium)

With the linear New Keynesian Phillips curve (K.27), linearized around $\pi_t = \bar{\pi}$ and $\hat{Y}_t = \mu$, and the policy rule (K.27) that accounts for long-run targets of inflation, output gap, and total variance: $\bar{\pi}$, μ, and $\bar{x}$, respectively, we can construct a rational expectations equilibrium characterized by $\hat{Y}_t$ in (O.1), π_t given in (O.2), and the σ_t^s process in (O.8). This "Ornstein-Uhlenbeck" equilibrium satisfies the following properties:

Property 1 *For $\theta > 0$ and $\bar{x} > \sigma^2$,*[53] *the process of σ_t^s in (O.8) is stable and admits a unique stationary distribution. If $\check{\phi}_2 \neq 0$, in the limit $\sigma \longrightarrow 0$, this stationary distribution coincides with the generalized gamma distribution $GGD(a, d, p)$, given by*[54]

$$a = \sqrt{\frac{2\left(\check{\phi}_2\right)^2}{\theta}}, \quad d = \frac{\theta\overline{(\sigma^s)^2}}{\left(\check{\phi}_2\right)^2}, \quad \text{and} \quad p = 2, \tag{O.10}$$

where $\check{\phi}_2$ is defined in (O.5). a is the scale parameter, d is the power-law shape parameter, and p is the exponential shape parameter.

Property 2 *For $\theta \geq 0$ and $\bar{x} = \sigma^2$, the process of σ_t^s in (O.8) is non-stationary and its distribution degenerates to the perfectly stabilized equilibrium with $\sigma_t^s = 0$ as $t \longrightarrow \infty$.*

Property 3 *The long-run expectations of the output gap $\hat{Y}_t$, aggregate variance x_t and inflation π_t are given by*

$$\lim_{t\to\infty} \mathbb{E}_0\left[\hat{Y}_t\right] = \mu, \quad \lim_{t\to\infty} \mathbb{E}_0\left[x_t\right] = \bar{x}, \quad \text{and} \quad \lim_{t\to\infty} \mathbb{E}_0\left[\pi_t\right] = \bar{\pi}.$$

Property 3 shows that under this policy, the long-run monetary policy targets coincide with the unconditional expectations of the variables, so the policy is effectively targeting deviations from the unconditional mean. Note also that from equation (O.4), the output gap is perfectly stabilized around its long-run mean when $\check{\phi}_2 \longrightarrow +\infty$, which occurs only when $\phi_{vol} = \frac{1}{2}$ or $\check{\phi} \longrightarrow +\infty$.[55]

O.1 Equilibrium Value Function for the Representative Household and the Transversality Condition with the Linearized Phillips Curve and Long-Run Targets of Monetary Policy

This section derives the equilibrium value function of households and proves the Transversality condition with the log-linearized New Keynesian Phillips curve with

[53] Due to the same reason as in Proposition 3, $\bar{x} < \sigma^2$ is not allowed. See Online Appendix F.

[54] See, e.g., Boukai (2022) for an application of the generalized gamma distribution as a benchmark risk-neutral distribution in stochastic volatility models.

[55] $\check{\phi} \rightarrow +\infty$ when $\phi \rightarrow +\infty$, where ϕ is the Taylor coefficient defined in (34).

long-run targets (K.29), when the monetary authority can choose its long-run target for output gap $\hat{Y}_t$, total output variance x_t, and inflation π_t.

The Hamilton-Jacobi-Bellman (HJB) equation of household h is given by:

$$\rho \cdot \Gamma^h = \max_{C_t^h, L_t^h} \left\{ \log C_t^h - \frac{\left(L_t^h\right)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \mu_t^{\Gamma,h} \right\}. \tag{O.11}$$

where we assume B_t^h as the only endogenous state variable, and $\left\{p_t, A_t, \sigma_t^s\right\}$ as exogenous state variables from individual household's perspective. Note that p_t follows (28) while π_t is a function of $\hat{Y}_t$, which is in turn a function of σ_t^s. We calculate $\mu_t^{\Gamma,h}$ as follows:

$$\begin{aligned}\mu_t^{\Gamma,h} =& \Gamma_t^h + \Gamma_B^h \cdot \left(i_t B_t^h - p_t \cdot C_t^h + w_t L_t^h + D_t\right) + \Gamma_A^h \cdot A_t g + \Gamma_\sigma^h \cdot \mu_t^\sigma \\ &+ \frac{1}{2}\Gamma_{AA}^h \cdot (A_t\sigma)^2 + \frac{1}{2}\Gamma_{\sigma\sigma}^h \cdot (\sigma_t^\sigma)^2 + \Gamma_{A\sigma}^h \cdot (A_t\sigma)(\sigma_t^\sigma) + \Gamma_p^h \cdot \pi_t p_t,\end{aligned}$$

where subscripts denote differentiation with respect to the corresponding variable, yielding the following first-order conditions:

$$\Gamma_B^h = \frac{1}{p_t C_t} = \frac{L_t^{\frac{1}{\eta}}}{w_t},$$

where we impose equilibrium conditions $C_t^h = C_t$ and $L_t^h = L_t$.

Under the Ornstein-Uhlenbeck equilibrium described in Section O, we can characterize the exact functional form of $\Gamma^h(B_t^h, p_t, A_t, \sigma_t^s, t)$. Again, based on the fact that $w_t L_t - p_t \cdot C_t + D_t = 0$, together with $\log(C_t) = \left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \log(A_t) + \hat{Y}_t$ and equation (A.2) to substitute for labor L_t, we can express the value function Γ^h of the household evaluated at the optimum as:

$$\begin{aligned}
\Gamma^h =& \frac{1}{\rho}\cdot\left(\frac{\eta}{\eta+1}\right)\log\left(\frac{\varepsilon-1}{\varepsilon}\right)+\frac{1}{\rho}\cdot\Gamma_B^h\cdot i_t B_t^h+\frac{1}{\rho}\cdot\log A_t \\
&+\frac{1}{\rho}\cdot\Gamma_A^h\cdot A_t g+\frac{1}{2\rho}\Gamma_{AA}^h\cdot(A_t\sigma)^2 \\
&-\frac{1}{\rho}\Gamma_{A\sigma}^h\cdot(A_t\sigma)\cdot\check{\phi}_2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)+\frac{1}{\rho}\cdot\Gamma_p^h\cdot\pi_t p_t+\frac{1}{\rho}\cdot\Gamma_t^h \\
&+\frac{1}{\rho}\cdot\left[\mu-\frac{1}{2\check{\phi}_2}\left[\left(\sigma+\sigma_t^s\right)^2-\bar{x}\right]\right] \\
&-\frac{1}{\rho}\cdot\left(\frac{\varepsilon-1}{\varepsilon}\right)\frac{e^{\left(\frac{\eta+1}{\eta}\right)\left[\mu-\frac{1}{2\check{\phi}_2}\left[\left(\sigma+\sigma_t^s\right)^2-\bar{x}\right]\right]}}{1+\frac{1}{\eta}} \\
&-\frac{1}{2\rho}\Gamma_\sigma^h\cdot\left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2-\bar{x}\right]+\left(\check{\phi}_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
&+\frac{1}{2\rho}\Gamma_{\sigma\sigma}^h\cdot\left(\check{\phi}_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2.
\end{aligned} \tag{O.12}$$

where $i_t-\pi_t$ is a sole function of σ_t^s in equilibrium, as shown in (N.3).

As in Online Appendix B, we adopt a guess-and-verify approach by assuming that the value function Γ^h takes the following form:

$$\Gamma^{guess}=\frac{1}{\rho}\cdot\log(A_t)+h\left(\sigma_t^s\right)\cdot\frac{B_t^h}{p_t A_t}+\tilde{\Gamma}\left(\sigma_t^s\right). \tag{O.13}$$

Therefore, we assume that the value function depends on $\log(A_t)$; on a term that is a multiple of a function of σ_t^s and the ratio $\frac{B_t^h}{p_t A_t}$; and on a function of σ_t^s. Differentiating yields:

$$\begin{aligned}
&\Gamma_B^{guess}=\frac{h\left(\sigma_t^s\right)}{p_t A_t},\ \Gamma_A^{guess}=\frac{1}{\rho A_t}-h\left(\sigma_t^s\right)\cdot\frac{B_t^h}{p_t A_t^2},\ \Gamma_p^{guess}=-h(\sigma_t^s)\cdot\frac{B_t^h}{p_t^2 A_t} \\
&\Gamma_{AA}^{guess}=-\frac{1}{\rho\left(A_t\right)^2}+2h\left(\sigma_t^s\right)\cdot\frac{B_t^h}{p_t A_t^3},\ \Gamma_{A\sigma}^{guess}=-h'\left(\sigma_t^s\right)\cdot\frac{B_t^h}{p_t A_t^2}, \\
&\Gamma_\sigma^{guess}=h'\left(\sigma_t^s\right)\cdot\frac{B_t^h}{p_t A_t}+\tilde{\Gamma}'\left(\sigma_t^s\right),\ \Gamma_{\sigma\sigma}^{guess}=h''\left(\sigma_t^s\right)\cdot\frac{B_t^h}{p_t A_t}+\tilde{\Gamma}''\left(\sigma_t^s\right).
\end{aligned}$$

Finding $h(\sigma_t^s)$ First, collecting all the terms that have $\frac{B_t^h}{A_t}$, we see that $h(\sigma_t^s)$ satisfies

$$
\begin{aligned}
h(\sigma_t^s) =& \frac{1}{\rho}\underbrace{(i_t - \pi_t)}_{=r_t} h(\sigma_t^s) - \frac{1}{\rho} g h(\sigma_t^s) + \frac{1}{\cancel{2}\rho}\sigma^2 \cdot \cancel{2} h(\sigma_t^s) \\
& - \frac{1}{\rho} h'(\sigma_t^s)\sigma \left[-\check{\phi}_2 \cdot \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s} \right) \right] \\
& - \frac{1}{2\rho} h'(\sigma_t^s) \left(\frac{1}{\sigma + \sigma_t^s} \right) \left[\theta \left[(\sigma + \sigma_t^s)^2 - \bar{x} \right] + \left(\check{\phi}_2 \right)^2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s} \right)^2 \right] \\
& + \frac{1}{2\rho} h''(\sigma_t^s) \left(\check{\phi}_2 \right)^2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s} \right)^2 .
\end{aligned}
\tag{O.14}
$$

We conjecture

$$
h(\sigma_t^s) \propto \exp\left[\left(\frac{1}{2\check{\phi}_2} \right) \left(\sigma + \sigma_t^s \right)^2 \right]. \tag{O.15}
$$

If conjecture (O.15) is true, we obtain

$$
\begin{aligned}
h'(\sigma_t^s) &= h(\sigma_t^s) \cdot \frac{1}{\check{\phi}_2} (\sigma + \sigma_t^s), \\
h''(\sigma_t^s) &= h(\sigma_t^s) \cdot \left[\frac{1}{\check{\phi}_2} + \left(\frac{1}{\check{\phi}_2} \right)^2 (\sigma + \sigma_t^s)^2 \right],
\end{aligned}
$$

which can be plugged into (O.14) and lead to

$$
\begin{aligned}
\rho =& (i_t - g + \sigma^2) + \sigma \sigma_t^s \\
& - \frac{1}{2} \frac{1}{\check{\phi}_2} \cancel{(\sigma + \sigma_t^s)} \left(\frac{1}{\cancel{\sigma + \sigma_t^s}} \right) \left[\theta \left[(\sigma + \sigma_t^s)^2 - \bar{x} \right] + \left(\check{\phi}_2 \right)^2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s} \right)^2 \right] \\
& + \frac{1}{2} \left[\frac{1}{\check{\phi}_2} + \left(\frac{1}{\check{\phi}_2} \right)^2 (\sigma + \sigma_t^s)^2 \right] \left(\check{\phi}_2 \right)^2 \left(\frac{\sigma_t^s}{\sigma + \sigma_t^s} \right)^2 ,
\end{aligned}
$$

leading to the equilibrium real interest rate (O.9), thereby confirming our conjectural functional form (O.15). Finally, from the optimality condition

$$\Gamma_B^h = \Gamma_B = h(\sigma_t^s)\frac{1}{p_t A_t} = \frac{1}{p_t C_t} = \frac{1}{p_t Y_t}, \tag{O.16}$$

and with the help of (A.2), we obtain

$$h\left(\sigma_t^s\right) = \frac{1}{\left(\frac{\varepsilon-1}{\varepsilon}\right)^{\left(\frac{\eta}{\eta+1}\right)}} \cdot e^{-\left[\mu-\frac{1}{2\check{\phi}_2}\left[\left(\sigma+\sigma_t^s\right)^2-\bar{x}\right]\right]}.$$

Finding $\tilde{\Gamma}(\sigma_t^s)$ The final term $\tilde{\Gamma}(\sigma_t^s)$ satisfies the following ordinary differential equation (ODE):

$$\begin{aligned}
\tilde{\Gamma}\left(\sigma_t^s\right) = {} & \frac{1}{\rho} \cdot \left(\frac{\eta}{\eta+1}\right) \log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{\rho^2}\left(g - \frac{1}{2}\sigma^2\right) \\
& + \frac{1}{\rho} \cdot \left[\mu - \frac{1}{2\check{\phi}_2}\left[\left(\sigma+\sigma_t^s\right)^2 - \bar{x}\right]\right] \\
& - \frac{1}{\rho}\left(\frac{\eta}{\eta+1}\right)\left(\frac{\varepsilon-1}{\varepsilon}\right) \cdot e^{\left(\frac{\eta+1}{\eta}\right)\left[\mu-\frac{1}{2\check{\phi}_2}\left[\left(\sigma+\sigma_t^s\right)^2-\bar{x}\right]\right]} \\
& - \frac{1}{2\rho}\tilde{\Gamma}'\left(\sigma_t^s\right) \cdot \left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2 - \bar{x}\right] + \left(\check{\phi}_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
& + \frac{1}{2\rho}\tilde{\Gamma}''\left(\sigma_t^s\right) \cdot \left(\check{\phi}_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2.
\end{aligned}$$

We can rearrange the previous equation as

$$\tilde{\Gamma}\left(\sigma_t^s\right) + \mathcal{A}\left(\sigma_t^s\right)\tilde{\Gamma}'\left(\sigma_t^s\right) + \mathcal{B}\left(\sigma_t^s\right)\tilde{\Gamma}''\left(\sigma_t^s\right) = \mathcal{R}\left(\sigma_t^s\right) \tag{O.17}$$

where:

$$\begin{aligned}
\mathcal{A}\left(\sigma_t^s\right) &= \frac{1}{2\rho}\left(\frac{1}{\sigma+\sigma_t^s}\right)\left[\theta\left[(\sigma+\sigma_t^s)^2 - \bar{x}\right] + \left(\check{\phi}_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2\right] \\
\mathcal{B}\left(\sigma_t^s\right) &= -\frac{1}{2\rho}\left(\check{\phi}_2\right)^2\left(\frac{\sigma_t^s}{\sigma+\sigma_t^s}\right)^2
\end{aligned}$$

$$
\begin{aligned}
\mathcal{R}\left(\sigma_t^s\right) = {} & \frac{1}{\rho} \cdot \left(\frac{\eta}{\eta+1}\right) \log\left(\frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{\rho^2}\left(g - \frac{1}{2}\sigma^2\right) \\
& + \frac{1}{\rho} \cdot \left[\mu - \frac{1}{2\check{\phi}_2}\left[\left(\sigma + \sigma_t^s\right)^2 - \bar{x}\right]\right] \\
& - \frac{1}{\rho}\left(\frac{\eta}{\eta+1}\right)\left(\frac{\varepsilon-1}{\varepsilon}\right) \cdot e^{\left(\frac{\eta+1}{\eta}\right)\left[\mu - \frac{1}{2\check{\phi}_2}\left[\left(\sigma+\sigma_t^s\right)^2 - \bar{x}\right]\right]}.
\end{aligned}
$$

Then, by Peano's Theorem (Walter, 1973), a function $\tilde{\Gamma}\left(\sigma_t^s\right)$ solving the ODE in (O.17) locally exists. It becomes global as $\mathcal{A}\left(\sigma_t^s\right)$, $\mathcal{B}\left(\sigma_t^s\right)$, and $\mathcal{R}\left(\sigma_t^s\right)$ are globally Lipschitz in $\mathcal{E}_t \equiv (\sigma + \sigma_t^s)^2 - \sigma^2$,[56] and σ_t^s converges to a stationary distribution from Proposition O.1, covering the full support up to ∞ on the equilibrium paths (Murray and Miller, 2007).

Several points are worth noting. First, the initial assumption that Γ depends on $\{B_t^h, p_t, A_t, \sigma_t^s, t\}$, i.e., $\Gamma = \Gamma(B_t^h, p_t, A_t, \sigma_t^s, t)$, can be simplified to $\Gamma(B_t, p_t, A_t, \sigma_t^s)$ by dropping the explicit time dependence (and hence eliminating Γ_t from the derivations). Second, with aggregate bonds in zero net supply, $B_t = 0$ for all t, the representative household's value function simplifies to:

$$
\Gamma\left(A_t, \sigma_t^s\right) = \frac{1}{\rho} \cdot \log(A_t) + \tilde{\Gamma}\left(\sigma_t^s\right). \tag{O.18}
$$

Transversality condition With the equilibrium value function given by (O.18) having the same form as (B.7), we can prove the transversality condition in the same way as in Online Appendix B.

[56]Note that $\mathcal{E}_t \geq 0$ almost surely for $\theta \geq 0$ and $\bar{x} > \sigma^2$.

P Ricardian Fiscal Policies and the Transversality Condition

In this section, we assume that the government can issue a positive amount of bonds starting from $B_0 > 0$ at $t = 0$, in an economy with the log-linearized Phillips curve of Section 5.[57] We derive conditions on bond issuance that satisfy the household transversality condition and a no-Ponzi-game restriction of the form used in textbook New Keynesian treatments, which discounts remaining debt by the cumulative short rate (Galí, 2015, 2020).

P.1 Households

Recall that the price process under sticky prices à la Rotemberg (1982) is given by:

$$dp_t = \pi_t p_t dt. \tag{P.1}$$

which leads to

$$p_t = p_0 \, e^{\bar{\pi}_t t}, \tag{P.2}$$

where $\bar{\pi}_t \equiv \frac{1}{t}\int_0^t \pi_s ds$ is the time-average inflation rate between time 0 and t.

The flow budget constraint of the household is given by:

$$\dot{B}_t = i_t B_t - p_t C_t + w_t L_t + D_t. \tag{P.3}$$

We consider the case where the initial bond issuance is positive, i.e., $B_0 > 0$. Combining (P.3) and (P.1) yields the flow budget constraint of the household in real terms:

$$d\left(\frac{B_t}{p_t}\right) = \left[(i_t - \pi_t)\frac{B_t}{p_t} - C_t + \frac{w_t L_t}{p_t} + \frac{D_t}{p_t}\right] dt. \tag{P.4}$$

Let $\bar{i}_t \equiv \frac{1}{t}\int_0^t i_s ds$ denote the time-average nominal interest rate between 0 and t. Multiplying both sides of (P.4) by the integrating factor $e^{-(\bar{i}_t - \bar{\pi}_t)t}$ and integrating

[57] In particular, we consider inflation whose dynamics is given by the linearized New Keynesian Phillips curve (31).

from 0 to t gives

$$\left(\frac{B_t}{p_t}\right) e^{-(\bar{i}_t - \bar{\pi}_t)t} - \frac{B_0}{p_0} = \int_0^t \left[\frac{w_s L_s}{p_s} + \frac{D_s}{p_s} - C_s\right] e^{-(\bar{i}_t - \bar{\pi}_t)t} \, ds.$$

Taking expectations:

$$\mathbb{E}_0 \left[\left(\frac{B_t}{p_t}\right) e^{-(\bar{i}_t - \bar{\pi}_t)t}\right] - \frac{B_0}{p_0} = \mathbb{E}_0 \left[\int_0^t \left[\frac{w_s L_s}{p_s} + \frac{D_s}{p_s} - C_s\right] e^{-(\bar{i}_s - \bar{\pi}_s)s} \, ds\right]. \tag{P.5}$$

Let us assume that transfers D_t are given by

$$D_t = p_t Y_t - w_t L_t - p_t T_t, \tag{P.6}$$

where T_t denotes (real) lump-sum taxes.

Substituting equation (P.6) into (P.5) yields

$$\mathbb{E}_0 \left[\left(\frac{B_t}{p_t}\right) e^{-(\bar{i}_t - \bar{\pi}_t)t}\right] - \frac{B_0}{p_0} = \mathbb{E}_0 \left[\int_0^t [Y_s - T_s - C_s] \, e^{-(\bar{i}_s - \bar{\pi}_s)s} \, ds\right]. \tag{P.7}$$

P.2 Government

The government's flow budget constraint is

$$\dot{B}_t = -p_t T_t + i_t B_t, \tag{P.8}$$

where $T_t > 0$ corresponds to taxes while $T_t < 0$ corresponds to fiscal transfers (or deficits).[58]

Equation (P.8), using equation (P.1), can be written in real terms as

$$d\left(\frac{B_t}{p_t}\right) = \left[(i_t - \pi_t)\frac{B_t}{p_t} - T_t\right] dt. \tag{P.9}$$

Multiplying by the same integrating factor and taking expectations as before, we

[58]For tractability, we abstract from government spending: government spending will affect both aggregate demand—dynamic IS equation (30)—and aggregate supply—linearized Phillips curve (31)—as in Galí (2020). All the findings in this section remain qualitatively valid when government spending is included.

obtain

$$\mathbb{E}_0\left[\left(\frac{B_t}{p_t}\right)e^{-(\bar{i}_t-\bar{\pi}_t)t}\right]-\frac{B_0}{p_0}=-\mathbb{E}_0\left[\int_0^t T_s\, e^{-(\bar{i}_s-\bar{\pi}_s)s}\,ds\right]. \tag{P.10}$$

Combining equations (P.7) and (P.10) yields

$$0=\mathbb{E}_0\left[\int_0^t [Y_s-C_s]\,e^{-(\bar{i}_s-\bar{\pi}_s)s}\,ds\right]. \tag{P.11}$$

Finally, applying Lemma I.3 to equation (P.11), we obtain the standard goods market clearing condition:

$$Y_t=C_t. \tag{P.12}$$

It is worth noting that the goods market clearing condition (P.12) remains unchanged. The constructed equilibria (the martingale equilibrium and the Ornstein-Uhlenbeck equilibria) still emerge even when government bond issuance is positive if fiscal policy is Ricardian and those equilibria satisfy the household transversality condition, which will be discussed more precisely below.

P.3 Equilibrium Behaviors in the Long Run

Given the same market clearing condition (P.12), we will show that the equilibrium characterized in Section 5 is feasible even when government bond supply is non-zero, under the conditions in Section P.4 and Section P.5.

If the Ornstein-Uhlenbeck equilibria in Section 5 are viable, we first characterize the long-run behavior of the economy under that equilibrium. By Property 3 of Proposition 5 with equation (40), the expected inflation is:

$$\lim_{t\to\infty}\mathbb{E}_0[\pi_t]=\frac{\kappa^y}{\kappa^\pi}\,\mu. \tag{P.13}$$

From Online Appendix C.2, $\{\sigma_t^s\}_{t\geq 0}$ process is stable and ergodic (Meyn and Tweedie, 1993, 1998, 2012; Duc et al., 2018), so do π_t and $\hat{Y}_t$ processes, given that both are functions of σ_t^s only, given by equations (40) and (41), respectively:

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t \pi_s \, ds = \lim_{t\to\infty} \mathbb{E}_0\left[\pi_t\right] = \frac{\kappa^y}{\kappa^\pi}\,\mu$$
$$\lim_{t\to\infty} \frac{1}{t}\int_0^t \hat{Y}_s \, ds = \lim_{t\to\infty} \mathbb{E}_0\left[\hat{Y}_t\right] = \mu.$$

P.4 No-Ponzi-Game Condition

Integrating the government's flow budget constraint yields a present-value identity in which remaining real debt is discounted by the money-market factor $e^{-(\bar{i}_t-\bar{\pi}_t)t}$. Following Galí (2015) and Galí (2020), we record the associated No-Ponzi-Game (NPG) restriction, which requires that this terminal term vanish. In those textbook settings the same discount factor $e^{-(\bar{i}_t-\bar{\pi}_t)t}$ is often used for the household and the government, so the government condition is implied by household transversality. That coincidence usually holds under perfect foresight and in a linearization around a deterministic steady state, where the Euler equation equates the cumulative real rate with the household's stochastic discount factor.

The economically relevant no-bubble restriction is different. Household optimality requires the transversality condition that values remaining debt at the stochastic discount factor (Acemoglu, 2009; Woodford, 2001a, 2003). In a representative-agent economy this is also the intertemporal government budget constraint at Arrow-Debreu prices, and it is the criterion of Santos and Woodford (1997). Bohn (1995) shows that, in a stochastic economy, that restriction is not equivalent to discounting debt at the safe or average short rate: debt discounted at the short rate may fail to vanish even when the stochastic-discount-factor condition holds. We show that this is the case under our constructed equilibria.

The NPG below is therefore the short-rate identity. We do not treat it as a second optimality condition, and we do not interpret its failure as a rational bubble.

Formally,

$$\lim_{t\to\infty} \mathbb{E}_0\left[\left(\frac{B_t}{p_t}\right) e^{-(\bar{i}_t-\bar{\pi}_t)t}\right] \le 0. \tag{P.14}$$

Using equation (P.2) in (P.14) we obtain

$$\lim_{t\to\infty} \mathbb{E}_0 \left[\left(\frac{B_t}{p_0} \right) e^{-\bar{i}_t t} \right] \leq 0. \tag{P.15}$$

To check whether the No-Ponzi-Game condition is satisfied, we must specify the process for taxes or transfers (T), or equivalently the process for government bond issuance (B). For example, suppose that government debt grows at rate $g_{B,t}$ instantaneously:

$$dB_t = g_{B,t} B_t dt. \tag{P.16}$$

Substituting (P.16) into equation (P.15) yields

$$\underbrace{\left(\frac{B_0}{p_0} \right)}_{>0} \lim_{t\to\infty} e^{-(\bar{i}_t - \overline{g}_{B,t})t} \leq 0, \tag{P.17}$$

where $\overline{g}_{B,t} \equiv \frac{1}{t} \int_0^t g_{B,s} ds$, which is equivalent to

$$\underbrace{\left(\frac{B_0}{p_0} \right)}_{>0} \lim_{t\to\infty} e^{-(\bar{i}_t - \overline{g}_{B,t})t} = 0.$$

Using (K.20) and the long-run expected excess variance $-2\mu\phi_{total}$ under that rule, the long-run average nominal interest rate is

$$\begin{aligned} \lim_{t\to\infty} \bar{i}_t &= r^n + \left(\phi_y + \frac{\kappa^y}{\kappa^\pi} \phi_\pi \right) \mu - \phi_{vol} \cdot \left(-2\mu\phi_{total} \right) \\ &= r^n + \frac{1}{1 - 2\phi_{vol}} \left[\phi_y + \left(\phi_\pi - 2\phi_{vol} \right) \frac{\kappa^y}{\kappa^\pi} \right] \mu. \end{aligned}$$

Therefore, when $B_0 > 0$, the No-Ponzi-Game condition that satisfies (P.17) requires

$$\lim_{t\to\infty} \overline{g}_{B,t} < r^n + \frac{1}{1 - 2\phi_{vol}} \left[\phi_y + \left(\phi_\pi - 2\phi_{vol} \right) \frac{\kappa^y}{\kappa^\pi} \right] \mu. \tag{P.18}$$

If (P.17) holds for any price-level path, and especially if (P.18) holds in equilibrium, we follow Galí (2015) and Galí (2020) in calling fiscal policy *Ricardian*, and we assume that restriction throughout this section. Woodford (2001a) and Woodford (2003) define Ricardian policy by the requirement that the stochastic-discount-factor transversality condition hold for all price and asset-price processes. In the linearized New Keynesian model the two definitions coincide. In the global equilibria studied here they need not, as we record below. Fiscal policy is Ricardian in the textbook sense only if there is a constraint on how quickly bond issuance can expand. Under that condition, fiscal transfer T_0 at time 0 has no effect on consumption, equivalently output, at $t = 0$.

P.5 Transversality Condition

The original transversality condition for households is given by

$$\lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\rho t} \Gamma_t \right] = 0 \tag{P.19}$$

where Γ_t is the equilibrium value function of the household given by (O.13) with $h(\sigma_t^s)$ given by (O.16). Given that the $\{\sigma_t^s\}$ process is stable and ergodic, it is sufficient to check

$$\lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\rho t} h(\sigma_t^s) \frac{B_t}{p_t A_t} \right] = \lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\rho t} \frac{1}{Y_t} \frac{B_t}{p_t} \right] = 0. \tag{P.20}$$

where we use $h(\sigma_t^s) = \frac{A_t}{C_t} = \frac{A_t}{Y_t}$ from (O.16). Note that equation (P.20) is of a usual form of the transversality condition (Acemoglu, 2009) given by:

$$\lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\rho t} u'(C_t) \frac{B_t}{P_t} \right] = 0.$$

Given that the $\{\sigma_t^s\}$ process converges to its stationary distribution for $\mu < 0$ and $\theta \geq 0$ in (39) while $h(\sigma_t^s)$ is a well-defined function, it is sufficient to prove that

$$\lim_{t\to+\infty} \mathbb{E}_0 \left[e^{-\rho t} \frac{B_t}{p_t A_t} \right] = 0. \tag{P.21}$$

Given that

$$\lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\rho t} \frac{B_t}{p_t A_t} \right] = \frac{B_0}{p_0 A_0} \cdot \lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\left(\rho + g - \sigma^2\right)t} \cdot e^{-\left(\overline{\pi}_t - \overline{g}_{B,t}\right)t} \right] \tag{P.22}$$

$$= \frac{B_0}{p_0 A_0} \cdot \lim_{t\to\infty} \mathbb{E}_0 \left[e^{-\left(r^n + \overline{\pi}_t - \overline{g}_{B,t}\right)t} \right], \tag{P.23}$$

we obtain the following sufficient condition for the transversality condition:

$$\lim_{t\to\infty} \overline{g}_{B,t} < r^n + \frac{\kappa^y}{\kappa^\pi}\mu. \tag{P.24}$$

Comparison Comparing inequalities (P.18) and (P.24), we find that (P.18) is sufficient for (P.24) since $\mu \le 0$ and $\phi_{vol} < \frac{1}{2}$. Therefore, the rate-compounded NPG requires

$$\lim_{t\to\infty} \overline{g}_{B,t} < r^n + \frac{1}{1 - 2\phi_{vol}} \left[\phi_y + \left(\phi_\pi - 2\phi_{vol}\right) \frac{\kappa^y}{\kappa^\pi} \right] \mu,$$

and in that case the household transversality condition (P.19) is satisfied.[59]

Interpretation We can restate the above two restrictions as follows. The household transversality condition (TVC) and the rate-compounded No-Ponzi-Game condition (NPG) reduce to

$$\lim_{t\to\infty} \bar{g}_{B,t} < r^n + \lim_{t\to\infty} \bar{\pi}_t \qquad \text{(TVC)},$$

$$\lim_{t\to\infty} \bar{g}_{B,t} < r^n + \lim_{t\to\infty} \bar{\pi}_t + \phi_{total}\,\mu \qquad \text{(NPG)},$$

where $\phi_{total} > 0$ as long as the Taylor principle is satisfied and $\phi_{vol} < 1/2$.

When $\mu = 0$, the long-run equilibrium of the model coincides with that of the log-linearized New Keynesian model. The two conditions then coincide, which is immediate from the expressions above. That is the environment in which textbook treatments equate the government NPG with household transversality (Galí, 2015,

[59] It has been proposed in the literature (Woodford, 2001a, 2003; Galí, 2015, 2020) that in a Ricardian fiscal regime, the equilibrium should be determined by active monetary policy. In that regard, we uncover that even active monetary policy, i.e., $\phi = \phi_y + (\phi_\pi - 1)\frac{\kappa^y}{\kappa^\pi} > 0$, does not guarantee determinacy under the Ricardian fiscal policy in a global environment. The only way to pin down equilibrium is to have $\phi_{vol} = \frac{1}{2}$.

2020). The TVC is an optimality condition of the household. It is also the restriction that corresponds to the no-bubble criterion of Santos and Woodford (1997) and to the stochastic-discount-factor transversality condition used to define Ricardian policy in Woodford (2001a) and Woodford (2003): remaining public debt is valued at the household's stochastic discount factor. With logarithmic utility and goods-market clearing, the present value of aggregate resources is finite, and government bonds are in positive net supply when $B_0 > 0$, so that criterion applies. A rational bubble on public debt is therefore ruled out whenever the TVC holds. The literature on rational bubbles in New Keynesian models accordingly uses overlapping-generations or perpetual-youth structures, in which the present value of aggregate resources need not be finite (Yaari, 1965; Blanchard, 1985; Galí, 2021; Asriyan et al., 2021).

The NPG discounts the same debt by the average nominal interest rate. Those operations differ by the wedge $\phi_{total}\,\mu$. $\phi_{total} > 0$ implies that $\phi_{total}\,\mu < 0$ whenever $\mu < 0$, so there is an interval of debt-growth rates for which the TVC holds and the NPG does not. As Bohn (1995) emphasizes, that configuration is possible in a stochastic economy: the state-price intertemporal budget constraint may hold even if debt discounted at the average short rate does not vanish. The interval is therefore a fiscal accounting restriction under money-market discounting. It is not a rational bubble in the sense of Santos and Woodford (1997), and we do not interpret it as a bubbly equilibrium.

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